

DNS Measurements in Space and Time

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Abstract

The DNS is a critical Internet infrastructure translating domain names into IP addresses. CDNs leverage this system to implement geo-routing, returning different IP addresses based on the client's geographic location. This thesis presents an empirical study of DNS geographic diversity for the top 10,000 most popular domains (Tranco list), measured daily over 90 days from ≈ 100 globally distributed RIPE Atlas vantage points. We quantify the prevalence of geographic diversity (Q1), its temporal stability (Q2), the impact of RIPE Atlas geographic bias (Q3), and the effect of resolver choice (Q4). Results reveal that $\approx 30\%$ of Tranco Top-10K domains exhibit significant geographic diversity, stable over 90 days, with measurable effects from platform bias and resolver choice.

Keywords: DNS, distributed measurements, geolocation, CDN, RIPE Atlas, geographic diversity, Tranco

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Acronyms

A: Address Record (IPv4 DNS record)

AAAA: IPv6 Address Record (IPv6 DNS record)

API: Application Programming Interface

AS: Autonomous System

ASN: Autonomous System Number

BGP: Border Gateway Protocol

CDN: Content Delivery Network

CNAME: Canonical Name Record

DNS: Domain Name System

DNSSEC: DNS Security Extensions

DoH: DNS over HTTPS

DoT: DNS over TLS

ECS: EDNS Client Subnet

EDNS: Extension Mechanisms for DNS

GeoIP: IP Geolocation Database

HTTP: HyperText Transfer Protocol

HTTPS: HyperText Transfer Protocol Secure

ICANN: Internet Corporation for Assigned Names and Numbers

IETF: Internet Engineering Task Force

IP: Internet Protocol

IPv4: Internet Protocol version 4

IPv6: Internet Protocol version 6

ISP: Internet Service Provider

MX: Mail Exchanger Record

NS: Name Server Record

NSID: Name Server Identifier

PoP: Point of Presence

QoS: Quality of Service

RFC: Request for Comments

RIPE: Réseaux IP Européens

RTT: Round-Trip Time

TCP: Transmission Control Protocol

TLD: Top-Level Domain

TLS: Transport Layer Security

TTL: Time To Live

TXT: Text Record

UDP: User Datagram Protocol

URL: Uniform Resource Locator

Chapter 1

Introduction

1.1 Context and Motivation

1.1.1 The DNS: Critical Infrastructure of the Internet

The Domain Name System (DNS), specified by Paul Mockapetris in [RFC1034, 1987; RFC1035, 1987], is the distributed directory service that underlies virtually every networked application on the Internet. Its function — translating human-readable domain names such as `www.example.com` into machine-routable information such as IPv4 and IPv6 addresses, mail server designations, and cryptographic key references — is so fundamental that van Rijswijk-Deij et al. [van Rijswijk-Deij et al, 2016] characterise it as a service on which "almost all networked services depend". The scale of this dependency is correspondingly enormous: the `.com` Top-Level Domain alone contained approximately 123 million registered names during the 2015–2016 period studied by van Rijswijk-Deij et al. [van Rijswijk-Deij et al, 2016], and OpenINTEL's measurement infrastructure must process at least 1.85 billion DNS queries per day for `.com` alone to achieve full daily coverage.

The criticality of the DNS is most visible when it fails. The Facebook BGP misconfiguration of October 4, 2021 — which withdrew the BGP routes to Facebook's authoritative name servers, making the domain unresolvable from any resolver worldwide — illustrates what Xu et al. [Xu et al, 2023] document empirically: when over 98% of all global domains depend on a single DNS provider, a provider-level failure propagates immediately and universally. The Akamai DNS outage of June 2021 and the Fastly CDN outage of the same month produced comparable effects, crippling access to large portions of the web for millions of users. In each case, the DNS was the critical failure point: not the application servers, not the network, but the name resolution layer that clients depend on before any application connection can be established.

Despite this criticality, the DNS has historically received less systematic measurement attention than other Internet infrastructure layers. Bortzmeyer [Bortzmeyer, 2017] observes that the DNS is "often forgotten in studies of Internet resilience and quality of service" — functioning so transparently in normal operation that its role is fully recognised only in failure. This relative neglect is reflected in the state of measurement infrastructure: as of this thesis's writing, the most comprehensive longitudinal DNS measurement system — OpenINTEL, developed at the University of Twente [van Rijswijk-Deij et al, 2016] — measures over 50% of the global DNS namespace daily, but does so from **a single vantage**

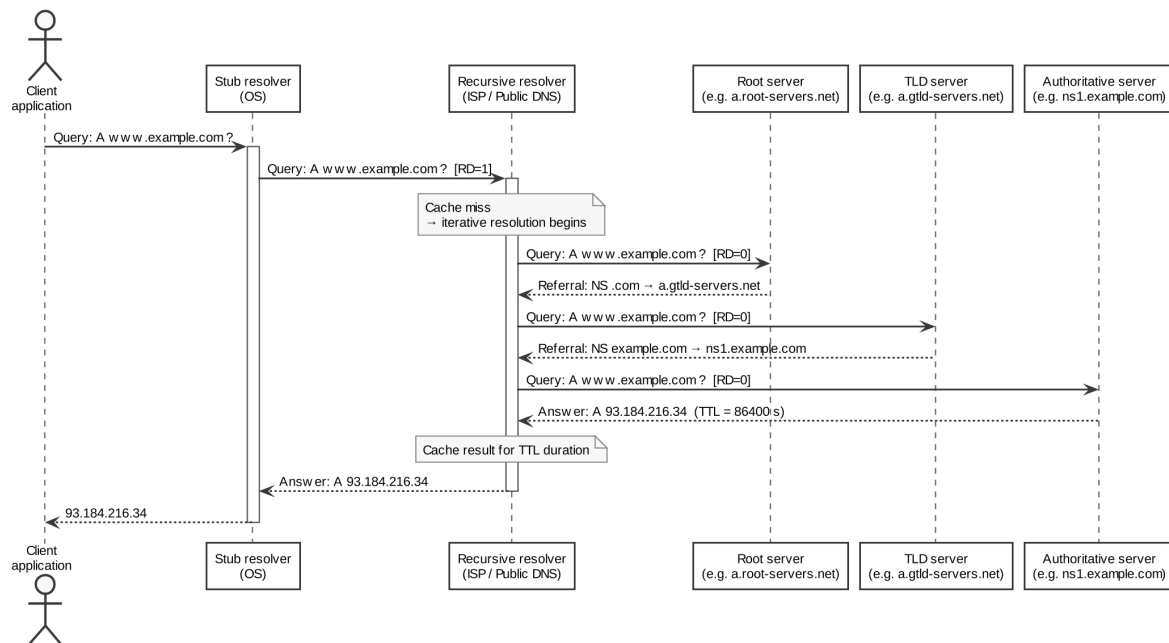


Figure 1.1: Iterative DNS resolution process. The stub resolver forwards the query to a recursive resolver, which performs three successive referral queries (root → TLD → authoritative) before returning the final answer. The TTL field in the authoritative response governs how long the recursive resolver caches the result.

point in the Netherlands. The geographic dimension of DNS responses is not captured. Figure 1.1 illustrates the standard iterative resolution process.

1.1.2 The Ephemeral Nature of DNS Data

A fundamental characteristic of the DNS that motivates archival measurement is the **ephemeral nature of DNS data**. DNS resource records — the atomic units of information in the DNS — carry a Time To Live (TTL) field that governs how long resolvers and caches may retain them [RFC1034, 1987; RFC1035, 1987]. When a TTL expires, the record is discarded; subsequent queries retrieve a fresh response from the authoritative server, which may contain different information. Zone administrators can modify records at any time: a domain that resolved to a server cluster in Europe this morning may resolve to a server cluster in Asia this afternoon, and the DNS infrastructure itself retains no trace of the previous configuration.

This ephemerality creates a structural gap in Internet research. As the thesis subject statement articulates: "in certain research domains, it is interesting to be able to obtain the DNS information provided at a given period." The use cases are varied but share a common dependency on historical DNS data:

Network simulation: Researchers developing Internet simulators — for evaluating routing protocols, security mechanisms, or traffic engineering policies — need realistic DNS data for their simulated networks. Simulating the Internet as it existed two years ago requires knowing what addresses the DNS returned for major domains at that time, from diverse geographic vantage points. Without archival DNS data, these simulations rely on present-day snapshots that do not reflect the historical state of the network.

Security forensics: Malicious infrastructure — command-and-control servers for botnets, phishing domains, spam networks — frequently changes its DNS configuration to evade detection and blacklisting. Van der Toorn et al. (2018) demonstrate that snowshoe spam domains are detectable from their DNS patterns as much as 100 days before appearing on blacklists, but only if historical DNS data is available for analysis. Without longitudinal archival, post-incident investigation cannot reconstruct the full temporal evolution of malicious DNS configurations.

Infrastructure evolution studies: Long-term trends in DNS configuration — the adoption of cloud email services [van Rijswijk-Deij et al, 2016], the migration of domains between DNS providers, the deployment of DNSSEC — are only observable through continuous longitudinal measurement. Point-in-time studies capture a static snapshot; only sustained active measurement captures the dynamics.

1.1.3 The Geographic Dimension: A Neglected Aspect

Beyond the temporal dimension, DNS responses exhibit a **geographic dimension** that single-vantage-point measurement cannot capture. The authoritative name server for a domain does not necessarily return the same response to all clients: it may return different IP addresses, or redirect through different CNAME chains, depending on the inferred geographic location of the querying resolver. Three distinct mechanisms produce this geographic variation:

CDN-based geographic routing: Large web services — streaming platforms, search engines, social networks — deploy Content Delivery Networks with server clusters distributed across multiple continents. DNS is the primary mechanism for directing clients to the appropriate cluster: when the CDN's authoritative server receives a DNS query, it infers the client's location from the querying resolver's IP address and returns the address of the nearest server cluster. Li and Huang [Li and Huang, 2025] find that this geographic routing operates at continent-level granularity for the Twitch CDN: distinct IP address sets are assigned to clients in Europe, North America, and Asia-Pacific. Hours et al. [Hours et al, 2016] causally confirm the performance rationale: clients using local ISP resolvers are assigned Akamai CDN servers that are, on average, 20 ms closer in round-trip time than servers assigned to clients using Google Public DNS, translating to a 14% throughput improvement. DNS-based CDN routing thus produces **systematically different** DNS responses from different geographic vantage points.

IP anycast routing: DNS root servers, TLD servers, and large public resolvers (Google 8.8.8.8, Cloudflare 1.1.1.1) employ IP anycast: a single global IP address is announced from multiple geographically distributed Points of Presence (PoPs), and BGP routing directs each client to the "nearest" PoP in BGP path-length terms. Koch et al. [Koch et al, 2021] demonstrate that more than 95% of users experience some geographic inflation when accessing DNS root servers — being routed to a PoP that is farther than the geographically closest one — because BGP metrics do not correspond to geographic distance. Bortzmeyer [Bortzmeyer, 2013] documents this counter-intuitively for the .fr TLD: North American probes route 55% of their queries to the Paris PoP rather than to any transatlantic alternative, due to BGP peering arrangements. Anycast thus creates **BGP attraction basins** — geographic regions whose traffic converges on the same physical instance — that do not respect continental boundaries.

EDNS Client Subnet (ECS): RFC-7871 [RFC7871, 2016] specifies the EDNS Client Subnet extension, which allows a recursive resolver to embed a truncated version of the client's IP prefix in its query

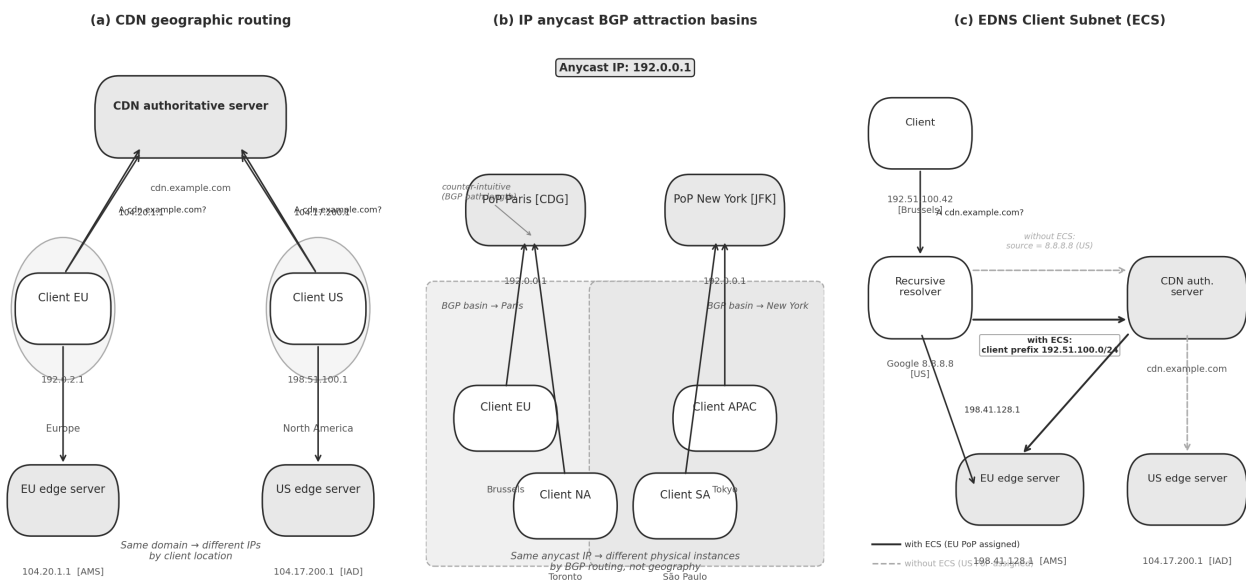


Figure 1.2: The three mechanisms that produce geographic variation in DNS responses. **(a) CDN geographic routing:** the CDN authoritative server infers client location from the resolver IP and returns region-specific edge server addresses. **(b) IP anycast BGP attraction basins:** the same anycast IP routes clients to different physical PoPs depending on BGP path-length topology — not geographic proximity. **(c) EDNS Client Subnet (ECS):** the recursive resolver embeds the client’s /24 prefix in the query, enabling the CDN to select the geographically appropriate replica even when using a remote resolver.

to the authoritative CDN server. ECS enables the CDN to perform geographic routing based on the end client’s network location rather than the resolver’s location — partially solving the remote DNS problem identified by Wang et al. [Wang et al, 2018], wherein centralized public resolvers (Google 8.8.8.8, Cloudflare 1.1.1.1) cause CDN servers to assign geographically inappropriate replicas. However, ECS carries significant privacy implications (the client prefix is exposed to the authoritative server and on-path observers) and is not universally deployed: Bortzmeyer [Bortzmeyer, 2017] reports that only a small fraction of RIPE Atlas probes in Belgium use resolvers that forward ECS options. Figure 1.2 provides a visual summary of all three mechanisms.

The practical consequence of these three mechanisms is that DNS responses for popular domains are **geographically heterogeneous**: clients at different locations receive different responses, reflecting different CDN server assignments, different anycast routing paths, and different resolver configurations. OpenINTEL, despite measuring 50% of the global DNS namespace daily, cannot observe this geographic variation because it operates from a single point in the Netherlands [van Rijswijk-Deij et al, 2016; van Rijswijk-Deij, 2018]. The fraction of popular domains that return geographically differentiated responses, the magnitude of that differentiation, the mechanisms that drive it, and its temporal stability remain empirically uncharacterised at population scale. This is the research gap that this thesis addresses.

1.1.4 Implications for Research and Practice

The absence of a systematic, longitudinal, spatially distributed DNS measurement dataset constrains multiple research communities:

Network performance research cannot retrospectively validate whether CDN operators actually

optimised their geographic routing policies over time, or identify which network regions consistently receive suboptimal server assignments. Calder et al. [Calder et al, 2015] show empirically that 20% of clients accessing Microsoft’s Bing CDN are directed to suboptimal front-end servers, with potential latency savings of 7-100 ms if DNS-based routing were applied. Identifying and quantifying such routing inefficiencies at scale requires distributed measurement data.

DNS centralisation research can document that 10 DNS providers host 48.5% of all gTLD domains [Xu et al, 2023], but cannot determine whether this centralisation has a geographic dimension — whether specific regions are more heavily dependent on specific providers, or whether provider failures would have asymmetric geographic impact. Addressing this requires measurement from diverse geographic vantage points.

Internet simulation requires historically accurate DNS data from multiple geographic perspectives to construct realistic network models. A simulation of Internet-circa-2023 from a Tokyo vantage point requires knowing what DNS responses a Tokyo client would have received at that time — data that is neither available from OpenINTEL (single Dutch vantage point) nor reconstructible from existing passive DNS datasets (which observe only queries that naturally flowed through specific observation points).

Policy and sovereignty debates around DNS infrastructure — including questions of which nations’ citizens depend on DNS infrastructure under foreign jurisdiction, and what the geographic exposure of national DNS infrastructure is — require geographically distributed empirical data. Xu et al. [Xu et al, 2023] document that over 60% of DNS provider pairs share underlying infrastructure, but the geographic distribution of this shared dependency has not been characterised.

1.2 Research Problem

1.2.1 Main Research Question

This thesis addresses the following main research question:

How can one design and deploy a distributed DNS measurement system that captures the geographic and temporal diversity of DNS responses for the most popular domains on the web, in a manner that is ethically sound, reproducible, and useful to the research community?

The question has two inseparable dimensions: the **engineering dimension** (how to build the system) and the **empirical dimension** (what the system reveals about DNS behaviour). The engineering choices — platform, domain corpus, measurement parameters, data format — are motivated by and evaluated against the empirical goals: producing data that accurately characterises the geographic and temporal structure of popular-domain DNS responses.

1.2.2 Secondary Research Questions

The main question decomposes into four operational sub-questions that the measurement campaign and analysis directly address:

Q1 — Geographic diversity: What proportion of Tranco top-ranked domains return different DNS responses depending on the geographic location of the querying client, and what infrastructure mechanisms (CDN-based routing, anycast, ECS) explain the observed differentiation?

This question quantifies the scale of the phenomenon. If a large proportion of popular domains return identical responses regardless of query origin, then single-vantage-point measurement captures the essential picture and geographically distributed measurement adds little. If, conversely, a substantial proportion return geographically differentiated responses, then the single-vantage-point approach of OpenINTEL systematically misses a critical dimension of the DNS. The answer determines how significant the research gap is in quantitative terms, and guides the interpretation of any DNS study that does not account for geographic variation.

Q2 — Temporal stability: What is the temporal stability of DNS responses for popular domains over a multi-month measurement period? Do changes occur at day, week, or month timescales, and are they correlated with identifiable infrastructure events?

This question characterises the dynamics of the DNS at the population level. DNS resource records carry TTLs that reflect administrator intent about how frequently records should be refreshed, but the actual rate of content change — when the *substance* of records changes, not merely their cached lifetime — has not been characterised at the scale of the Tranco top list. Understanding temporal dynamics informs the appropriate measurement frequency for longitudinal archival: if records change rarely, weekly measurements suffice; if records change daily, daily measurement is required.

Q3 — Geographic bias of RIPE Atlas: Do the documented geographic biases of the RIPE Atlas probe distribution — 91% of probes in Europe and North America [Bajpai et al, 2017] — significantly affect the ability to observe geographic DNS response variation?

RIPE Atlas is the only platform with sufficient probe count (approximately 12,892 probes in 178 countries as of Nosyk et al., 2024) and built-in DNS measurement support to enable the distributed measurement campaign this thesis requires. However, its geographic distribution is heavily skewed toward Europe and North America. This question evaluates whether probes from underrepresented regions (Africa, Latin America, Southeast Asia, Oceania) produce systematically different DNS observations — indicating that the platform bias limits the validity of global inferences — or whether they corroborate patterns observed in the majority regions, indicating that the bias is acceptable for the study’s purposes.

Q4 — Impact of resolver choice: What is the measurable impact of resolver choice — local ISP resolver versus public DNS service — on the DNS responses observed, and does it interact with the geographic location of the probe?

The remote DNS problem [Wang et al, 2018] predicts that probes using centralized public resolvers (Google 8.8.8.8, Cloudflare 1.1.1.1) should receive geographically inappropriate CDN assignments relative to probes using local ISP resolvers. This question empirically tests this prediction in the RIPE Atlas context. Xu et al. [Xu et al, 2023] further show that over 90% of forwarding resolvers are backed by fewer than 5% of indirect recursive resolvers, meaning that resolver centralisation may cause probes at geographically diverse locations to receive convergent DNS responses — not because the DNS is geographically uniform, but because their resolvers all converge on the same infrastructure. Measuring the impact of this convergence is both practically important (for interpreting the measurement results) and scientifically significant (validating the centralisation thesis of Xu et al.,

2023 from a client perspective).

1.3 Thesis Objectives

1.3.1 General Objective

The general objective of this thesis is to design, implement, and operate a **geographically distributed active DNS measurement system** that:

- Measures DNS responses for a representative, stable, and reproducible corpus of popular domains (the Tranco Top list, Le Pochat et al., 2019);
- Collects measurements from a globally distributed set of RIPE Atlas probes selected to maximise geographic coverage while controlling for known platform biases [Bajpai et al, 2017; Nosyk et al, 2025];
- Repeats measurements over a multi-month period to enable temporal analysis;
- Stores results in structured, archivable formats conforming to FAIR data principles (Findable, Accessible, Interoperable, Reusable);
- Analyses the collected data to provide quantitative answers to questions Q1–Q4.

1.3.2 Specific Objectives

Objective 1 — Design an optimal measurement methodology

The measurement methodology must balance competing constraints: geographic coverage (more probes = better coverage), domain coverage (more domains = more representative corpus), measurement frequency (more frequent = better temporal resolution), and resource consumption (all three dimensions are bounded by the finite RIPE Atlas credit budget). Designing the optimal point in this multidimensional space requires understanding the cost model of RIPE Atlas measurements (Holterbach et al., 2015; Nosyk et al., 2024), the geographic distribution of available probes [Bajpai et al, 2017], and the stability characteristics of the Tranco list [Pochat et al, 2019]. The measurement methodology must also specify EDNS options (including NSID for anycast instance identification, following Bortzmeyer, 2013 and Finnegan, 2018) and resolver targeting (direct authoritative queries to bypass resolver centralization effects).

Objective 2 — Implement a robust collection and processing pipeline

The engineering implementation must reliably collect measurement results via the RIPE Atlas API, parse and validate DNS responses (filtering invalid responses, identifying in-path proxy effects following Jones et al., 2016), and store results in a two-tier format: Apache Avro for long-term archival and Apache Parquet for efficient columnar analytics, following the storage architecture of OpenINTEL [van Rijswijk-Deij et al, 2016]. The pipeline must be sufficiently automated to operate over a three-month measurement period with minimal manual intervention.

Objective 3 — Quantitatively analyse the collected data

The analysis must produce empirical answers to Q1–Q4: quantifying the proportion of domains with geographically differentiated responses, characterising the distribution and magnitude of geographic variation, measuring the rate and pattern of temporal change, evaluating the effect of geographic probe bias and resolver choice, and identifying the DNS mechanisms (CDN routing, anycast, ECS) responsible for observed variation.

Objective 4 — Publish data and methodology

To fulfil the scientific contribution goals of the thesis, the measurement dataset must be published with a permanent identifier (DOI), the methodology must be documented to enable independent replication, and the analysis code must be made openly available. These commitments follow the FAIR data principles and the reproducibility norms of Internet measurement research (van Rijswijk-Deij et al., 2016; Le Pochat et al., 2019).

1.4 Challenges and Constraints

1.4.1 Technical Challenges

Data volume: A measurement campaign of 10,000 Tranco domains $\times$ 100 RIPE Atlas probes $\times$ one measurement per day $\times$ 90 days produces approximately 90 million individual DNS query-response records. Each record includes the queried domain, record type, DNS response (potentially multiple resource records), query and response timing, probe metadata (geographic location, ASN, resolver), and EDNS option results (NSID string, ECS scope). After parsing and enrichment, this produces on the order of 20–30 GB of compressed data. The pipeline must handle this volume with sufficient efficiency to process each day’s measurements before the next measurement cycle begins.

Measurement quality and reliability: Holterbach et al. [Holterbach et al, 2015] demonstrate that simultaneous measurements from multiple RIPE Atlas probes can be desynchronised by up to one hour under high platform load, and that concurrent measurements from other users degrade timing precision by more than 1 ms in the median and more than 7 ms at the 95th percentile. Bortzmeyer [Bortzmeyer, 2017] identifies additional quality issues: lying resolvers that return incorrect responses, DNS interception by in-path proxies [Jones et al, 2016], DNSSEC validation failures that produce SERVFAIL responses misidentified as unavailability, and cache divergence between probes that confounds geographic and temporal variation. Rigorous filtering and validation are required to distinguish genuine DNS variation from measurement artefacts.

Infrastructure heterogeneity: RIPE Atlas probes span three hardware generations with substantially different computing power [Holterbach et al, 2015]: Versions 1 and 2 are resource-constrained devices (167 MHz CPU, 8–16 MB RAM) susceptible to timing degradation under concurrent measurement load; Version 3 devices (400 MHz CPU, 32 MB RAM) are more reliable but not immune. Probes also use heterogeneous local resolvers — ISP resolvers, corporate gateways, misconfigured devices — with different caching policies, DNSSEC validation support, and ECS forwarding behaviour [Bajpai et al, 2017]. This heterogeneity must be characterised and controlled in the analysis.

1.4.2 Operational Constraints

RIPE Atlas credit budget: The measurement campaign is bounded by a finite RIPE Atlas credit allocation. The credit cost of each measurement depends on the measurement type, the number of probes, and the scheduling mode: scheduled recurring measurements cost approximately half as much as equivalent one-off measurements [Holterbach et al, 2015]. The credit constraint imposes a three-way trade-off between domain coverage (number of domains), geographic coverage (number of probes), and temporal resolution (measurement frequency). Optimising this trade-off within budget while respecting the RIPE Atlas measurement quotas is a central design challenge of the methodology (described in Chapter 3).

Geographic distribution bias: As documented by Bajpai et al. [Bajpai et al, 2017] and confirmed by Nosyk et al. [Nosyk et al, 2025], approximately 91% of dual-stacked RIPE Atlas probes are concentrated in the RIPE (Europe/Middle East) and ARIN (North America) regions. Africa, Latin America, Central Asia, and Oceania are substantially underrepresented relative to their shares of the global Internet user population. This concentration means that results will be statistically robust for Europe and North America but are based on smaller samples for other regions. Geographic stratification — selecting probes to ensure minimum coverage of each inhabited continent — mitigates but cannot eliminate this structural limitation.

Measurement duration: A Master 60 thesis at the University of Namur imposes a strict time constraint. The measurement period is limited to approximately three months of continuous collection, compared to the ten-plus years of continuous operation achieved by OpenINTEL [van Rijswijk-Deij et al, 2016; van Rijswijk-Deij, 2018]. This three-month window enables meaningful temporal analysis at day and week timescales but cannot capture annual seasonality, multi-year trends, or the long-term evolution of DNS infrastructure.

1.4.3 Ethical Constraints

Active DNS measurements using volunteer-hosted infrastructure raise ethical responsibilities that must be explicitly built into the research design. Kisteleki et al. [Kisteleki, 2016] establish the primary ethical framework for RIPE Atlas measurements, grounded in the Menlo Report (2012) and the Belmont Report (1978): RIPE Atlas probe hosts are volunteers whose internet connections generate the measurement traffic, and researchers bear responsibility for ensuring that traffic is benign, proportionate, and does not expose probe hosts to legal or social risk.

For this thesis, three ethical design choices follow from this framework:

Impact on DNS infrastructure: van Rijswijk-Deij et al. [van Rijswijk-Deij et al, 2016] demonstrate that OpenINTEL’s measurements — 1.85 billion queries per day — represent only 0.3%–1.6% of total DNS traffic at the measured authoritative servers, confirming that well-paced active measurement has negligible infrastructure impact. This thesis’s measurement campaign is orders of magnitude smaller and must similarly be paced to avoid creating an undue burden on measured authoritative servers.

Query target selection: Following the principles of Kisteleki et al. [Kisteleki, 2016], the domain corpus is drawn exclusively from the Tranco list of popular, publicly accessible domains. No politically sensitive, censored, or jurisdictionally restricted domains are included: measuring access to blocked content

from probes in restrictive jurisdictions could expose probe hosts to legal risk, as illustrated by the "Ebonia" case described in Kisteleki et al. [Kisteleki, 2016].

Probe host privacy: Raw probe IP addresses are not published in the dataset; results are aggregated at country and ASN level for public release, preserving the statistical value of the data while protecting the network location of individual volunteer probe hosts.

1.5 Methodological Approach

1.5.1 Choice of Domain Corpus: The Tranco List

The thesis uses the **Tranco Top list** [Pochat et al, 2019] as its domain corpus. This choice is motivated by three requirements identified by the research design:

Stability for longitudinal analysis: The domain corpus must remain stable over the three-month measurement period, so that the same domains are measured across all campaign repetitions and temporal comparisons are meaningful. Tranco's 30-day aggregation window produces a list that changes by only 0.6% per day, compared to approximately 50% for the Alexa list after its methodology change in 2018 [Pochat et al, 2019]. This stability is indispensable: a corpus that changes by 50% daily cannot serve as the foundation for temporal comparison.

Manipulation resistance for result validity: Le Pochat et al. [Pochat et al, 2019] demonstrate that every major commercial domain ranking list can be manipulated at low cost — a single HTTP request suffices to alter a domain's Alexa rank. A manipulated domain list would bias the DNS measurement results toward adversarially placed domains, compromising the scientific validity of the study. Tranco's multi-source aggregation over a 30-day window requires at least four times more effort to manipulate than single-source lists.

Reproducibility for scientific replication: Historical Tranco lists are archived and retrievable via a stable URL that includes the list generation parameters, enabling other researchers to obtain the exact domain set used in this study and replicate the measurements. This archivability is a prerequisite for scientific reproducibility that commercial lists — which overwrite their previous version daily — cannot provide.

This thesis focuses on the Tranco Top 10,000 domains, balancing representativeness (capturing the major web services responsible for the largest fraction of Internet traffic) with feasibility within the available RIPE Atlas credit budget.

1.5.2 Choice of Measurement Platform: RIPE Atlas

The thesis uses **RIPE Atlas** as its distributed measurement infrastructure, a choice that Bortzmeyer [Bortzmeyer, 2017] explicitly recommends as the preferred platform for distributed DNS measurements. Four properties make RIPE Atlas uniquely suitable:

Geographic distribution: With approximately 12,892 active probes across 178 countries [Nosyk et al, 2025], RIPE Atlas provides a globally distributed vantage point network that cannot be replicated with

academic infrastructure. No alternative platform — CAIDA Archipelago (~170 monitors), SamKnows (~70,000 probes but ISP-deployed), PlanetLab (~300 vantage points) — offers comparable geographic coverage with built-in DNS measurement support (Bajpai et al., 2017; Cicalese et al., 2015).

Native DNS measurement support: RIPE Atlas provides built-in DNS measurement types supporting all standard query types (A, AAAA, NS, MX, SOA), DNSSEC validation flags, and EDNS options including NSID (for anycast instance identification) and ECS (for client-subnet-aware routing analysis). This native support avoids the complexity and potential inconsistency of implementing DNS measurement from scratch.

Community validation: RIPE Atlas has been used in over 600 academic publications, ensuring that its measurement artefacts, biases, and limitations are well characterised (Nosyk et al., 2024; Holterbach et al., 2015; Bajpai et al., 2017). Results obtained from RIPE Atlas are directly comparable with the existing literature.

Accessible resource model: RIPE Atlas operates on a credit economy in which probe hosts earn credits that can be shared with the research community. Researchers can obtain measurement credits by describing their study to the RIPE Atlas community, making the platform financially accessible to academic research projects.

1.5.3 System Architecture Overview

The measurement system follows an architecture inspired by OpenINTEL [van Rijswijk-Deij et al, 2016] but adapted to the constraints and capabilities of RIPE Atlas:

Domain corpus preparation: The Tranco Top 10,000 list is retrieved via the Tranco API using a fixed version identifier for reproducibility; domains are filtered for validity and pre-resolved to identify their authoritative name servers.

Measurement scheduling: RIPE Atlas user-defined DNS measurements are configured to query each domain's authoritative name servers directly — bypassing local resolvers to control for resolver centralization effects [Xu et al, 2023] — from a geographically stratified probe set. Measurements include the NSID EDNS option to capture anycast instance information [Bortzmeyer, 2013; Finnegan, 2018] and are scheduled as recurring daily campaigns to minimise credit consumption [Holterbach et al, 2015].

Data collection: Measurement results are retrieved via the RIPE Atlas REST API and validated. Probes with known system failures (identified via RIPE Atlas system tags following Bajpai et al., 2017) are excluded; in-path proxy signatures (as described by Jones et al., 2016) are flagged.

Storage: Results are serialised in Apache Avro format for archival and converted to Apache Parquet for efficient columnar analysis, directly following the data architecture of OpenINTEL [van Rijswijk-Deij et al, 2016].

Analysis: The collected dataset is analysed to answer questions Q1–Q4 using geographic aggregation (by continent, country, and ASN), temporal comparison (daily, weekly, monthly), and mechanism attribution (NSID-based anycast instance mapping, CNAME chain analysis, ECS scope analysis).

1.5.4 Ethical and Scientific Compliance

Ethics: The measurement design complies with the RIPE Atlas ethical guidelines [Kisteleki, 2016] and the Menlo Report (2012) framework: all query targets are publicly accessible domains; measurement frequency is set to respect DNS TTLs and avoid server overloading; and no sensitive or politically controversial content is queried.

Reproducibility: The measurement methodology and analysis code are documented and published following FAIR data principles. The specific Tranco list version, RIPE Atlas measurement identifiers, probe selection criteria, and analysis scripts are included in the supplementary materials to enable independent replication.

1.6 Thesis Structure

This thesis is organised in five chapters:

Chapter 1 — Introduction (this chapter): Motivates the research by identifying the gap between existing DNS measurement infrastructure and the geographic variation question; states the research questions; describes the objectives, challenges, and methodological approach; and previews the thesis structure.

Chapter 2 — State of the Art: Reviews the existing literature across five technical domains directly relevant to the thesis: (1) DNS measurement paradigms — active versus passive, and the OpenINTEL infrastructure; (2) the RIPE Atlas distributed measurement platform and its capabilities, biases, and limitations; (3) anycast routing in DNS infrastructure and the NSID-based instance identification methodology; (4) CDN DNS-based geographic routing, the remote DNS problem, and EDNS Client Subnet; (5) domain ranking lists, from the limitations of commercial lists to the Tranco methodology; (6) DNS infrastructure centralisation; and (7) DNS security dimensions that affect measurement validity. The chapter concludes with a gap analysis that formally positions the thesis contribution.

Chapter 3 — Methodology: Describes the measurement methodology in full operational detail: domain corpus selection (Tranco Top 10K, version identifier, filtering criteria); RIPE Atlas probe selection (system tag filters, geographic stratification protocol, probe count per region); DNS measurement configuration (query types, EDNS options, target selection — authoritative versus resolver, scheduling); data collection pipeline (API integration, validation, error handling); storage architecture (Avro schema, Parquet conversion, metadata schema); and analysis methods for each of the four research questions.

Chapter 4 — Results (*to be completed after data collection*): Presents the empirical findings of the measurement campaign — descriptive statistics of the collected dataset; quantitative results for Q1 (geographic diversity), Q2 (temporal stability), Q3 (RIPE Atlas bias assessment), and Q4 (resolver impact); geographic and temporal visualisations; and illustrative case studies of domains exhibiting extreme geographic variation or notable temporal dynamics.

Chapter 5 — Discussion and Conclusion (*to be completed after analysis*): Interprets the results in the context of the state of the art; discusses the limitations of the study (three-month temporal window, RIPE Atlas geographic bias, authoritative query methodology); evaluates the generalisation of findings

beyond the Tranco Top 10K corpus; identifies future research directions; and summarises the thesis's contributions to the measurement community.

1.7 Expected Contributions and Impact

1.7.1 Scientific Contributions

Primary empirical contribution: This thesis will produce the first systematic, population-scale characterisation of geographic DNS response variation for popular domains, validated over a multi-month longitudinal measurement campaign. Where existing literature contains individual case studies of anycast routing (Bortzmeyer, 2013 for d.nic.fr with ~500 probes; Finnegan, 2018 for UncensoredDNS with 500 probes; Calder et al., 2015 for Microsoft Bing) and individual CDN routing studies (Hours et al., 2016; Li et al., 2025), none has characterised **what proportion of the popular-domain corpus returns geographically differentiated DNS responses**, which mechanisms account for that differentiation, and how the differentiation distributes across geographic regions. Answering these questions at the scale of 10,000 domains and 100+ global vantage points constitutes the novel empirical contribution.

Methodological contribution: The measurement methodology developed in this thesis — combining direct authoritative server queries (to bypass resolver centralization), NSID-based anycast tracking, geographically stratified probe selection, and a reproducible Tranco-based domain corpus — addresses three gaps identified in the existing literature (Section 2.10 of Chapter 2) and can be adopted by future measurement studies as a validated protocol.

Dataset contribution: The collected dataset — approximately 90 million DNS measurement records covering 10,000 Tranco domains from 100+ globally distributed RIPE Atlas probes over three months, archived in Avro/Parquet format with full metadata — will be published with a permanent DOI on Zenodo. This makes the dataset the first publicly available longitudinal, geographically distributed DNS dataset for the popular-domain tier, directly addressing the data availability gap motivating this thesis.

1.7.2 Impact for the Research Community

Network simulation: Researchers developing Internet simulators will be able to use the collected dataset to configure realistic DNS responses for simulations representing Internet states from the measurement period and from multiple geographic perspectives.

CDN and routing research: Researchers studying CDN performance, anycast routing efficiency, and geographic routing optimisation will have empirical ground truth for the DNS dimension of client routing at population scale.

DNS security: Security researchers investigating large-scale DNS manipulation, geographic censorship patterns, or DNS infrastructure resilience will have a baseline dataset of correct DNS responses from diverse vantage points against which anomalous behaviour can be detected.

DNS infrastructure policy: Policymakers and regulators addressing digital sovereignty, Internet resilience, and the geographic concentration of critical Internet infrastructure will have empirical data

on the geographic structure of DNS dependency at the level of popular domains.

1.8 Work Organisation

1.8.1 Project Phases

The work is organised in five successive phases aligned with the thesis chapter structure:

Phase 1 — Literature review and familiarisation (weeks 1–6, January–February 2026): Reading and annotation of the 22 primary sources; construction of the bibliography and source fiches; drafting of Chapter 2 (State of the Art).

Phase 2 — Design and pilot (weeks 7–10, February–March 2026): Design of the measurement methodology; implementation of the data collection pipeline; pilot campaign (500 domains, 10 probes, 7 days) to validate technical parameters and estimate credit consumption; drafting of Chapter 3 (Methodology).

Phase 3 — Full measurement campaign (weeks 11–22, March–June 2026): Daily automated measurements; monitoring of credit consumption and probe availability; adjustment of measurement parameters if anomalies are detected; incremental validation of collected data.

Phase 4 — Analysis (weeks 23–26, June 2026): Complete analysis addressing Q1–Q4; generation of visualisations and statistical results; drafting of Chapter 4 (Results).

Phase 5 — Discussion, revision, and defence (weeks 27–30, June–July 2026): Drafting Chapter 5 (Discussion and Conclusion); complete thesis revision; dataset and code publication; preparation of the oral defence. Table 1.1 summarises this work plan.

#	Phase	Period		Primary deliverable
1	Literature review and familiarisation	Jan–Feb	2026	Chapter 2 (State of the Art)
2	Methodology design and pilot campaign	Feb–Mar	2026	Chapter 3 (Methodology)
3	Full measurement campaign (90 days)	Mar–Jun	2026	Raw dataset + Avro/Parquet archive
4	Data analysis (Q1–Q4)	Jun 2026	(weeks 23–26)	Chapter 4 (Results)
5	Discussion, revision, and oral defence	Jun–Jul	2026 (weeks 27–30)	Final thesis + defence

Table 1.1: Work organisation: five successive phases aligned with the thesis chapter structure. The measurement campaign (Phase 3) runs from mid-March to mid-June 2026 and produces the dataset analysed in Phase 4.

1.8.2 Success Criteria

The thesis will be considered successful against the following measurable criteria:

Technical: At least 85% of scheduled measurements produce valid, parseable DNS responses; at least

95% of Tranco Top 10K domains are covered in the dataset; at least four inhabited continents are represented by RIPE Atlas probes in the final probe selection.

Scientific: Quantitative answers to all four research questions (Q1–Q4) are provided with appropriate statistical characterisation (confidence intervals, statistical tests where applicable); the methodological contribution is sufficiently documented to enable independent replication.

Academic: A complete five-chapter thesis is submitted in conformance with the University of Namur Master 60 format requirements; the dataset is published with a permanent DOI before the oral defence; ethical guidelines (Kisteleki et al., 2016; Menlo Report, 2012) are demonstrably respected in the measurement design.

Chapter 2

State of the Art

2.1 The DNS System: Background and Fundamental Concepts

The Domain Name System (DNS) is the distributed directory that translates human-readable domain names into machine-readable information — primarily IP addresses. As van Rijswijk-Deij et al. [van Rijswijk-Deij et al, 2016] observe, "almost all networked services depend on the DNS to store information about the service", from IP address resolution to mail server designation to cryptographic key publication. Understanding its architecture, resolution model, caching behaviour, and resolver ecosystem is a prerequisite for interpreting any DNS measurement study.

2.1.1 Hierarchical Structure and Resolution

DNS is organised as a hierarchy whose structure reflects the delegation of administrative authority over portions of the namespace [RFC1034, 1987]. The root zone, maintained by ICANN/IANA and operated by twelve root-server organisations, forms the apex of this hierarchy: it contains delegations for every Top-Level Domain (TLD) on the Internet. Below the root, TLDs such as *.com*, *.net*, *.org*, *.fr*, and over 1,500 others each maintain their own authoritative zone. Authoritative name servers at the domain level contain the definitive resource records for specific domains and their subdomains.

Name resolution is carried out by a **recursive resolver** (also called a full-service resolver), which performs the iterative query sequence on behalf of the client [RFC1034, 1987]. When a user or application requests the address of *www.example.com*, the recursive resolver contacts a root server to obtain the address of the *.com* TLD servers, then the TLD servers to obtain the addresses of the authoritative servers for *example.com*, and finally the authoritative server to obtain the A or AAAA record for *www.example.com*. From the client's perspective, this entire process appears as a single request-response exchange.

The recursive resolver may be operated by the client's ISP, by a corporate network gateway, or by a commercial public DNS service such as Google Public DNS (8.8.8.8) or Cloudflare (1.1.1.1). The geographic location of the recursive resolver is a critical parameter for DNS-based CDN routing, as discussed in Section 2.6. Van Rijswijk-Deij et al. (2016) note that the *.com* TLD alone contained approximately 123 million registered names during their measurement period (March 2015 – January

2016), and that *.com*, *.net*, and *.org* together constitute roughly 50% of the global DNS name space. The root zone is protected by DNSSEC since July 2010, establishing the trust anchor for the cryptographic chain that can extend downward to authoritative name servers for individual domains. Twelve of the thirteen logical root server addresses use IP anycast to distribute query load across hundreds of globally distributed physical instances: Jones et al. [Jones et al, 2016] note that the B root is the sole exception, hosted at a single physical location in Los Angeles — a property exploited in their detection methodology for DNS root manipulation (Section 2.9).

2.1.2 Resource Record Types Relevant to Measurement Studies

DNS stores information in typed **Resource Records** (RR), each serving a distinct purpose [RFC1035, 1987]. The record types most relevant to distributed measurement studies are:

A and AAAA records map a domain name to an IPv4 or IPv6 address respectively. A single name may have multiple A or AAAA records, enabling load distribution through round-robin responses and geographic routing through location-aware DNS (different addresses returned depending on the querying resolver's location). This geographic differentiation in A/AAAA responses is one of the primary phenomena this thesis investigates.

NS records designate the authoritative name servers responsible for a DNS zone and are essential for identifying the DNS infrastructure provider responsible for a domain. Van Rijswijk-Deij et al. (2016) include NS records in the OpenINTEL standard query set because changes in NS delegation reveal infrastructure migrations and provider switches.

MX records designate the mail-handling servers for a domain. Van Rijswijk-Deij et al. (2016) use longitudinal MX analysis as one of two case studies for OpenINTEL, tracking the adoption of cloud email services (Google Workspace, Microsoft Exchange Online, Yahoo Mail) across tens of millions of domains over a ten-month period.

CNAME records create an alias pointing from one name to another canonical name. CDNs make extensive use of CNAME chains: a customer domain (`www.shop.example`) points to a CDN-managed name (`shop.cdn-provider.net`), allowing the CDN to update routing without requiring the customer to change their DNS configuration.

DNSKEY, DS, and RRSIG records are the DNSSEC record types carrying public keys, delegation signers, and cryptographic signatures respectively. Van Rijswijk-Deij et al. (2016) include all three in the OpenINTEL query set specifically to monitor DNSSEC deployment and key management practices at scale. Their longitudinal data revealed systemic key rollover failures invisible to point-in-time measurements.

TXT records carry free-text information and are used for SPF (Sender Policy Framework), DKIM (DomainKeys Identified Mail), DMARC configuration, and domain validation tokens for certificate authorities. Table 2.1 summarises the record types relevant to this thesis's measurement campaign.

Type	Content	Measured	Relevance for distributed DNS measurement
A	IPv4 address(es)	Primary	Core observable: geographic variation in returned IPs indicates CDN routing or anycast
AAAA	IPv6 address(es)	Primary	Parallel to A; anycast routing may differ between IPv4 and IPv6 [Bortzmeyer, 2013]
NS	Authoritative name server names	Weekly	Identifies DNS provider; enables provider migration tracking and centralisation analysis [Xu et al, 2023]
CNAME	Canonical name alias	Derived	CDN delegation chain; resolving the chain reveals the CDN operator behind a domain
MX	Mail server hostname(s)	Quarterly	Cloud email adoption tracking [van Rijswijk-Deij et al, 2016]; not geographically differentiated
SOA	Zone authority record	Quarterly	Zone serial number enables change detection; TTL field reveals operator update policy
TXT	Free-form text	Quarterly	SPF/DKIM/DMARC policy; domain ownership verification
DNSKEY/DSDNSSEC	public key / delegation	Quarterly	DNSSEC deployment monitoring; broken chains cause SERVFAIL [van Rijswijk-Deij et al, 2016]

Table 2.1: *DNS Resource Record types relevant to distributed measurement studies, with their measurement frequency in this thesis’s campaign. Primary types are included in the daily measurement campaign; weekly and quarterly types are measured in separate batches. References: [RFC1034, 1987; RFC1035, 1987].*

2.1.3 DNS Caching and TTL: Implications for Active Measurements

Every DNS resource record carries a **Time To Live** (TTL) value specifying how long a recursive resolver or cache may retain the record before treating it as stale and re-querying the authoritative server [RFC1034, 1987; RFC1035, 1987]. TTL values reflect a trade-off between server load (longer TTLs mean fewer queries to authoritative servers) and responsiveness (shorter TTLs allow faster propagation of changes). CDNs routinely use short TTLs — often 60 seconds or less — to enable rapid traffic redirection, while stable infrastructure records (mail server delegations, authoritative NS records) may carry TTLs of 24 hours or more.

Caching operates at multiple levels: the recursive resolver shared by many clients, the operating system stub resolver, and — in some implementations — the application itself. Koch et al. [Koch et al, 2021] emphasise that DNS root records have TTLs measured in days (typically 518,400 seconds = 6 days), meaning that recursive resolvers rarely contact the root in the steady state; most user queries are served from cache.

Two direct implications follow for active DNS measurement studies:

First, if measurements query through a recursive resolver rather than directly at the authoritative server, the response may reflect a cached value from an earlier resolution, not the current authoritative answer. This introduces **temporal staleness** that must be controlled by querying authoritative servers directly or by accounting for TTL values when interpreting results. Van Rijswijk-Deij et al. (2016) and Bortzmeyer [Bortzmeyer, 2017] both explicitly recommend direct authoritative queries when studying DNS record content rather than the user experience.

Second, different probes use different local resolvers with different cache states and different cache populations, meaning the same domain queried "simultaneously" from multiple probes may yield responses that differ due to cache divergence rather than geographic DNS routing intent. Separating these two sources of variation — intentional geographic routing and incidental caching — is a method-

ological challenge that the measurement design of this thesis directly addresses.

2.1.4 DNS Resolver Infrastructure and Its Impact on Measurements

The recursive resolver occupies a pivotal position in the DNS resolution chain: it is the infrastructure that CDN authoritative servers query to infer the client's location, and its behaviour determines which cached responses are returned to clients. For distributed DNS measurement studies, understanding the resolver ecosystem is therefore essential.

Xu et al. [Xu et al, 2023] introduce a four-layer taxonomy of client-side DNS infrastructure that reveals the hidden complexity behind what appears to be a simple client-resolver interaction:

- **Forwarding DNS resolvers (FDNS):** Resolvers deployed close to clients (e.g., ISP edge resolvers) that forward queries to upstream recursive resolvers rather than performing full resolution themselves. They are the entry point for client queries.
- **Recursive DNS resolvers (RDNS):** Perform full iterative resolution from root to authoritative server. May be collocated with FDNS or geographically distant.
- **Indirect RDNS (iRDNS):** Higher-level recursive resolvers that serve multiple RDNS instances. Operated by major public DNS providers (Google, Cloudflare) at a small number of locations.
- **Direct RDNS (dRDNS):** Recursive resolvers that directly contact authoritative servers without going through iRDNS.

This layered architecture has a critical implication for measurements: when a RIPE Atlas probe queries through its local resolver (the FDNS), that query may ultimately be resolved by an iRDNS cluster operated by a major provider whose IP address — not the probe's address — is presented to the CDN authoritative server. The geographic location of the iRDNS thus determines the CDN's routing decision, not the probe's location. This problem is exacerbated by the extreme concentration documented by Xu et al. [Xu et al, 2023]: over 90% of forwarding resolvers are backed by fewer than 5% of indirect recursive resolvers (Section 2.8 expands on this).

In practice, three categories of recursive resolver are commonly encountered by RIPE Atlas probes (Bortzmeyer, n.d., tutorial):

- **ISP resolvers:** Operated by the probe's Internet Service Provider, typically close to the probe geographically. Their IP address is generally a good proxy for the probe's network region, making them preferable for CDN-routing-aware measurements.
- **Public DNS services:** Google Public DNS (8.8.8.8/8.8.4.4), Cloudflare (1.1.1.1/1.0.0.1), Quad9 (9.9.9.9), and others. Their anycast IP addresses resolve to geographically distributed clusters, and the specific cluster that receives a probe's query depends on BGP routing — which may or may not correlate with the probe's geographic location.
- **Open resolvers:** Resolvers inadvertently left accessible to the public Internet. Bortzmeyer [Bortzmeyer, 2017] advises against using open resolvers for structured measurement, both for ethical reasons

(they are open by negligence, not intent) and for reliability reasons (their behaviour is unpredictable).

For this thesis, these distinctions motivate the design choice to query authoritative name servers directly — bypassing the resolver layer entirely — in order to control for resolver-location effects and measure the geographic DNS responses that would be returned to clients regardless of their resolver infrastructure.

2.2 DNS Measurements: Active and Passive Paradigms

The literature broadly distinguishes two fundamental measurement paradigms: **passive** measurements, which analyse DNS traffic as it flows through existing infrastructure without injecting artificial queries, and **active** measurements, which deliberately generate queries from controlled vantage points to study the DNS system directly. Each paradigm offers complementary trade-offs in terms of coverage, geographic control, reproducibility, and ethical impact.

2.2.1 Passive DNS Measurement

Passive DNS (pDNS) systems capture DNS queries and responses at existing resolvers or network transit points — Internet Exchange Points, ISP recursive resolvers, or authoritative name servers — without generating artificial traffic. Van der Toorn et al. (2018) describe passive DNS as providing a "live" view of actual DNS usage: observations reflect queries generated by real users, making passive DNS well suited for studying organic DNS behaviour, detecting anomalies in query patterns, and identifying malicious domains that are actively used. Commercial passive DNS services such as Farsight Security's DNSDB aggregate billions of DNS responses daily from voluntarily participating resolvers worldwide; these datasets are widely used in security research for botnet tracking, phishing detection, and malware infrastructure analysis.

The advantages of passive DNS measurement are its non-intrusiveness (no artificial traffic is injected, eliminating any concern about overloading queried servers) and its ability to capture real traffic volumes and timing. These properties make it the preferred approach for traffic analysis and security forensics.

However, passive DNS carries fundamental limitations for the geographic DNS response variation research question at the centre of this thesis:

Access constraints: Passive observation requires privileged access to DNS traffic transit points — ISP resolvers, IXPs, or authoritative servers — which is typically available only to operators of that infrastructure or through specific research partnerships. Independent academic researchers face significant logistical barriers to deploying passive observation at geographically diverse points.

Geographic blindness: A passive sensor observes only the traffic that passes through its observation point. Van Rijswijk-Deij et al. (2016) explicitly identify this as the core limitation of passive approaches: single-vantage-point passive DNS cannot reveal how DNS responses differ across geographic locations, which is precisely the question distributed active measurement is designed to answer.

Domain selection bias: Passive DNS observes only domains that are actually queried by the local user population. A controlled comparative study of a predefined domain set — such as measuring all 10,000 Tranco top domains — is impossible with passive measurement, since many domains in any fixed set will receive no organic traffic from any given observation point.

Privacy constraints: DNS traffic contains implicit information about user browsing behaviour. Collection and retention of DNS traffic raises significant privacy concerns under data protection regulations such as the GDPR, constraining the scope of permissible passive DNS research and the granularity at which data can be stored.

2.2.2 Active DNS Measurement

Active measurement deliberately sends DNS queries to predefined domain targets from controlled vantage points. Van Rijswijk-Deij et al. (2016) motivate this paradigm by observing that "studying the state of large parts of the DNS over time reveals valuable information about the evolution of the Internet", a goal that only active measurement can achieve comprehensively and reproducibly.

The core advantages of the active approach are:

Full domain coverage: Any domain in a TLD zone file, or any domain on a research list such as Tranco, can be measured regardless of whether it has organic traffic. This enables systematic, bias-free coverage of predefined domain sets (van Rijswijk-Deij et al., 2016, Goal G1).

Reproducibility: The same set of domains, record types, vantage points, and measurement frequency can be reproduced identically across multiple time periods and by different research groups. This is a fundamental criterion of scientific research that passive measurements — tied to organic traffic patterns — cannot guarantee.

Geographic control: Vantage points can be explicitly selected to cover specific geographic regions or network positions of interest, enabling the comparison of DNS responses across locations. This is the central capability leveraged by this thesis.

Parameter control: The experimenter fully controls technical measurement parameters: DNS query type (A, AAAA, NS, MX, SOA), use of DNSSEC validation, inclusion of EDNS options (NSID, ECS), whether to query through a local resolver or directly at an authoritative server, and measurement frequency.

Infrastructure impact quantification: Van Rijswijk-Deij et al. (2016) empirically quantify the impact of OpenINTEL's active measurements on the global DNS infrastructure. Despite generating at least 1.85 billion queries per day for the .com TLD alone (123 million names $\times$ 14 query types per name, plus recursion overhead), the traffic generated represents only **0.3% to 1.6%** of total DNS traffic observed on the authoritative servers analysed. No correlation between OpenINTEL measurement peaks and server performance degradation is detectable. This result establishes that well-designed active measurements — appropriately paced and sized — have a negligible impact on the infrastructure being studied.

The scientific value of active measurements is further illustrated by van der Toorn et al. [van der Toorn et al, 2018], who apply OpenINTEL data (measuring more than 60% of the global DNS namespace daily) to detect snowshoe spam domains using supervised machine learning. Snowshoe spam distributes

sending across many domains, each with low individual traffic — precisely the pattern that makes these domains invisible to passive DNS (which only observes domains with organic query traffic). The active measurement approach reveals the structural DNS patterns (many A records per domain, SPF configurations with many hosts) that characterise snowshoe spam infrastructure even before any spam has been sent. The result is a detection precision of over **93%** with a **100-day lead** over existing blacklists — a capability exclusive to the active measurement paradigm.

2.2.3 Ethical Dimensions of Active Measurement

Active measurements using volunteer-hosted infrastructure raise ethical responsibilities that must be explicitly addressed in the research design. Kistelevi et al. [Kistelevi, 2016] provide the primary ethical framework for RIPE Atlas measurements, grounding it in the **Menlo Report** (2012) — which extends the Belmont Report (1978) principles of beneficence, justice, and respect for persons to ICT research — and in the work of Allman and Partridge (2016), who propose formal ethics review sections in measurement papers.

The core principle articulated by Kistelevi et al. [Kistelevi, 2016] is that RIPE Atlas probe hosts are volunteers whose internet connections are used to generate measurement traffic. Researchers bear responsibility for the traffic those connections produce, not only in terms of volume but in terms of the legal and social risk that traffic creates for the probe host in their jurisdiction. This is illustrated by the "Ebonia" case: a researcher planned to use probes in a politically volatile region to measure access to socially sensitive websites, until colleagues highlighted that the probe hosts — not the researcher — would bear the legal risk of that traffic. The study was redesigned with less sensitive query targets. Kistelevi et al. [Kistelevi, 2016] further describe a case where RIPE NCC required that a Google search measurement study use only "risk-free search terms such as 'cat' and 'dog'" rather than the politically sensitive terms initially proposed.

RIPE NCC enforces these principles institutionally: HTTP measurements are restricted to RIPE Atlas anchors (trusted, known-good destinations) by default, and access to the platform can be terminated for research that generates complaints about unethical behaviour [Kistelevi, 2016].

For this thesis, these principles translate to concrete methodological design choices: the domain corpus is drawn from the Tranco list of popular, publicly accessible domains; no politically sensitive, jurisdictionally censored, or legally ambiguous content is queried; measurement frequency is set to respect DNS TTLs; and all measurement traffic is identifiable as RIPE Atlas-sourced, allowing authoritative server operators to filter if needed.

2.2.4 Specific Technical Challenges of Distributed Active DNS Measurements

Beyond the general advantages and limitations, distributed active DNS measurements from geographically dispersed probes introduce specific technical challenges that must be addressed in the measurement design.

Measurement desynchronisation: Holterbach et al. [Holterbach et al, 2015] demonstrate that when a "simultaneous" measurement is launched from multiple RIPE Atlas probes, the actual DNS queries are not sent at the same instant. Under heavy concurrent platform load, the spread between the first and

last probe completing a one-off measurement can reach **up to one hour**. This matters when studying dynamic DNS properties (CDN traffic shifts, DNS record changes) that evolve on timescales comparable to the measurement spread. Mitigation strategies include using scheduled periodic measurements (rather than one-off) and repeating measurements to identify transient anomalies.

Heterogeneous resolver behaviour: Each RIPE Atlas probe uses the resolver configured by its host — typically the probe’s ISP resolver. These resolvers have heterogeneous characteristics: different TTL caching policies, different DNSSEC validation support, different EDNS compliance, and potentially different filtering policies. Bajpai et al. [Bajpai et al, 2017] expose this through the RIPE Atlas tag system: tags such as `system-resolves-a-correctly` and `system-resolver-mangles-case` identify probes whose resolvers exhibit abnormal behaviour. Measurements that rely on consistent resolver behaviour across all probes must filter by these system tags. Alternatively, specifying an explicit resolver (e.g., a known public resolver) in the measurement definition bypasses local resolver variability.

Cache state divergence: Because each probe’s resolver independently caches DNS responses, the cache state of two probes measuring the same domain at the same time may differ substantially — one resolver may have cached an older version of a record while another has a fresh response. This source of variation can mimic geographic routing differences. Querying authoritative servers directly, rather than through local resolvers, eliminates this ambiguity by bypassing resolver caches entirely.

IPv4/IPv6 measurement parallelism: Anycast routing tables for IPv4 and IPv6 are maintained independently, and the same anycast service may direct IPv4 and IPv6 queries to different physical instances (Section 2.5.3). Bortzmeyer [Bortzmeyer, 2013] documents this empirically for `d.nic.fr`: the dominant anycast instance reverses between the two protocols. Measurement campaigns that combine IPv4 and IPv6 probes without distinguishing between them will obtain a conflated view of anycast instance distribution. Dual-stack measurements should be conducted and analysed separately.

2.3 OpenINTEL: Large-Scale Forward DNS Measurement

OpenINTEL, developed at the University of Twente in collaboration with SURFnet, SIDN (the `.nl` registry), and (since 2018) NLnet Labs, is the most comprehensive active DNS measurement infrastructure for longitudinal studies of the global DNS [van Rijswijk-Deij et al, 2016; van Rijswijk-Deij, 2018]. Its architecture and operational experience directly inform the design of this thesis’s measurement system.

2.3.1 Design Goals, Challenges, and Architecture

Van Rijswijk-Deij et al. (2016) define seven research goals (G1–G7) and five technical challenges (C1–C5) that shaped OpenINTEL’s design. The core goals are: measuring **every domain in a TLD** once per day (G1, G4), covering even the largest TLD — `.com` with 123 million names (G2) — querying a **fixed set of relevant resource records** per domain (G3), storing at least one year of data for longitudinal analysis (G5), and supporting efficient analysis at scale using modern data processing tools (G6, G7).

The fixed query set (Table I of van Rijswijk-Deij et al., 2016) includes A, AAAA, NS, MX, SOA, DNSKEY, DS, and RRSIG record types. This selection was chosen to cover the most common DNS uses with the

minimum number of queries: IPv4 and IPv6 addressing, mail routing, zone delegation, and DNSSEC cryptographic infrastructure. Each domain thus requires 14 queries (8 record types, some requiring multiple query forms), producing the primary challenge: querying 123 million *.com* names $\times$ 14 queries = **at least 1.85 billion DNS queries per day** for *.com* alone.

Three architectural choices address this challenge:

DNS software selection: Rather than a bare-metal approach that would require re-implementing DNS recursion (risking bugs that compromise measurement correctness), OpenINTEL uses off-the-shelf DNS software: LDNS as a stub resolver for sending individual queries, and Unbound as a full recursive resolver. This prioritises correctness and robustness (Challenge C4) over maximum throughput.

Distributed worker architecture: A central orchestration component coordinates a swarm of worker nodes implemented as virtual machines, enabling horizontal scaling. The distributed design allows the system to handle both the current query volume and future TLD growth (Goal G7 and Challenge C5), and in principle permits geographic distribution of workers for measurements requiring local vantage points.

Query pacing: To avoid overloading authoritative DNS servers for individual TLDs — particularly the root servers and TLD servers that receive a query for every measured domain — queries are distributed over time rather than batched. This pacing is essential for meeting Challenge C2: the traffic OpenINTEL generates must not be an "unacceptable burden" on the global DNS infrastructure.

Data storage: Results are serialised in **Apache Avro** format (self-describing, compressed) and subsequently converted to **Apache Parquet** for efficient columnar analysis within the Hadoop ecosystem (Goal G6). The estimated storage volume for *.com* alone is $123\text{M} \times 14 \text{ queries} \times \sim 150 \text{ bytes/query} \approx 240 \text{ GB/day}$. For multi-year longitudinal studies, this necessitates petabyte-scale storage, addressed by the Hadoop cluster added in September 2015 [van Rijswijk-Deij, 2018].

2.3.2 Scale, Operational Reach, and Published Results

The first complete OpenINTEL measurement day was **21 February 2015**, with regular daily measurements beginning 1 March 2015 [van Rijswijk-Deij, 2018]. Coverage expanded progressively: initial deployment covered *.com*, *.net*, and *.org* (50% of the global DNS namespace); subsequent phases added major country-code TLDs and additional gTLDs; by 2018, all generic TLDs were included. As of this thesis's writing, the platform has accumulated over ten years of continuous daily DNS measurements, constituting a unique longitudinal dataset for Internet infrastructure research.

Three award-winning publications illustrate the platform's research value:

- **DDoS protection adoption** (Jonker et al., IMC 2016): Using 1.5 years of OpenINTEL data, Jonker et al. study how domains adopt DDoS mitigation services over time, revealing concentration patterns in DDoS protection infrastructure [van Rijswijk-Deij, 2018].
- **DNSSEC key management failures** (van Rijswijk-Deij et al., USENIX Security 2017, Distinguished Paper Award): A 21-month longitudinal analysis reveals that many DNSSEC-signed domains suffer from key rollover failures — domains whose signing keys expire but are not replaced, leaving them

cryptographically unsigned despite appearing in signed zones. These findings, which demonstrate systemic operational failures in DNSSEC implementation, are only possible through continuous active measurement: a point-in-time study would capture each domain's state at a single moment without revealing the temporal dynamics of key management [van Rijswijk-Deij, 2018].

- **Snowshoe spam detection** (van der Toorn et al., NOMS 2018, Best Paper Award): 180 days of OpenINTEL data, combined with supervised machine learning, detect snowshoe spam domains with 93%+ precision and a 100-day lead over traditional blacklists. The approach is validated through three-month production deployment in a major Dutch network operator's mail filter [van der Toorn et al, 2018].

The **Research Data Netherlands Prize**, received in November 2018, further validates OpenINTEL's position as a premier research infrastructure [van Rijswijk-Deij, 2018].

OpenINTEL case study — cloud email adoption: Van Rijswijk-Deij et al. (2016) use OpenINTEL's longitudinal MX record data as a concrete demonstration of the platform's capacity to track large-scale changes in Internet infrastructure. Over a ten-month period (March 2015 to January 2016), they observe the MX records of tens of millions of domains in the *.com*, *.net*, and *.org* TLDs, identifying which mail server infrastructure each domain delegates its inbound email to. Three cloud email services — Google Workspace (G Suite at the time), Microsoft Exchange Online, and Yahoo Mail — are each identifiable by the distinctive MX hostname patterns they assign to their customers (e.g., *aspmx.l.google.com* for Google Workspace, **.protection.outlook.com* for Microsoft). The longitudinal daily time series reveals the adoption trajectory of each service: which TLD registrant communities adopt each service first, how adoption velocity differs across TLDs, and whether adoption events are correlated with external announcements (product launches, pricing changes). This application demonstrates a category of research entirely unavailable to point-in-time measurement or passive DNS: the ability to observe and quantify the timing and pace of large-scale infrastructure migration decisions by millions of domain operators worldwide. For this thesis, the same principle applies to DNS resolver infrastructure: longitudinal active measurement can reveal when CDN operators reconfigure their DNS-based routing policies, when new anycast PoPs are deployed, and when domain operators migrate between DNS providers — events that are instantaneous from a point-in-time perspective but become legible through continuous active measurement.

Critical limitation — geographic monoculture: All OpenINTEL measurements are performed from a **single vantage point in the Netherlands**. This architecture provides exhaustive, unbiased TLD-wide domain coverage but makes **geographic variation in DNS responses completely invisible**. A CDN authoritative server responding to OpenINTEL queries always sees the same source IP address: the Dutch measurement server's address. Whether that domain returns different IP addresses to probes in Japan, Brazil, or South Africa — the core question of this thesis — cannot be answered from OpenINTEL data alone. Van Rijswijk-Deij (2018) explicitly identifies the single-vantage-point design as a limitation of the platform. This gap, the absence of spatial diversity in existing longitudinal DNS measurement infrastructure, is the primary motivation for the geographically distributed RIPE Atlas approach developed in this thesis.

2.4 Distributed Measurement Platform: RIPE Atlas

RIPE Atlas is an Internet measurement platform operated by RIPE NCC, consisting of a globally distributed network of hardware probes hosted by volunteers in home networks, ISPs, IXPs, and academic institutions. It provides the spatial diversity that OpenINTEL lacks and serves as the primary measurement infrastructure for this thesis.

2.4.1 Platform Overview, Hardware, and Comparison with Alternatives

Scale and hardware: As of the analysis by Nosyk et al. [Nosyk et al, 2025], RIPE Atlas comprises **12,892 active probes** distributed across **178 countries**, generating approximately **1.3 billion measurement results per day**. The platform has grown substantially since its launch: Holterbach et al. [Holterbach et al, 2015] report 6,700 public probes in 197 countries as of April 2015, and Bajpai et al. [Bajpai et al, 2017] report 9,100 probes as of January 2017. Three hardware generations of probes have been deployed [Holterbach et al, 2015]: Versions 1 and 2 are Lantronix XPort Pro devices with 167 MHz CPU and 8–16 MB RAM; Version 3 is a TP-Link TL-MR3020 router with a 400 MHz CPU, 32 MB RAM, and 4 MB NAND flash, making it substantially more powerful. By 2015, v3 probes hosted approximately two-thirds of all measurements, though v1 and v2 probes remained non-negligible [Holterbach et al, 2015].

Credit system: Platform access is governed by a credit economy. Probe hosts earn credits for each hour their probe is connected and available for measurements; these credits can be spent to create user-defined measurement campaigns. Holterbach et al. [Holterbach et al, 2015] describe the cost model: traceroutes are the most expensive measurement type; scheduled repeated measurements cost approximately half as much as equivalent one-off measurements. Built-in measurement types available to all users include ping, traceroute, DNS, and SSL certificate retrieval.

Comparison with alternative measurement platforms: Bajpai et al. [Bajpai et al, 2017] survey the landscape of Internet measurement platforms active as of January 2017. **CAIDA Archipelago (Ark)** consists of approximately 170 hardware monitors coordinated for Internet topology mapping; it is smaller and more specialised than RIPE Atlas. **SamKnows** deploys approximately 70,000 hardware probes in collaboration with ISPs and regulators to assess broadband performance; probes are ISP-deployed rather than volunteer-hosted, limiting independence from the monitored infrastructure. **DIMES** and **iPlane** are software-based platforms focused on topology mapping. **BISmark** from Georgia Tech deploys approximately 420 probes for home network characterisation. For anycast census studies, Cicalese et al. [Cicalese et al, 2015] use approximately 300 active PlanetLab vantage points — fewer than RIPE Atlas but with greater flexibility for scanning-style experiments. Bajpai et al. [Bajpai et al, 2017] conclude that RIPE Atlas is "the largest open platform today" and plays "a critical role in not only providing operational support to network operators but also facilitating measurement-based research". For DNS-focused studies specifically, RIPE Atlas's built-in DNS measurement type and the geographic diversity of its probe population make it the preferred platform. Table 2.2 summarises the key characteristics of the main alternatives.

Platform	Probes	Countries	DNS native	Notes
RIPE Atlas	≈12,900	178	Yes	Volunteer hardware probes; credit economy; open academic access [Nosyk et al, 2025]
CAIDA Ark	≈170	60+	No	Topology-focused; not volunteer-based; limited DNS support [Bajpai et al, 2017]
SamKnows	≈70,000	50+	No	ISP-deployed; broadband QoS focus; not openly accessible [Bajpai et al, 2017]
PlanetLab	≈300	60+	No	Software VMs; flexible but low reliability; largely inactive [Cicalese et al, 2015]
BISmark	≈420	40+	No	Home-network focus (Georgia Tech); limited geographic diversity [Bajpai et al, 2017]

Table 2.2: Comparison of distributed Internet measurement platforms relevant to DNS studies. Probe counts as of 2024 for RIPE Atlas [Nosyk et al, 2025] and approximately 2017 for others [Bajpai et al, 2017]. RIPE Atlas is the only platform with native DNS measurement support and open academic access at global scale.

2.4.2 Probe Selection and the Tag System

Selecting appropriate vantage points is a critical methodological decision that directly affects the validity and interpretability of measurement results. Bajpai et al. [Bajpai et al, 2017] analyse the tag-based filtering system introduced by RIPE Atlas in July 2014 (available for probe selection since October 2014) and identify important differences in tag reliability.

System tags are automatically assigned by the RIPE Atlas infrastructure based on the results of continuous built-in measurements. Table I of Bajpai et al. [Bajpai et al, 2017] lists the default measurements run by every probe, including ping and traceroute to root servers and Atlas infrastructure, DNS queries to all root servers, and SSL certificate retrieval. From these measurements, RIPE Atlas derives and updates system tags every four hours. Examples include `system-ipv4-works` (IPv4 connectivity verified), `system-ipv6-works` (IPv6 connectivity verified), `system-resolves-a-correctly` (probe's local resolver returns correct A records), and `system-resolver-mangles-case` (resolver modifies the case of query names, indicating potential anomalous behaviour). Because system tags are derived from actual measurements and refreshed frequently, they reliably reflect the current network state of the probe.

User tags are applied voluntarily by probe hosts to describe their deployment context. Bajpai et al. [Bajpai et al, 2017] find that only **~2.8% of probe hosts ever update their user tags** after initially setting them. User tags therefore frequently become stale — a probe initially tagged as "home DSL" may now be on a fibre connection, but the tag remains unchanged. Researchers relying on user tags for probe selection risk drawing on outdated classification information. The practical recommendation from Bajpai et al. [Bajpai et al, 2017] is to prioritise system tags over user tags whenever system tag equivalents exist.

Using the `system-ipv6-works` tag to select dual-stacked probes, Bajpai et al. [Bajpai et al, 2017] identify approximately **2,300 probes** with working IPv6 connectivity as of January 2017, spanning **88 countries** and **822 Autonomous Systems**. Of these, 83% are connected within access networks (DSL, cable, or fibre), and 91% are concentrated in the RIPE (Europe/Middle East) and ARIN (North America) regions. Countries such as Belgium and Japan are explicitly identified as underrepresented for IPv6 measurement studies relative to their actual IPv6 user populations.

Bajpai et al. [Bajpai et al, 2017] propose a practical probe selection protocol applicable to any RIPE Atlas study requiring geographic diversity and connectivity reliability. The recommended sequence is: (1) apply the relevant connectivity system tags (`system-ipv4-works`, `system-ipv6-works`, `system-resolves-a-correctly`) to exclude probes with known connectivity or resolver problems; (2) apply any study-specific system tags (e.g., `system-ipv4-stable-1d` for probes with stable 24-hour IPv4 connectivity); (3) stratify the remaining probe pool geographically to achieve the desired regional balance, accounting for the known 91% Europe/North America concentration; (4) draw a random sample from each geographic stratum. This stratified-sampling protocol avoids the alternative of accepting RIPE Atlas’s natural distribution, which would produce a dataset dominated by European and North American probes and yield unreliable results for Africa, Latin America, and Southeast Asia. For studies of geographic DNS variation specifically, the geographic stratification step is the most critical: ensuring that each inhabited continent has a minimum number of probes guarantees that regional DNS responses — which may be qualitatively different from European responses — are captured with statistical confidence. Bajpai et al. [Bajpai et al, 2017] further note that selecting probes by country (rather than by geographic region) provides finer granularity but risks selecting all probes from a single ASN in small countries, reducing network diversity; the recommended trade-off is geographic stratification at the continent level with AS diversity as a secondary constraint.

2.4.3 DNS Measurement Capabilities and Notable Applications

DNS is one of RIPE Atlas’s primary built-in measurement types. Every probe continuously runs default DNS measurements to all thirteen DNS root server addresses, querying for `SOA`, `version.bind`, `hostname.bind`, and `id.server` in both standard (IN) and CHAOS classes, as well as over TCP [Bajpai et al, 2017]. User-defined DNS measurements extend this to any domain, any record type, any target resolver (the probe’s local resolver or a specified IP address), and optionally include EDNS options such as NSID or DNSSEC OK flags.

Nosyk et al. [Nosyk et al, 2025] quantify the DNS measurement activity of RIPE Atlas as of 2024: approximately **88,000 user-defined DNS measurement instances are executed daily**, in addition to the continuous built-in measurements. The platform generates approximately **1.3 billion results per day** across all measurement types.

Nosyk et al. [Nosyk et al, 2025] further characterise the composition of RIPE Atlas DNS measurements. They distinguish between measurements targeting **public resolvers** (querying 8.8.8.8, 1.1.1.1, or similar services), measurements querying **local resolvers** (the probe’s own ISP resolver), and measurements querying **authoritative servers** directly. Each category provides a different perspective on DNS infrastructure: public resolver measurements capture the performance and correctness of centralised resolver infrastructure; local resolver measurements reflect the DNS experience of typical end users in that network; and authoritative server measurements reveal the DNS responses returned at the authoritative layer, independent of resolver mediation. Nosyk et al. [Nosyk et al, 2025] observe that the distribution of measurement types varies substantially by user: some researchers deploy systematic targeted campaigns while others run exploratory single-run measurements. The operational analysis also characterises measurement temporal patterns: most user-defined measurements run during peak working hours in Europe and North America, creating temporal concentration that interacts with the

interference effects documented by Holterbach et al. [Holterbach et al, 2015]. Nosyk et al. [Nosyk et al, 2025] further derive practical guidelines for measurement design: selecting probes using system tags (not user tags), scheduling campaigns at off-peak times to reduce interference, validating probe geolocation against external databases, and cross-referencing DNS non-responses against connectivity indicators to distinguish censorship from measurement failure.

Several published applications illustrate the analytical reach of RIPE Atlas for DNS research:

DNS root manipulation detection [Jones et al, 2016]: Using approximately 8,000 Atlas probes in 2,755 Autonomous Systems across 189 countries, Jones et al. detect in-path DNS proxies by comparing ICMP ping RTT to the B root server (the only non-anycasted root, physically located in Los Angeles) with the DNS query RTT observed by the same probe. A Kenyan ISP’s transparent proxy is identified by the anomaly: DNS query RTT is 14 ms while the ping RTT to the actual B root in Los Angeles is 318 ms — a factor of 23, which is physically impossible unless the DNS response originates from a local proxy. Eleven probes in total return unexpected `HOSTNAME.BIND` identities, indicating ISP-level DNS interception or local root mirrors. The study further identifies that 1.7% of Atlas probes have inconsistent geolocation data, highlighting the importance of probe metadata validation.

Internal domain names and name collision risk [Boswell and Perkins, 2024]: This study leverages the concentration of RIPE Atlas probes in home networks — typically treated as a bias to be corrected — as a deliberate research feature. The authors design a four-step active measurement methodology combining traceroute-based gateway discovery, CHAOS TXT queries for `hostname.bind` to fingerprint gateway models, and reverse DNS queries to detect internal names resolved by home routers. Applied to all IPv4 RIPE Atlas probes in early 2024, the study discovers 3,092 distinct internal domain names used by 4,305 probes (50.86% of probes tested). The top ten internal names are all related to AVM FRITZ!Box routers (e.g., `fritz.box`, `myfritz.box`), reflecting the gateway’s dominance in the European market. Of the internal names discovered, 34.51% use a TLD not currently delegated in the public DNS — a significant name collision risk if those TLDs are later delegated, as demonstrated by the concrete `.box` TLD delegation in August 2023.

Anycast instance characterisation (Bortzmeyer, 2013; Finnegan, 2018; Edgecast, 2017): described in Section 2.5.

2.4.4 Geographic Bias and Measurement Interference

Two systematic platform-level limitations must be accounted for in any RIPE Atlas measurement methodology.

Geographic concentration: Nosyk et al. [Nosyk et al, 2025] quantify the geographic distribution of RIPE Atlas probes: **Germany and the USA together account for 28%** of all active probes, and Europe and North America together account for the majority of the platform. The Bajpai et al. [Bajpai et al, 2017] analysis confirms that 91% of dual-stacked probes are concentrated in the RIPE (Europe/Middle East) and ARIN (North America) regions. Africa, Latin America, and Central/Southeast Asia are substantially underrepresented relative to their shares of the global Internet user population. This geographic concentration means that any global inference from RIPE Atlas data must account for regional sampling bias: results will be statistically robust for Europe and North America but less reliable

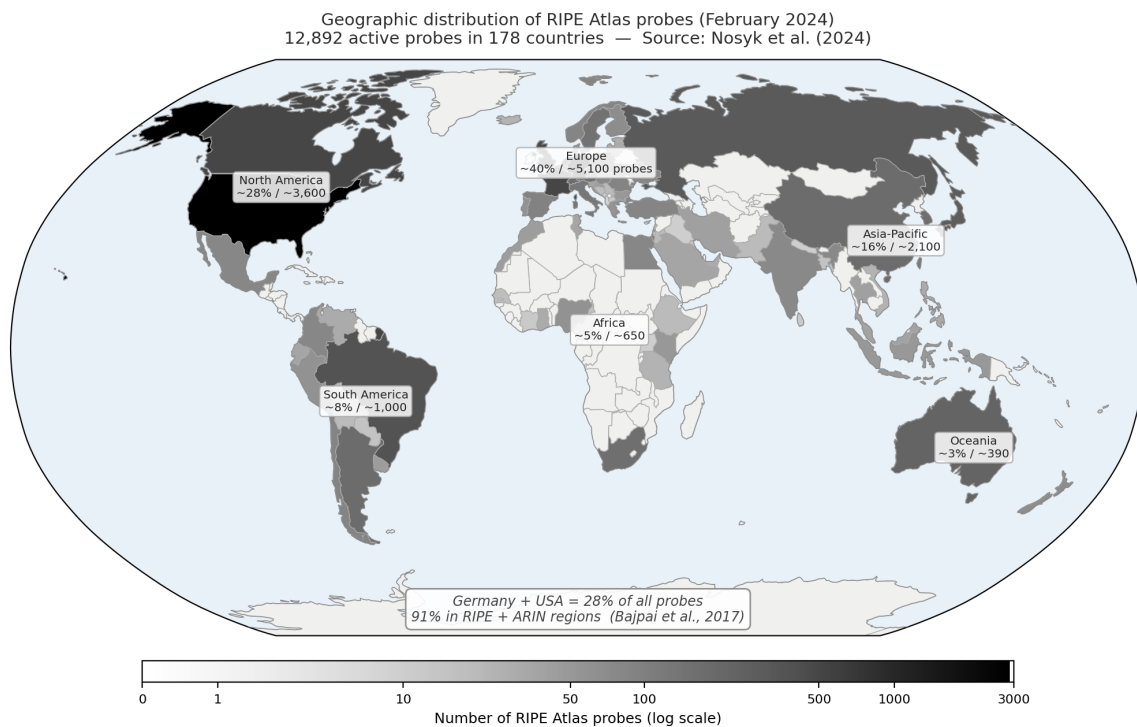


Figure 2.1: Geographic distribution of RIPE Atlas probes (February 2024). Choropleth on a Robinson projection; probe counts per country on a logarithmic scale. Germany and the USA together account for 28% of all active probes; 91% of dual-stacked probes are concentrated in the RIPE (Europe/Middle East) and ARIN (North America) regions [Bajpai et al, 2017; Nosyk et al, 2025]. Country probe counts are approximate, derived from published regional statistics.

for underrepresented regions. Figure 2.1 maps this distribution.

Measurement interference: Holterbach et al. [Holterbach et al, 2015] provide the definitive empirical characterisation of interference between concurrent measurements on RIPE Atlas. The platform schedules measurements from multiple users concurrently on the same probes, which creates two forms of interference. First, **precision degradation**: when a probe is under concurrent measurement load, timing measurements increase by more than 1 ms in the median case and more than 7 ms at the 95th percentile, even for the newer, more powerful v3 probe hardware. Second, **temporal desynchronisation**: measurements launched nominally simultaneously from multiple probes can complete up to **one hour apart** under heavy concurrent load. Critically, Holterbach et al. [Holterbach et al, 2015] show that upgrading probe hardware reduces precision degradation but does not resolve desynchronisation — the latter is a scheduling and network propagation problem, not a computation problem. The authors further demonstrate that historic RIPE Atlas datasets may have been affected by interference, and recommend two mitigation strategies: scheduling measurements at off-peak periods to reduce concurrent load, and using multiple repeated measurements to identify interference-induced outliers.

2.5 Anycast Routing in the DNS Infrastructure

IP anycast is the dominant routing architecture for DNS root servers, TLD servers, large public resolvers (Google 8.8.8.8, Cloudflare 1.1.1.1), and many authoritative name servers. Understanding anycast — and its implications for geographic routing variation — is essential context for interpreting distributed DNS measurements: probes at different geographic locations may contact physically different servers behind the same anycast IP address, and the resulting DNS responses may differ not only in which IP is returned, but in response latency and even content.

2.5.1 Internet-Wide Anycast Deployment

Cicalese et al. [Cicalese et al, 2015] conduct the first Internet-wide census of IPv4 anycast adoption, scanning the entire routable IPv4 address space from approximately 300 PlanetLab vantage points spread across research institutions. The detection principle is the **speed-of-light constraint**: if two geographically separated vantage points observe round-trip times to the same IP address that are inconsistent with a single physical server — that is, the latency "disks" (circles of radius $RTT/2$ centred on each VP) do not intersect — the target must be anycast, with each VP contacting a different replica.

Anycast replica enumeration uses the **Maximum Independent Set** (MIS) algorithm on the set of non-overlapping latency disks, which conservatively lower-bounds the number of distinct replicas. Geolocation of each replica applies a maximum-likelihood estimator biased toward city population within the smallest bounding latency disk, validated against known CDN ground truths (Cloudflare, Edgecast). Cicalese et al. [Cicalese et al, 2015] report approximately 75% city-level accuracy with a median geolocation error of 350 km for incorrectly located replicas.

Key census findings: approximately **10^3 IPv4 /24 subnets** are anycast in the current Internet; each deployment serves on average approximately **10 replicas** (a conservative lower bound given the 300-VP limitation); anycast is used not only by DNS operators but by tier-1 ISPs, CDNs, cloud providers, network equipment vendors, and DDoS mitigation services. Cicalese et al. [Cicalese et al, 2015] note an important calibration result for the thesis: studies targeting individual anycast deployments (e.g., DNS root servers) use $O(10^4)$ – $O(10^5)$ vantage points to achieve approximately 90% replica recall — a figure that motivates the use of RIPE Atlas's 12,000+ probes for high-recall anycast characterisation.

2.5.2 Anycast Instance Identification Techniques

Two complementary techniques allow active DNS measurements to identify which physical anycast instance responded to a query, without requiring access to BGP routing tables:

NSID option (RFC 5001): The NSID EDNS option is appended to a DNS query in the OPT record's Additional section. The responding server includes a server-defined identifier string in the OPT record of its response. The NSID string typically encodes the instance's location (e.g., `dns.th2.nic.fr` for the Paris instance of `d.nic.fr`, `dns.fra.nic.fr` for Frankfurt). NSID can be added to any DNS query type (A, AAAA, SOA, etc.), making it universally applicable to any anycast service that implements it. Bortzmeyer [Bortzmeyer, 2013] uses NSID to characterise the instance distribution of `d.nic.fr` across approximately 500 RIPE Atlas probes in six measurement runs.

"id.server" CHAOS query (RFC 4892): Querying for `id.server` in the DNS CHAOS class (TXT `id.server CHAOS`) returns a text string identifying the responding instance — typically an IATA airport code (e.g., `ams` for Amsterdam, `fra` for Frankfurt) or a city abbreviation. CHAOS class records are not cached by resolvers, because the CHAOS class is outside the normal IN (Internet) class caching scope, guaranteeing that each query reaches the actual BGP-routed anycast instance rather than a cached response from a different resolution path. Finnegan [Finnegan, 2018] identifies this cache-bypassing property as the mechanism's key advantage for anycast characterisation.

As Finnegan [Finnegan, 2018] explains, `id.server` is the preferred technique when supported by a service, since it is specifically designed for instance identification; NSID serves as the universal fallback applicable to any query type. A third complementary approach used by Edgecast (2017) is **penultimate-hop traceroute analysis**: traceroutes from the 12 RIPE Atlas anchors hosted by Edgecast are analysed for the penultimate hop (the last router hop before the CDN server), whose IP address encodes the CDN PoP identity and can be used to map the traceroute's routing path to a specific Point of Presence.

2.5.3 BGP Attraction Basins and IPv4/IPv6 Divergence

Two empirical findings from the anycast DNS measurement literature have direct implications for interpreting geographic DNS variation in this thesis.

BGP attraction basins [Bortzmeyer, 2013; Finnegan, 2018]: BGP routing directs each anycast query to the "nearest" replica in BGP path-length terms, where "nearest" is determined by the AS-path topology of peering agreements and transit relationships — not by geographic proximity. Bortzmeyer [Bortzmeyer, 2013] demonstrates this counter-intuitive effect strikingly for `d.nic.fr`, the `.fr` TLD's anycast authoritative name server: in IPv4, the **Paris instance** (`dns.th2.nic.fr`) and the **Frankfurt instance** (`dns.fra.nic.fr`) each attract approximately **36% of global queries**. But North American probes route **55% of their traffic to Paris** and only 33% to Frankfurt — despite Paris being approximately 6,000 km farther from North America than a theoretically well-placed US instance would be. Asia-Pacific probes similarly route approximately 38% to Paris. This pattern reflects BGP peering arrangements that make European PoPs the BGP-shortest path for large portions of non-European network space — a phenomenon Bortzmeyer [Bortzmeyer, 2013] terms "BGP attraction basins."

Finnegan [Finnegan, 2018] confirms the same pattern at operational scale for UncensoredDNS: 500 RIPE Atlas probes (250 globally distributed, 250 US-focused) reveal strong European coverage but very limited US West Coast presence, identified as a gap that a Fremont Internet Exchange Point PoP would address. The 250/250 probe split strategy — a global component for worldwide distribution characterisation plus a region-focused component for fine-grained regional analysis — is directly applicable to the measurement design of this thesis. Figure 2.2 depicts the BGP attraction basins documented by Bortzmeyer [Bortzmeyer, 2013] for `d.nic.fr`.

IPv4/IPv6 anycast divergence [Bortzmeyer, 2013]: The IPv4 and IPv6 BGP routing tables are maintained independently; they may propagate different prefixes and reflect different peering relationships for the same anycast service. Bortzmeyer [Bortzmeyer, 2013] documents a striking reversal for `d.nic.fr`: the dominant instance **reverses between protocols** — Frankfurt receives **36% of global IPv6 queries** (dominant) but only 17% in IPv4 (subordinate); Paris receives 36% in IPv4 but only 17% in IPv6. This

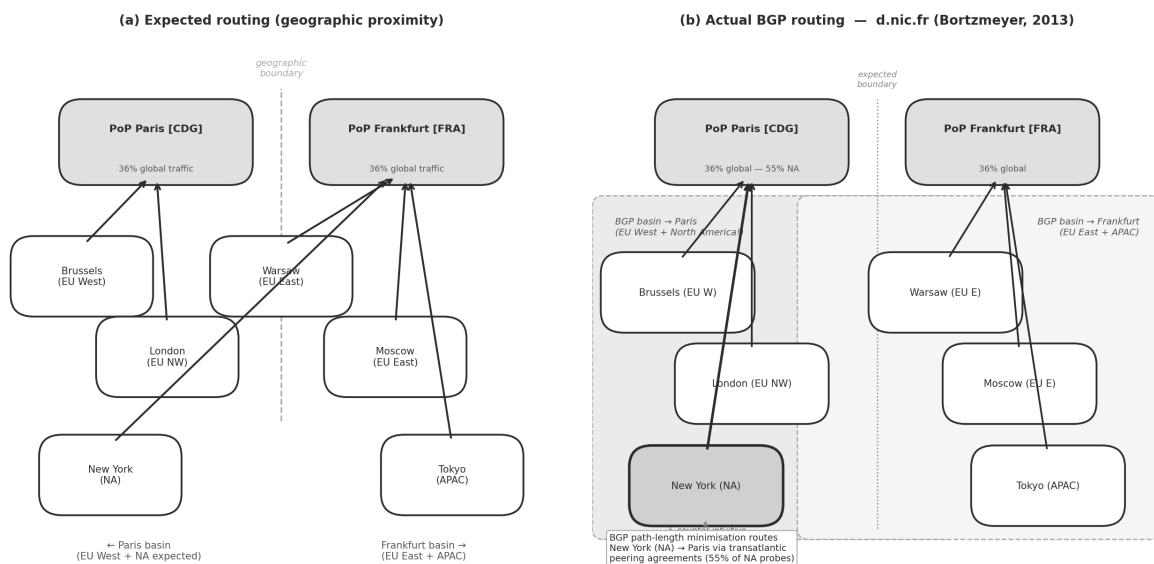


Figure 2.2: BGP attraction basins in anycast DNS routing, illustrated for `d.nic.fr` [Bortzmeyer, 2013]. **(a) Expected routing:** geographic proximity would route EU West clients to Paris and EU East / APAC clients to Frankfurt. **(b) Actual BGP routing:** North American probes route 55% of their queries to Paris due to BGP peering arrangements, despite the 6,000 km additional distance. BGP path-length topology, not geography, determines attraction basin boundaries.

result demonstrates that dual-stack measurements must be run and analysed separately: combining IPv4 and IPv6 probes in a single analysis would obscure a qualitative difference in anycast routing.

For CDN PoP optimisation, Edgecast (2017) test six different probe grouping strategies — by country, by geolocation, by ASN, by RIR region, by continent, and by clustering — and find that **geolocation-based grouping** provides the best trade-off between routing accuracy and interpretability. They also propose a logistic-normalised scoring framework (+1 for improved routing, -1 for degraded routing) that quantifies the net impact of anycast BGP configuration changes on per-PoP traffic distribution across probe groups. This scoring framework is adaptable to this thesis’s analysis of geographic DNS response variation.

2.5.4 Anycast Routing Quality and Application Context

The performance cost of BGP-based anycast routing — its potential to deliver clients to geographically or topologically suboptimal replicas — was systematically quantified by Koch et al. [Koch et al, 2021] in the largest comparative study of anycast inflation to date, covering both the DNS root server system and Microsoft’s global CDN.

Koch et al. [Koch et al, 2021] distinguish **geographic inflation** (the assigned replica is farther than the closest deployed site) from **latency inflation** (the measured RTT exceeds the theoretical minimum from the closest site). For root DNS, they find that more than **95% of users** experience some geographic inflation for at least one root letter, and up to 40% experience more than 100 ms of inflation to individual root letters. However, when averaged across all root letters — exploiting the fact that recursive resolvers preferentially route queries to their best-performing letter among the 13 root addresses — only **10% of users** experience more than 100 ms of inflation.

Critically, Koch et al. [Koch et al, 2021] argue that root DNS inflation is **practically irrelevant to users**.

Root DNS records have TTLs of approximately 518,400 seconds (six days), meaning recursive resolvers contact the root DNS at most once per root record per six days. The additional latency from inflation is amortised across thousands of queries served from cache, making its per-user-query cost negligible. By contrast, Microsoft's CDN serves directly user-facing content where each HTTP connection involves multiple RTTs to the front-end server. Here, inflation directly degrades user experience: the authors estimate that if Microsoft's CDN suffered the same inflation as individual root letters, each page load would incur hundreds of additional milliseconds. Microsoft has therefore invested heavily in peering agreements to limit CDN inflation to approximately **35% of users experiencing any inflation**.

For this thesis, these findings provide important context for interpreting DNS timing variations across RIPE Atlas probes. Observed RTT differences between probes querying the same anycast DNS server do not necessarily reflect the geographic routing variation relevant to DNS response content — the IP address returned in the A record. The routing of the DNS query (which anycast instance answers) may differ from the routing of subsequent data connections to the IP address returned. Both dimensions require separate characterisation.

Koch et al. [Koch et al, 2021] use RIPE Atlas directly in their study: 7,000 ping measurements from 1,000 probes (3 measurements per probe per CDN ring) serve as supplementary calibration data for the CDN latency analysis, illustrating how RIPE Atlas can complement server-side logs to provide client-perspective ground truth.

2.6 Content Delivery Networks and DNS-Based Geographic Routing

CDNs distribute content across geographically dispersed server clusters and use DNS as the primary mechanism to direct each client to an appropriate replica. The interaction between DNS resolution and CDN routing is central to our thesis: understanding it explains why different geographic vantage points may receive different DNS responses for the same domain, and what those differences mean in terms of content accessibility.

2.6.1 DNS as the Dominant CDN Redirection Mechanism

Wang et al. [Wang et al, 2018] provide a systematic survey of DNS-based CDN server redirection, positioning it as "the most popular means of deploying CDNs" due to its compatibility with existing DNS infrastructure and ease of deployment without requiring client-side software changes. The CDN request routing problem involves two sub-problems: a **server selection mechanism** (choosing the optimal replica based on network distance, server load, latency, and availability) and a **server redirection mechanism** (directing the client to the selected replica). DNS-based redirection uses the DNS resolution process as the redirection channel: the CDN's authoritative name server returns a different A/AAAA record or CNAME depending on the inferred location of the querying resolver. Major CDNs using this approach include Akamai (which pioneered DNS-based CDN redirection), Limelight Networks, and Mirror Image [Wang et al, 2018].

Calder et al. [Calder et al, 2015] empirically evaluate a production anycast CDN — Microsoft's Bing front-end network — and find that anycast-based redirection (where BGP routing, rather than DNS,

directs clients to replicas) directs approximately **20% of clients to suboptimal front-end servers** relative to the geographically closest available front-end. For this 20% of affected clients, switching to DNS-based unicast redirection — where a geolocation-aware DNS system selects the closest front-end server and returns its IP directly — could improve latency by **7 to 100 ms** depending on the client’s region and routing context. This result confirms that BGP’s performance-agnostic routing creates measurable quality-of-service degradation even at a well-engineered commercial CDN, and motivates the hybrid DNS-anycast architectures adopted by major CDNs: anycast for proximity and resilience, DNS for fine-grained routing corrections. To investigate the causes, Calder et al. [Calder et al, 2015] use RIPE Atlas traceroutes as a diagnostic complement to server-side logs and JavaScript beacon measurements — illustrating the platform’s role as a flexible supplementary measurement tool for validating client-perspective routing hypotheses derived from server-side data. The geographic distribution of the 20% suboptimal clients is non-uniform: clients in regions where Bing operates few anycast PoPs (e.g., parts of South America, Africa, and Southeast Asia) experience higher inflation rates than clients in densely covered regions (North America, Western Europe). This geographic heterogeneity in anycast routing quality parallels the heterogeneity in DNS response geography that this thesis sets out to characterise.

Li et al. (2025), studying Twitch’s global CDN for live video streaming, find that DNS-based CDN routing for Twitch operates at **continent-level granularity**: distinct groups of CDN server IP addresses are assigned to clients in Europe, North America, Asia-Pacific, and other regions, with the assignment determined by the querying resolver’s inferred continent. This coarse geographic partitioning simplifies CDN management but creates sharp geographic boundaries in DNS response content — boundaries detectable through distributed active measurement.

2.6.2 The Remote DNS Problem

DNS-based CDN routing critically assumes that the recursive resolver’s IP address is a reliable proxy for the end user’s geographic location — specifically, that the resolver is topologically close to its clients. Wang et al. [Wang et al, 2018] identify the **remote DNS problem**: this assumption fails when clients use a centralised public resolver (Google 8.8.8.8, Cloudflare 1.1.1.1, OpenDNS) that is geographically or topologically distant from the client. When the CDN authoritative server receives a DNS query from a public resolver in California, it may return a California-optimised CDN replica even though the actual client is in Brazil. The result is suboptimal server assignment and degraded content delivery performance.

Hours et al. [Hours et al, 2016] quantify this effect causally using passive TCP traffic traces from a large European ISP and a Bayesian causal network (PC algorithm + do-calculus). Their analysis confirms a causal pathway: clients using **Google DNS** (GDNS) are assigned geographically farther Akamai CDN servers than equivalent clients using their **local ISP DNS** (LDNS), resulting in higher ISP-side RTT and reduced download throughput. A secondary finding — not predictable from correlation analysis alone — is that servers assigned to GDNS users are configured with a smaller TCP initial congestion window than servers assigned to LDNS users, amplifying the throughput difference beyond the server-distance effect.

Wang et al. [Wang et al, 2018] quantify the scale of the remote DNS problem: public DNS service adoption among Internet users is growing at approximately **27% per year**, meaning that a rapidly

increasing fraction of DNS queries to CDN authoritative servers originate from a small number of centralised resolver IP addresses rather than the diverse set of ISP resolvers that would accurately reflect client geography. The study measures this as a distance mismatch between the CDN server assigned by the remote resolver and the CDN server geographically closest to the actual client, finding mismatches of up to **113%** — meaning the assigned server is more than twice as far as the optimal server. At these mismatch levels, the impact on throughput and latency is measurable even for users with high-bandwidth connections. Wang et al. [Wang et al, 2018] frame this as the "remote DNS paradox": the very features that make public DNS services popular with users (reliability, performance, privacy) are precisely the features that degrade CDN routing quality, because those features require operating a centralised resolver fleet rather than a geographically distributed one. Hours et al. [Hours et al, 2016] confirm the causality of this effect for the Akamai CDN from the perspective of a large French ISP: clients using Google DNS are assigned servers that are, on average, **20 ms farther** in RTT than servers assigned to equivalent clients using the ISP's own resolver, with a secondary server configuration effect that reduces download throughput by approximately **14%**.

The remote DNS problem is directly relevant to RIPE Atlas measurements: probes that use a public resolver (rather than their local ISP resolver) will obtain CDN-assigned IP addresses based on the public resolver's location, not the probe's location. This must be accounted for when interpreting geographic variation in DNS A/AAAA responses across probes with heterogeneous resolver configurations.

2.6.3 EDNS Client Subnet: Mechanism, Deployment, and Privacy Trade-offs

The **EDNS Client Subnet** (ECS) extension, specified in RFC~7871 [RFC7871, 2016], addresses the remote DNS problem by embedding a truncated version of the client's IP address in the DNS query forwarded from the recursive resolver to the authoritative CDN name server. The ECS EDNS option carries four fields:

- **FAMILY:** Address family (1 = IPv4, 2 = IPv6).
- **SOURCE PREFIX-LENGTH:** The number of bits of the client address included in the option. RFC~7871 [RFC7871, 2016] recommends /24 for IPv4 and /48 for IPv6, balancing geographic precision with privacy protection.
- **SCOPE PREFIX-LENGTH:** Set by the authoritative server in the response; indicates how many bits of the client address were actually used in selecting the response. A scope of /0 means the response applies globally and is not location-specific.
- **ADDRESS:** The truncated client IP prefix.

An ECS-aware CDN authoritative server uses the ADDRESS field — the client's IP prefix — rather than the resolver's source IP to select the appropriate CDN replica, then returns the SCOPE in the response. Intermediate caches must store separate entries keyed by (question, ECS scope prefix), since the same domain may resolve differently for clients in different /24 subnets. This **per-scope caching** inflates resolver cache size substantially, a concern acknowledged in RFC~7871 [RFC7871, 2016] but not quantified.

ECS introduces a fundamental privacy trade-off (Wang et al., 2018; RFC~7871 [RFC7871, 2016]): forwarding the client's IP prefix to the authoritative CDN server — and to any on-path observer — makes the client's network location visible to infrastructure that would otherwise only see the resolver's IP. RFC~7871 [RFC7871, 2016] explicitly acknowledges this: "if we were just beginning to design this mechanism, and not documenting existing protocol, it is unlikely that we would have done things exactly this way." The RFC recommends that ECS be **disabled by default** and enabled only when the performance benefit is clear. A further privacy risk identified by Wang et al. [Wang et al, 2018] is **redirection enumeration**: an adversary with ECS-enabled queries can systematically probe the CDN's mapping from client IP prefixes to server assignments, exposing the CDN's traffic engineering policies. For RIPE Atlas measurements, the practical deployment of ECS is limited. Bortzmeyer [Bortzmeyer, 2017] reports that a test of 100 Belgian Atlas probes found that only a small number used resolvers that forwarded ECS options. This means that, for most probes, the DNS response is determined by the resolver's IP address (the pre-ECS default behaviour) rather than the probe's IP. Researchers wishing to disable ECS explicitly — to obtain resolver-IP-determined responses consistently — can include an ECS option with SOURCE PREFIX-LENGTH = 0 in their queries; Google Public DNS, for instance, respects this opt-out (Bortzmeyer, n.d., tutorial). Figure 2.3 illustrates the three ECS scenarios.

2.6.4 Alternative Solutions to the Remote DNS Problem

Wang et al. [Wang et al, 2018] catalogue and compare three main approaches to the remote DNS problem, each with different deployment complexity, privacy implications, and effectiveness:

ECS (EDNS Client Subnet): Described above. Effective for CDNs that implement it; major performance benefit over plain remote DNS; significant privacy cost; caching complexity; not universally deployed. Wang et al. [Wang et al, 2018] note that ECS is a "recognised solution" but its privacy cost and the conditional deployment recommendation of RFC~7871 [RFC7871, 2016] limit its reach.

Name extension: The client's IP address prefix is encoded in the domain name itself — for example, `<client-prefix>.cdn.example.com` — and the CDN authoritative server selects a response based on the prefix encoded in the queried name. This approach avoids the ECS privacy problem (the prefix is visible in the query name, not in an EDNS option, but both are observable) and does not require EDNS support. However, it requires the recursive resolver to forward the original query name unmodified, and it inflates the DNS namespace with per-prefix subdomains, complicating zone management.

Direct resolution: The client bypasses recursive resolvers entirely and contacts the CDN authoritative server directly, or the CDN provides a resolver co-located with its edge servers. This eliminates the LDNS-client distance problem at the cost of losing resolver-level privacy protection, requiring client software changes, and potentially exposing the client's IP directly to the CDN operator. Wang et al. [Wang et al, 2018] consider this approach impractical for general deployment, though it is implemented in specialised scenarios (e.g., mobile CDN clients that contact CDN-provided resolvers).

For this thesis, the implication of this landscape is that no single DNS measurement protocol can capture all aspects of CDN routing: ECS-enabled queries may return CDN-assigned addresses different from those returned by non-ECS queries for the same probe. Controlling for ECS behaviour — by either consistently disabling ECS (subnet=0/0) or consistently enabling it at /24 granularity — is an important

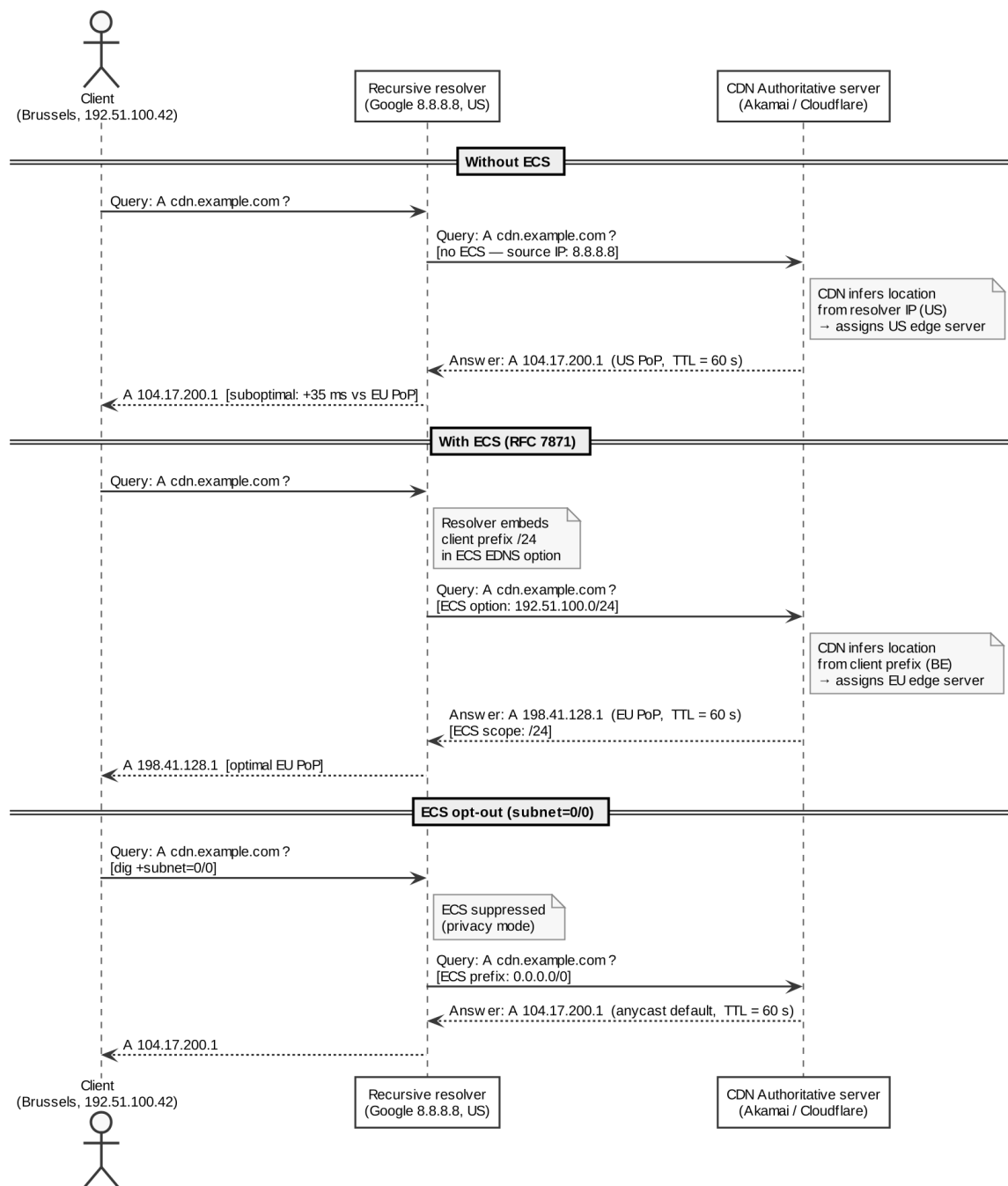


Figure 2.3: Effect of EDNS Client Subnet (ECS, RFC 7871) on CDN geographic routing. Without ECS, the CDN authoritative server infers client location from the recursive resolver’s IP (8.8.8.8, US), assigning a suboptimal US edge server to a Belgian client. With ECS, the resolver embeds the client’s /24 prefix; the CDN selects the geographically appropriate EU PoP. ECS opt-out (*subnet=0/0*) suppresses the prefix entirely, reverting to resolver-IP-based routing.

measurement design choice documented in Chapter 4.

2.7 Domain Ranking Lists: From Alexa to Tranco

Studies of DNS behaviour at scale require a representative and reproducible corpus of domain names. The choice of domain list has significant implications for result validity, research reproducibility, and resistance to adversarial manipulation.

2.7.1 Problems with Commercial Ranking Lists

Le Pochat et al. [Pochat et al, 2019] systematically evaluate the four main freely available domain ranking lists across five properties: similarity, stability, representativeness, responsiveness, and benignness. The four lists analysed are **Alexa** (Amazon subsidiary, based on browser extension traffic), **Cisco Umbrella** (based on DNS traffic to OpenDNS), **Majestic** (based on web crawl backlinks), and **Quantcast** (based on US-centric traffic instrumentation).

Similarity: Despite all four lists purporting to rank "the most popular domains on the Internet", they agree on only approximately 70,000 domains out of a combined pool of ~2.82 million unique domains. The rank-biased overlap (RBO) — a similarity measure that weights highly-ranked domains more heavily — remains between 24% and 33% even when the top 100 domains receive most of the weight. This means that choosing Alexa rather than Umbrella for a study of the top 10,000 domains yields a substantially different set of domains, potentially changing study conclusions [Pochat et al, 2019].

Stability: Majestic's list changes by at most 1% per day, making it the most stable. Cisco Umbrella changes by approximately 10% daily. Alexa, which was similarly stable to Majestic through January 2018, underwent an undisclosed change in its averaging methodology that caused daily turnover to reach approximately **50% of the list** — meaning half the domains on Alexa's "top million" change every day. This extreme volatility makes Alexa-based longitudinal studies unreliable, since the domain corpus changes faster than the phenomena being studied.

Representativeness and benignness: Cisco Umbrella includes approximately 51% entries that are not actual web domains — EC2 internal hostnames, typo domains (e.g., `google.com`), and internal corporate names that happen to be visible in OpenDNS query traffic. Majestic's list, widely used as a whitelist, contains 2,162 malicious domains — a dangerous characteristic for security research that treats list membership as an indicator of legitimacy.

Manipulability: Le Pochat et al. [Pochat et al, 2019] empirically demonstrate that every list has at least one low-cost manipulation technique. The most striking finding is that **a single HTTP request from a browser with the Alexa extension** suffices to increment a domain's request count for that day — and the authors experimentally reach a rank as good as **28,798** with minimal effort. For Cisco Umbrella, injecting DNS queries from a single recursive resolver that participates in the Umbrella data collection suffices to inflate a domain's apparent popularity. For Majestic, creating links from a small set of domains already in the Majestic index is sufficient. Le Pochat et al. [Pochat et al, 2019] note that none of the commercial list operators publishes the full details of their ranking methodology,

making it impossible to audit the lists for manipulation or to reproduce their ranking calculations independently. This opacity is not merely an academic concern: security researchers who use these lists as whitelists (treating top-ranked domains as legitimate) are exposed to adversarial list manipulation; and researchers who compare results across studies may be comparing datasets whose domain compositions differ substantially due to methodology differences rather than actual changes in Internet usage. The security dimension is particularly acute: van der Toorn et al. [van der Toorn et al, 2018] note that malicious actors have strong incentives to inflate their domains' ranks on traffic-based lists, since a high rank provides social legitimacy. This makes Alexa trivially manipulable by adversaries seeking to insert malicious domains into whitelists or to bias research results toward domains they control. Table 2.3 compares the four lists across the key properties identified by Le Pochat et al. [Pochat et al, 2019].

Property	Alexa	Cisco Umbrella	Majestic	Tranco
Data source	Browser extension traffic	DNS queries to OpenDNS	Web crawl backlinks	Multi-source aggregation
Daily change rate	≈50% (post-2018)	≈10%	≈1%	≈0.6%
Manipulation effort	1 HTTP request from browser ext.	1 DNS query injection	Few backlinks from indexed domains	4× effort vs. single-source lists
Non-domain entries	Low	≈51% (EC2 host-names, typos...)	Low	Filtered (HTTP 200 + safe browsing)
Malicious domains	Some	Some	2,162 documented	Filtered
Reproducibility	Daily overwrite (no archive)	Daily overwrite	Daily overwrite	Stable permalink per list version
Recommended for	Legacy reference only	DNS-traffic studies (with caution)	Backlink analysis	DNS/security research

Table 2.3: Comparison of freely available domain ranking lists. Stability, manipulation resistance, and reproducibility data from Le Pochat et al. [Pochat et al, 2019]. Tranco is the only list designed with research reproducibility and manipulation resistance as explicit objectives.

2.7.2 Tranco: Aggregation, Stability, and Manipulation Resistance

Le Pochat et al. [Pochat et al, 2019] propose **Tranco** as a research-oriented alternative that addresses all four identified limitations. Tranco is publicly available at <https://tranco-list.eu> and operates as follows:

Multi-source aggregation: Tranco aggregates multiple ranking lists — originally Alexa, Cisco Umbrella, Majestic, and Quantcast — using either the **Borda count** or the **Dowdall rule**. In the Borda count, each domain receives a score equal to $N - \text{rank}_i$ for each list where it appears (N is the maximum rank considered; rank_i is the domain's rank in list i), and scores are summed across lists. The Dowdall rule assigns a score of $1/\text{rank}_i$ per list. Both methods give higher weight to domains that rank highly across multiple lists, and assign nonzero scores to domains absent from some lists (based on their worst-case rank). Aggregation across multiple data sources reduces the influence of any single source's biases and requires that an adversary manipulate multiple independent lists to affect Tranco rankings.

Temporal smoothing: Aggregation is performed over a **30-day rolling window** of daily list snapshots. Each domain's rank is derived from the aggregated scores over these 30 days, smoothing out day-to-day volatility. The result: Tranco changes by only **0.6% per day** (versus ~50% for Alexa), providing a stable

corpus for longitudinal studies.

Manipulation resistance: Because an adversary must manipulate multiple lists over 30 days to shift a Tranco ranking, Le Pochat et al. [Pochat et al, 2019] show that manipulating Tranco to achieve the same rank as in existing lists requires at least **quadrupled effort**.

Reproducibility: Historical Tranco lists are archived and accessible via a stable URL including the list generation parameters. A researcher can retrieve the exact list version used in a previous study, enabling reproduction and replication of results — a fundamental requirement of scientific research that commercial lists cannot guarantee (they overwrite their previous version daily).

For this thesis, Tranco is the appropriate choice for the domain corpus: stability ensures that the same set of domains is measured across multiple campaign repetitions, enabling meaningful temporal comparison; manipulation resistance limits the risk that adversarially placed domains distort the DNS measurement results; and archive access supports reproducibility of the measurement dataset.

2.8 DNS Infrastructure Centralization

Despite its designed-in distributed architecture, the DNS ecosystem has undergone progressive centralization driven by commercial consolidation and economies of scale. Xu et al. [Xu et al, 2023] provide the most comprehensive recent characterisation of this phenomenon, revealing concentration levels that are not visible from DNS query counts or market share statistics alone.

2.8.1 Client-Side Resolver Centralization

Xu et al. [Xu et al, 2023] introduce the **NS chain reflection** technique: a crafted DNS query sequence that causes resolvers to reveal their internal resolution chain through the sequence of NS referrals they generate at the authoritative level. Unlike the CNAME-based resolver pool discovery method proposed by Schomp et al. — which Xu et al. [Xu et al, 2023] show is no longer effective against modern resolver architectures — NS chain reflection works from a single vantage point and reveals the full resolver pool structure including indirect recursive resolvers.

Using this technique applied Internet-wide, Xu et al. [Xu et al, 2023] find that over **90% of all forwarding resolvers (FDNSes)** are backed by fewer than **5% (4,071) of indirect recursive resolvers (iRDNSes)** — an extreme concentration in the actual DNS resolution infrastructure even when the visible surface (the set of FDNS IP addresses) appears diverse. This means that the geographic diversity of RIPE Atlas probes — each using a different ISP resolver — may be illusory: many of those ISP resolvers ultimately forward queries to the same small set of iRDNS clusters operated by major public DNS providers.

The practical consequence for distributed DNS measurement: querying through local resolvers provides geographic diversity in the FDNS layer but not necessarily in the resolution layer that actually contacts authoritative servers. Direct querying of authoritative servers is required to bypass this invisible centralization.

2.8.2 Server-Side Authoritative DNS Centralization

Analysing zone files from all 1,138 gTLDs covering 210,446,494 domain names, Xu et al. [Xu et al, 2023] find dramatic concentration on the authoritative side as well:

- Just **10 DNS providers** — operating 0.45% (12,679) of all name servers across 1,138 gTLDs — provide authoritative resolution for **48.5% of all gTLD domain names**, representing over 100 million domains.
- More than **98% of all domain names** rely on a **single authoritative name server provider** for all their NS records. A failure of that provider causes complete DNS unavailability for those domains.
- **60% of combinations of DNS provider pairs** share underlying infrastructure, meaning enterprises that distribute their NS records across multiple providers may still implicitly share a common failure point.

The **shared infrastructure risk** is arguably more insidious than single-provider dependence. Xu et al. [Xu et al, 2023] find that 60% of combinations of DNS provider pairs — enterprises or domain operators who distribute their NS records across two ostensibly independent providers — share underlying physical infrastructure. This means that a failure of a shared upstream network component (a shared transit provider, a shared IXP interconnect, or shared anycast routing infrastructure) can simultaneously affect both nominal providers, defeating the intended redundancy. Xu et al. [Xu et al, 2023] attribute this to the consolidation of Internet backbone infrastructure among a small number of global tier-1 providers, which creates hidden dependencies even among nominally independent DNS operators.

Real-world outages confirm the operational stakes: the Akamai DNS outage in June 2021, the Fastly CDN outage in June 2021, and the Facebook DNS misconfiguration in October 2021 all caused widespread Internet inaccessibility for millions of domains and users, consistent with the concentration levels documented by Xu et al. [Xu et al, 2023]. The Facebook outage of October 4, 2021 is particularly instructive from a DNS perspective: a misconfigured BGP route withdrawal caused Facebook’s own authoritative name server addresses to become unreachable, making the domain unresolvable from any resolver in the world simultaneously. For a domain that relies on a single DNS provider — which Xu et al. [Xu et al, 2023] show is over 98% of all gTLD domains — a provider-level BGP event of this type has no fallback. Distributed active measurement at the time of such outages can capture the propagation and recovery of DNS reachability in real time, providing empirical ground truth on outage geography and duration that server-side operator logs cannot provide from the client perspective.

For distributed DNS measurement campaigns, server-side centralization implies that measuring 10,000 Tranco domains will not yield measurements of 10,000 independent DNS infrastructures. Rather, the measurements will cluster around a small number of major DNS providers, each responding similarly for their client domains. Understanding this structure is important both for measurement design (aggregating results at provider level may be more meaningful than at domain level) and for interpreting geographic variation (variation may reflect provider-level anycast routing rather than domain-specific configuration).

2.9 DNS Security and Measurement Validity

The integrity of DNS responses received by RIPE Atlas probes is not guaranteed: several forms of DNS manipulation can cause probes to receive responses from unexpected sources or with altered content. Three dimensions of DNS security are directly relevant to the measurement validity of this thesis.

2.9.1 DNSSEC Deployment and Its Limitations

DNSSEC protects the authenticity and integrity of DNS responses through a cryptographic chain of trust extending from the signed root zone (since July 2010) to individual domains. Van Rijswijk-Deij et al. (2016) include DNSKEY, DS, and RRSIG record types in the OpenINTEL standard query set specifically to monitor DNSSEC adoption at the TLD zone and domain level. The longitudinal DNSSEC study (van Rijswijk-Deij et al., USENIX Security 2017, via van Rijswijk-Deij 2018) reveals systemic key management failures: many domains that appear in signed zones suffer from expired signing keys or incomplete key rollovers, leaving them with broken DNSSEC chains that validating resolvers would reject. Bajpai et al. [Bajpai et al, 2017] note through the system tag `system-resolves-a-correctly` that DNSSEC validation is far from universal: many ISP resolvers do not validate DNSSEC signatures, passing invalid signed responses to clients without error.

For this thesis, DNSSEC matters in two ways: (1) domains with broken DNSSEC chains may return `SERVFAIL` responses from validating resolvers, which can be misidentified as measurement errors rather than security failures; and (2) DNSSEC-signed domains provide an additional data dimension for characterising the DNS infrastructure of Tranco domains. When a RIPE Atlas probe uses a validating resolver and queries a domain with a broken DNSSEC chain, the probe returns `SERVFAIL` — the same response code returned when the authoritative server is unreachable. These two failure modes are indistinguishable without additional information. One mitigation is to include both the `DO` bit (DNSSEC OK) and the `CD` bit (Checking Disabled) in measurement queries: with `CD` set, the resolver returns the response despite DNSSEC validation failure, allowing the measurement to distinguish a DNSSEC failure (response exists with `CD` overriding validation) from a genuine unavailability (no response regardless of `CD`). Alternatively, querying authoritative servers directly — as proposed in this thesis — bypasses DNSSEC validation entirely, since the query is not mediated by a validating recursive resolver; the raw DNS response from the authoritative server is returned regardless of whether the DNSSEC chain is complete. This is another argument in favour of the direct-authoritative-query design choice: it eliminates both the resolver-location bias (Section 2.1.4) and DNSSEC-induced `SERVFAIL` confusion as sources of measurement artefact.

2.9.2 DNS Root Manipulation and In-Path Proxies

Jones et al. [Jones et al, 2016] demonstrate that DNS responses observed by RIPE Atlas probes may not originate from the expected authoritative servers. Using approximately 8,000 Atlas probes in 2,755 ASes across 189 countries, they identify ten ISPs that deploy in-path transparent DNS proxies: these proxies intercept root DNS queries from clients and answer them locally, without forwarding to the actual root servers. Detection uses two independent methods: (1) anomalous RTT — if DNS query RTT to the B root is a factor of $23\times$ lower than ICMP ping RTT to the same IP (14 ms vs. 318 ms for the

Kenyan example), a local proxy is almost certainly involved; (2) identity mismatch — `HOSTNAME.BIND` CHAOS queries return identifiers that do not match the expected pattern for the B root (e.g., returning the ISP’s own server name rather than a B-root-consistent string).

Jones et al. [Jones et al, 2016] also find that 1.7% of Atlas probes have inconsistent geolocation data between Atlas’s own metadata and the MaxMind geolocation database, complicating latency-based detection for those probes and motivating careful probe metadata validation in any measurement campaign.

For this thesis, the DNS root manipulation findings have two implications: (1) measurement campaigns should include data quality checks (e.g., RTT sanity checks) to identify probes whose DNS responses likely originate from in-path proxies rather than the intended authoritative servers; (2) the existence of ISP-level DNS proxies means that some probes’ responses reflect locally cached or filtered content rather than the global DNS, and treating such responses as geographically representative would be incorrect.

2.9.3 Censorship and Geographic Filtering

DNS manipulation for censorship purposes is detectable at scale through RIPE Atlas measurements. Nosyk et al. [Nosyk et al, 2025] find that **69% of Chinese RIPE Atlas probes** fail to resolve domain names operated by Meta (Facebook, Instagram, WhatsApp), a result consistent with China’s documented DNS-based blocking of these services. This finding has two implications: first, DNS non-resolution from certain probes is not measurement failure but a meaningful observation of censorship or geographic filtering; second, such probes must be handled explicitly in the analysis rather than discarded as unreliable — their responses (or lack thereof) are genuine geographic observations.

Any measurement methodology operating at global scale will encounter jurisdictional DNS filtering of this type. The measurement design must accommodate these probes and distinguish intentional DNS non-resolution (censorship) from measurement failures (timeouts due to network connectivity issues, probe hardware failures, or measurement platform problems).

Three indicators help distinguish censorship from measurement failure in the analysis of distributed DNS measurement data. First, **temporal persistence**: censorship-induced non-resolution is persistent across repeated measurement campaigns, while measurement failures are typically transient and affect a random subset of probes. Second, **geographic clustering**: censorship affects probes in specific network jurisdictions (Autonomous Systems under state control, or all probes in specific countries), creating geographically clustered failure patterns rather than the random geographic distribution expected from hardware or connectivity failures. Third, **cross-domain comparison**: if a probe correctly resolves unblocked domains (e.g., it correctly returns the IP for a neutral domain like `example.com`) but fails to resolve specific domains (e.g., `facebook.com`), the failure is likely jurisdictional rather than a general connectivity or hardware problem. Jones et al. [Jones et al, 2016] use a combination of these indicators — specifically cross-referencing DNS response identity with ping RTT to the expected authoritative server — to distinguish legitimate anycast routing from in-path proxy interception. For the thesis’s measurement campaign, RIPE Atlas system tags (`system-resolves-a-correctly`) provide a platform-managed baseline for probe connectivity health that can be used as the first filter when investigating DNS non-responses: probes that fail the system resolution check are likely experiencing

general connectivity problems, while probes that pass the system check but fail to resolve specific domains in the study corpus are exhibiting behaviour consistent with geographic filtering.

2.10 Synthesis and Research Positioning

The preceding nine sections review the state of the art across the four technical domains that jointly constitute the foundation of this thesis: DNS measurement methodology (Sections 2.2–2.4), anycast routing and its geographic implications (Section 2.5), CDN DNS-based routing and its associated resolver effects (Section 2.6), domain ranking methodology (Section 2.7), DNS infrastructure centralization (Section 2.8), and DNS security dimensions that affect measurement validity (Section 2.9). This synthesis identifies the structural gaps in the existing literature and positions this thesis’s contribution relative to those gaps.

The two infrastructure pillars of the existing measurement landscape — OpenINTEL and RIPE Atlas — are complementary in their design but leave a well-defined gap in the intersection of their respective strengths. OpenINTEL provides longitudinal, TLD-wide, daily measurement coverage of the DNS namespace from a single vantage point in the Netherlands, enabling temporal analysis of global DNS behaviour but providing no spatial differentiation. RIPE Atlas provides globally distributed measurement from nearly 13,000 probes across 178 countries, enabling fine-grained geographic comparison, but existing RIPE Atlas DNS studies are predominantly single-campaign, domain-limited case studies (a few hundred domains, a single time point) or platform characterisation studies (Nosyk et al., 2024; Bajpai et al., 2017). The gap — systematic, longitudinal, geographically distributed measurement of a large popular-domain corpus — is the precise combination that neither infrastructure has yet provided, and that this thesis sets out to implement.

2.10.1 Identified Gaps in the Literature

The preceding review reveals a coherent landscape of DNS measurement capabilities and gaps. Table 2.4 summarises the key dimensions:

Dimension	OpenINTEL	RIPE Atlas (existing studies)	Gap addressed by this thesis
Domain coverage	Full TLD zone files (all domains)	Experimenter-defined (per study)	Tranco Top N (popular, stable, reproducible)
Geographic vantage	Single point (Netherlands)	≈12,900 probes, 178 countries	Systematic multi-continent stratification
Temporal depth	Daily since March 2015	Single campaigns	Longitudinal repeated campaigns (90 days)
DNS response types	A, AAAA, NS, MX, SOA, DNSKEY	Any type	A, AAAA, NS (geolocation-relevant) + NSID
Anycast tracking	Not applicable	Case studies (d.nic.fr, UncensoredDNS)	Systematic across Tranco corpus
Resolver control	Fixed central resolver	Local probe resolver (default)	Direct authoritative query (resolver bypass)

Table 2.4: Comparison of OpenINTEL, existing RIPE Atlas DNS studies, and the contribution of this thesis across six measurement dimensions. Each row identifies a gap that the present work explicitly addresses.

Three specific gaps motivate the contribution of this thesis:

Gap 1 — Absence of systematic geographic DNS variation measurement at scale: The literature contains individual anycast case studies (d.nic.fr with ~500 probes by Bortzmeyer [Bortzmeyer, 2013]; UncensoredDNS with 500 probes by Finnegan [Finnegan, 2018]; Bing CDN by Calder et al. [Calder et al, 2015]) and individual CDN routing studies (Hours et al., 2016; Li et al., 2025). However, no study has systematically characterised **what fraction of domains in a popular-domain corpus return geographically differentiated DNS responses**, which infrastructure mechanisms (anycast, DNS-based CDN routing, ECS, centralized providers) drive that differentiation, and how differentiation varies by domain category and geographic region.

Gap 2 — Resolver centralization undermines vantage-point diversity: Xu et al. [Xu et al, 2023] show that over 90% of forwarding resolvers are backed by fewer than 5% of indirect recursive resolvers. This means that using local probe resolvers in RIPE Atlas measurements may produce convergent DNS responses across geographically dispersed probes — not because the DNS is geographically uniform, but because all resolvers converge on the same iRDNS infrastructure. Bypassing resolver centralization by querying authoritative servers directly is a methodological requirement that none of the case studies cited above explicitly document as a design choice.

Gap 3 — Temporal snapshots miss BGP dynamics: Single-campaign studies provide point-in-time anycast catchment measurements. Koch et al. [Koch et al, 2021] document that CDN operators actively engineer their anycast routing through peering agreements, and Bortzmeyer [Bortzmeyer, 2013] acknowledges that BGP routing changes over time. Anycast catchment boundaries shift as operators add or decommission PoPs, change peering, or reconfigure prefix announcements. A longitudinal measurement campaign is required to characterise the temporal stability or volatility of geographic DNS response variation.

2.10.2 Contribution of This Thesis

This thesis addresses the identified gaps by designing, implementing, and deploying a distributed active DNS measurement system with the following properties:

Domain corpus: Domain targets are drawn from the **Tranco Top list** [Pochat et al, 2019] — stable (0.6% daily change), manipulation-resistant (quadrupled manipulation effort), representative of actual Internet usage, and reproducible via archived list versions. This choice prioritises research validity over exhaustive TLD coverage.

Distributed vantage points: Measurements are performed from **geographically distributed RIPE Atlas probes** selected using system tags (following Bajpai et al., 2017) to ensure working IPv4/IPv6 connectivity and resolver correctness (`system-resolves-a-correctly`). Probe selection is stratified to provide coverage across all inhabited continents, accounting for the platform’s documented geographic bias toward Europe and North America [Nosyk et al, 2025].

Authoritative server querying with NSID: Measurements query authoritative name servers directly — bypassing the resolver layer and the centralization effects documented by Xu et al. [Xu et al, 2023] — and include the **NSID option** (RFC 5001) in all DNS queries, following the methodology of Bortzmeyer [Bortzmeyer, 2013] and Finnegan [Finnegan, 2018]. This enables per-probe, per-domain identification of which anycast instance responded, providing the anycast routing dimension of geographic DNS variation alongside the response content dimension.

Longitudinal design: Measurement campaigns are repeated on a regular schedule, enabling the temporal analysis of DNS response variation — whether geographic differentiation is stable or varies over time as CDN routing evolves, BGP configurations change, or new PoPs are deployed.

Data archiving: Results are structured and archived following FAIR data principles — Findable, Accessible, Interoperable, Reusable — building on the data model of OpenINTEL [van Rijswijk-Deij et al, 2016] while providing the spatial diversity that OpenINTEL explicitly lacks and that constitutes the primary motivation for this thesis.

The ethical framework for this measurement system follows Kisteleki et al. [Kisteleki, 2016] and the Menlo Report (2012): all query targets are publicly accessible domains from a research-grade list; no politically sensitive, censored, or jurisdictionally risky domains are included; measurement frequency is calibrated to avoid overloading authoritative DNS infrastructure [van Rijswijk-Deij et al, 2016]; and the RIPE Atlas credit system and terms of service are respected throughout.

Chapter 3

Methodology

Note on sourcing. This chapter describes the technical methodology of the measurement campaign as actually implemented and deployed. Sources from the bibliography are cited where the methodology is directly informed by or adapted from prior work. General software engineering choices (API usage, data format parameters, statistical tests) are not attributed to sources.

3.1 Overview

3.1.1 Methodological Objective

The overarching methodological objective of this thesis is to capture the **geographic and temporal diversity of DNS responses** for a representative corpus of popular domains, using geographically distributed active measurements, in a manner that is reproducible, ethically sound, and practically feasible within the available resource budget. The system must answer four empirical research questions:

- **Q1:** What proportion of Tranco top-ranked domains return geographically differentiated DNS responses, and what infrastructure mechanisms explain the differentiation?
- **Q2:** What is the temporal stability of DNS responses for popular domains at day, week, and month timescales?
- **Q3:** Do the geographic biases of the RIPE Atlas probe distribution significantly limit the ability to observe geographic DNS variation?
- **Q4:** What is the measurable impact of resolver type — local ISP resolver versus public DNS service — on the DNS responses observed?

3.1.2 System Architecture

The measurement system is organised in five sequential stages, inspired by the architecture of OpenINTEL [van Rijswijk-Deij et al, 2016] but adapted to the constraints and capabilities of the RIPE Atlas platform:

1. **Domain corpus preparation:** Retrieval and filtering of the Tranco Top 10,000 domain list.
2. **Measurement scheduling:** Configuration and deployment of RIPE Atlas DNS measurement campaigns.
3. **Data collection:** Automated daily retrieval of measurement results via the RIPE Atlas REST API.
4. **Storage and preprocessing:** Parsing, validation, and archival in Parquet format.
5. **Analysis:** Quantitative analysis addressing Q1–Q4.

The key design trade-off throughout is the three-way tension between **domain coverage** (how many domains are measured), **geographic coverage** (how many geographically distributed probes), and **temporal resolution** (measurement frequency), all bounded by the finite RIPE Atlas credit budget. The system is fully configurable via `-max-domains` and `-total-probes` parameters; the default operating point is **500 domains** $P^{69}MATH_{50probes}$, generating approximately 250,000 credits per day. Figure 3.1 presents the five-stage pipeline architecture.

3.1.3 Infrastructure

The entire pipeline runs as a Docker container on a Raspberry Pi 5 (ARM64 architecture), operating continuously and autonomously. This choice reflects the need for a low-cost, always-on measurement platform that can sustain a three-month measurement campaign without manual intervention.

The container image is based on `python:3.11-slim-bookworm` and is compiled natively for the ARM64 architecture, ensuring full compatibility with the Raspberry Pi 5 hardware. The image contains only the dependencies required for the pipeline (`dnspython`, `pandas`, `pyarrow`, `scipy`, `matplotlib`, `rclone`), keeping its footprint under 1 GB.

Data persistence is managed through Docker named volumes, which survive container restarts and image updates. Collected results are automatically synchronised daily to cloud storage via `rclone`, which is configured to target two destinations simultaneously: Microsoft OneDrive (UNamur institutional account) and Google Drive (personal account, 2 TB available). This dual-destination strategy ensures redundancy: if one service is temporarily unavailable, the other retains a complete copy of all results.

The pipeline is orchestrated by a cron daemon running inside the container, with the following schedule (all times UTC):

Time (UTC)	Day	Action
05:00	Monday	Weekly Tranco list refresh
06:00	Daily	Fetch RIPE Atlas results + Parquet parsing
07:00	Daily	Cloud sync (Parquet + raw JSON → OneDrive + Google Drive)
08:00	Sunday	Q1–Q4 analysis + figure generation
09:00	Sunday	Cloud sync (reports + figures → OneDrive + Google Drive)
04:00	1st of month	Log rotation

The pipeline source code is available at <https://github.com/ogautier-unam/dns-pipeline> and implements the six Python scripts described in the following sections.

3.2 Domain Corpus Selection

3.2.1 Tranco List Configuration

The domain corpus is the Tranco Top 10,000 list, retrieved via the Tranco API at ‘<https://tranco-list.eu>’. The choice of Tranco over commercial ranking lists (Alexa, Cisco Umbrella, Majestic) is motivated in Chapter 2 (Section 2.7) and rests on three properties established by Le Pochat et al. [Pochat et al, 2019]: stability (0.6% daily change versus approximately 50% for post-2018 Alexa), manipulation resistance (quadrupled manipulation effort relative to single-source lists), and archivability (each generated list is assigned a unique identifier retrievable indefinitely via permalink).

The `fetch_tranco.py` script retrieves the most recent available list, falling back up to seven days if the current day’s list has not yet been published. The exact Tranco list identifier and generation date are recorded in `tranco_meta.json` alongside the corpus, ensuring full reproducibility.

The list is updated weekly (each Monday at 05:00 UTC) to capture the slow evolution of the popular-domain corpus while limiting the rate of change introduced by list updates. Each weekly update computes the delta of added and removed domains relative to the previous week’s list. Domains that exit the Top 10K during the measurement campaign are retained in active measurement for two additional weeks to capture any transitional DNS behaviour.

3.2.2 Domain Corpus Size

The measurement corpus targets the top 500 domains from the Tranco Top 10,000 list. Although Tranco provides a 10K list, the RIPE Atlas credit system imposes a hard practical constraint: at 10 credits per probe per measurement instance, running 10,000 domains daily with 50 probes would consume 5,000,000 credits per day, draining a typical research allocation within days. The selection of 500 domains represents a deliberate balance between three competing requirements:

Representativeness: The Tranco Top 500 concentrates the domains responsible for the large

majority of global web traffic — precisely the CDN-backed services (streaming platforms, search engines, social networks, cloud providers) that exhibit the richest geographic DNS variation. Studies by Calder et al. [Calder et al, 2015] and Wang et al. [Wang et al, 2018] confirm that CDN routing diversity is most pronounced in this top-ranked tier.

Feasibility: At 50 probes per measurement, one A record query per domain per day, the Top 500 corpus generates approximately 25,000 measurement results per day and consumes 250,000 RIPE Atlas credits. This is sustainable over a 90-day measurement period.

Configurability: The pipeline exposes `-max-domains` and `-total-probes` parameters. If the available credit balance allows, the corpus can be extended to 1,000 or 2,000 domains without any other modification.

3.2.3 Domain Filtering and Validation

Before the measurement campaign begins, each domain in the Tranco Top 10K list is pre-validated by `fetch_tranco.py` in two steps:

Technical validity: A DNS A record query is issued against Google Public DNS (8.8.8.8) and Cloudflare DNS (1.1.1.1) as reference resolvers. Domains returning `NXDOMAIN` or `SERVFAIL` are excluded. Based on the filtering rates reported by Le Pochat et al. [Pochat et al, 2019] for the Tranco Top 10K, approximately 500–600 domains are expected to be excluded (5–6% of 10K), yielding a final measurement corpus of approximately 9,400–9,500 active domains.

Authoritative server pre-resolution: The authoritative name servers for each domain are identified by resolving the domain’s NS records from the same reference resolvers. This pre-resolution step maps each domain to its authoritative server set and is repeated weekly in synchrony with the Tranco list update. The resulting corpus is saved as `tranco_corpus.csv` with fields `rank`, `domain`, and `ns` (pipe-separated list of authoritative nameserver hostnames).

3.3 RIPE Atlas Measurement Configuration

3.3.1 Probe Selection

Probe selection is implemented in `create_ripe_measurements.py` and follows the stratified-sampling protocol recommended by Bajpai et al. [Bajpai et al, 2017], adapted to the geographic diversity requirements of this thesis. The selection proceeds in three steps.

Step 1 — System tag filtering: All probes must satisfy the following RIPE Atlas system tags:

- `system-ipv4-works`: IPv4 connectivity verified by built-in platform measurements.
- `system-resolves-a-correctly`: The probe’s local resolver returns correct A records.

Hardware version v3 or later is required. Holterbach et al. [Holterbach et al, 2015] show that v1 and v2 probes experience timing precision degradation under concurrent measurement load,

while v3 probes experience negligible degradation.

Step 2 — Geographic stratification: To compensate for RIPE Atlas’s documented geographic concentration (91% of dual-stacked probes in the RIPE and ARIN regions, per Bajpai et al., 2017), probe allocation is stratified by continent with inverse-weighted over-representation of underrepresented regions:

Region	Atlas distribution	Allocated probes	This study
Europe	~40%	15	30%
North America	~28%	12	25%
Asia-Pacific	~16%	10	20%
South America	~8%	6	10%
Africa	~5%	5	10%
Oceania	~3%	2	5%
Total	100%	50	100%

The proportional allocation preserves the relative weighting from prior work [Bajpai et al, 2017] while scaling the total to 50 probes for budget compatibility. The probe count is configurable via `-total-probes`; the proportions are applied dynamically.

Step 3 — AS diversity: Within each geographic stratum, probes are selected using a round-robin algorithm over distinct Autonomous Systems, preferring one probe per AS over multiple probes within the same AS. The selected probe set is saved as `selected_probes_<date>.json` for full reproducibility.

3.3.2 Primary Campaign: Direct Authoritative DNS Queries

The primary measurement campaign queries authoritative name servers directly, bypassing recursive resolvers. This design choice is motivated by two findings from the literature:

1. **Resolver centralization** [Xu et al, 2023]: Over 90% of forwarding resolvers are backed by fewer than 5% of indirect recursive resolvers. Querying through probe resolvers would collapse the geographic diversity of probe vantage points into a small number of centralised resolution paths, making geographic DNS variation invisible.
1. **Cache state divergence:** Probes using local resolvers have heterogeneous cache states, introducing variation that mimics geographic routing differences but is actually an artefact of cache timing. Direct authoritative queries bypass resolver caches entirely.

For each domain in the corpus, the primary authoritative name server identified during pre-resolution is used as the measurement target. The RIPE Atlas DNS measurement is configured as follows:

```
1 {  
2   "type": "dns",  
3   "af": 4,
```

```

4  "target": "<authoritative_ns_ip>",
5  "query_type": "A",
6  "query_class": "IN",
7  "query_argument": "<domain>",
8  "use_probe_resolver": false,
9  "set_rd_bit": false,
10 "set_nsid_bit": true,
11 "set_do_bit": false,
12 "set_cd_bit": false,
13 "protocol": "UDP",
14 "udp_payload_size": 1024,
15 "include_abuf": true,
16 "retry": 2,
17 "timeout": 5000,
18 "is_oneoff": false,
19 "interval": 86400,
20 "description": "DNS geo-diversity - auth - <domain> - <YYYY-MM-DD>"
21 }

```

Key parameter justifications:

‘set_nsid_bit: true’: The NSID option (RFC 5001) is included in every query, following the methodology of Bortzmeyer [Bortzmeyer, 2013] and Finnegan [Finnegan, 2018]. The NSID string returned by the authoritative server identifies which physical anycast instance responded, providing the anycast routing dimension of geographic DNS variation.

‘include_abuf: true’: The raw DNS response packet (base64-encoded) is included in the result, enabling full parsing of all DNS sections, extraction of TTL values, and NSID option decoding. The `parse_dns_results.py` script decodes the ABUF binary structure directly to extract OPT record NSID data (RFC 5001, option code 3).

‘interval: 86400’ (86,400 seconds = 24 hours): Daily measurements provide the temporal resolution needed to detect day-to-day DNS changes while remaining within the credit budget.

3.3.3 Supplementary Campaign for Q4: Resolver Comparison

To address Q4, a supplementary one-off measurement campaign is conducted on a subset of 100 domains drawn from the top of the primary corpus. This campaign provides four simultaneous resolver-type observations from the same 50 probes:

Campaign	Target	use_probe_resolver	Resolver
isp_local	—	true	Probe’s local ISP resolver
google	8.8.8.8	false	Google Public DNS
cloudflare	1.1.1.1	false	Cloudflare DNS
quad9	9.9.9.9	false	Quad9

The Q4 campaign uses `set_rd_bit: true` (recursion desired) and a larger UDP payload size

(4096 bytes) appropriate for recursive resolver queries. Measurements are configured as one-off (`is_oneoff: true`) to avoid continuous credit consumption.

This four-way comparison allows direct testing of whether the choice of resolver affects the CDN-assigned IP addresses observed from each geographic vantage point, and whether the local ISP resolver produces results closer to the authoritative response than centralised public DNS services.

3.3.4 Credit Budget

The RIPE Atlas credit consumption is estimated as follows (10 credits per probe per measurement instance):

Campaign	Domains	Probes	Frequency	Credits
Primary A record (direct auth)	500	50	daily	250,000 / day
Q4 resolver comparison (one-off)	100	50	once	200,000 total
90-day primary campaign total	—	—	—	~22,500,000

For context, RIPE Atlas free research allocations are typically in the order of 1,000,000–10,000,000 credits; a RIPE NCC research credit grant (available to academic researchers) can cover the full 90-day budget. The pipeline parameters `-max-domains` and `-total-probes` allow scaling to match the available balance: for example, `-max-domains 100 -total-probes 20` reduces daily consumption to 20,000 credits while retaining full pipeline functionality.

The credit budget is monitored by the log output of `create_ripe_measurements.py`, which prints the estimated consumption at each invocation before any measurements are created.

3.4 Data Collection and Processing

3.4.1 Daily Collection Pipeline

Measurement results are collected from the RIPE Atlas REST API by `fetch_ripe_atlas.py` on a daily schedule at 06:00 UTC, retrieving results for the previous calendar day. The script reads the measurement plan from `measurements.json` (produced by `create_ripe_measurements.py`) and iterates over all active measurement identifiers.

The collection is fully paginated, following RIPE Atlas API `next` links until all results for the time window are retrieved. Error handling includes automatic retry with exponential backoff (up to 5 attempts, base delay 2 seconds, doubling on each retry) and explicit handling of HTTP 429 rate-limit responses (waits for the `Retry-After` header duration before retrying). Failed measurement IDs are recorded in `fetch_errors_<date>.json` for post-hoc investigation. To avoid redundant downloads, already-fetched files are skipped unless the `-force` flag is specified.

Raw results are saved as JSON files in `data/raw/msm_<id>_<date>.json`.

3.4.2 Parsing and Field Extraction

Each raw RIPE Atlas DNS result is parsed by `parse_dns_results.py`, which extracts the following fields into a structured Parquet record:

Field	Type	Description
<code>msm_id</code>	<code>int64</code>	RIPE Atlas measurement ID
<code>prb_id</code>	<code>int32</code>	RIPE Atlas probe ID
<code>timestamp</code>	<code>int64</code>	Actual measurement time (Unix epoch)
<code>date</code>	<code>string</code>	ISO date (YYYY-MM-DD)
<code>country_code</code>	<code>string</code>	ISO 3166-1 alpha-2 country code
<code>asn_v4</code>	<code>int32</code>	Probe Autonomous System Number
<code>continent</code>	<code>string</code>	Continent code (EU/NA/SA/AF/AP/OC)
<code>query_domain</code>	<code>string</code>	Queried domain name
<code>rcode</code>	<code>int16</code>	DNS response code (0 = NOERROR)
<code>answer_count</code>	<code>int16</code>	Number of A/AAAA records returned
<code>answer_ips</code>	<code>string</code>	Pipe-separated list of returned IP addresses
<code>ttl_min</code>	<code>int32</code>	Minimum TTL across answer records
<code>rt_ms</code>	<code>float32</code>	Response time in milliseconds
<code>nsid_str</code>	<code>string</code>	NSID value (ASCII, if returned)
<code>nsid_hex</code>	<code>string</code>	NSID value (hex encoding)
<code>abuf_size</code>	<code>int32</code>	DNS response packet size in bytes
<code>resolver_ip</code>	<code>string</code>	Target resolver IP (Q4 campaign only)
<code>use_probe_resolver</code>	<code>bool</code>	True for ISP local resolver campaign

The NSID value is extracted by decoding the raw ABUF binary field: the script walks the DNS packet structure to locate the OPT record (type 41) in the Additional section, then parses EDNS0 options to extract option code 3 (NSID, RFC 5001).

3.4.3 Data Validation

The following validation filters are applied during parsing:

RCODE filtering: Only results with `rcode = 0` (NOERROR) and `answer_count > 0` are used for Q1 and Q2 content analysis. Results with `rcode != 0` are retained in the Parquet dataset with their specific error codes for reliability and censorship analysis.

Deduplication: The cumulative Parquet file (`dns_results.parquet`) is deduplicated on (`msm_id`, `prb_id`, `timestamp`) on each daily append to prevent double-counting in case of re-fetch.

3.4.4 Storage Architecture

Results are stored in a single-tier Parquet architecture optimised for the analytical query patterns of the analysis phase:

Cumulative Parquet file (`data/processed/dns_results.parquet`): All parsed results are appended daily to a single Parquet file using `pyarrow` with a fixed schema. This file grows at approximately 3–5 MB/month for the primary campaign (500 domains × 50 probes × 30

days, after columnar compression with dictionary encoding for repeated string values such as `country_code`, `continent`, and `answer_ips`).

Daily archives (`data/processed/dns_results_<date>.parquet`): Individual daily Parquet files are also retained for point-in-time analysis and as a recovery mechanism in case of cumulative file corruption.

Raw JSON files (`data/raw/msm_<id>_<date>.json`): Raw RIPE Atlas JSON results are retained locally for 7 days, then deleted after cloud synchronisation. This rolling window allows re-parsing with updated schema if needed, while controlling local storage consumption on the Raspberry Pi.

Cloud backup: All Parquet files, daily archives, and reports are synchronised daily to two cloud destinations via `rclone`: Microsoft OneDrive (UNamur institutional account) and Google Drive (personal account). The `sync_cloud.sh` script detects which remotes are configured and synchronises to all available destinations in parallel. The `rclone` configuration is persisted in a Docker named volume, surviving container updates.

Storage estimate for a three-month campaign ($500 \text{ domains} \times 50 \text{ probes}$):

Tier	Volume	Location
Raw JSON (7-day rolling)	~350 MB	Pi local
Cumulative Parquet	~15 MB	Pi + OneDrive + Google Drive
Daily Parquet archives	~45 MB	Pi + OneDrive + Google Drive
Reports and figures	~50 MB	Pi + OneDrive + Google Drive

3.5 Analysis Methods

3.5.1 Q1 — Geographic Diversity of DNS Responses

Objective: For each domain in the corpus, determine whether the set of IP addresses returned varies systematically by the geographic location of the querying probe.

Step 1 — Compute per-domain IP diversity: For each domain and each measurement day, the probe set is partitioned by continent. For each continent, the set of unique IP addresses returned is computed from all probes in that continent. A domain is classified as geographically diverse if at least two continents receive distinct IP address sets on the same day.

Step 2 — Aggregate over time: A domain is classified as exhibiting geographic diversity if it returns geographically differentiated responses on at least one measurement day. The `pct_geo_diverse` metric gives the percentage of days on which geographic differentiation is observed, characterising the consistency of differentiation over time.

Step 3 — NSID analysis: For domains with geographic IP variation, the presence of NSID values is used to distinguish between anycast instance selection (different NSID per region, indicating different anycast PoP) and DNS-based CDN routing (uniform NSID with varying A

records, indicating CDN geographic database lookup).

Key metrics:

- Total proportion of domains exhibiting geographic diversity
- Distribution of the number of distinct IP address sets across continents
- Top domains by geographic diversity score
- Proportion of geographically diverse domains returning NSID values

3.5.2 Q2 — Temporal Stability

Objective: Quantify the rate of change of DNS responses for each domain over time.

Step 1 — Per-probe change detection: For each (domain, probe) pair, consecutive daily observations are compared. A change event is recorded when the `answer_ips` value differs between two consecutive measurement days.

Step 2 — Change rate computation: The per-domain change rate is the proportion of consecutive-day pairs for which at least one probe observes a change. This is computed at three temporal windows: daily (consecutive days), weekly (observations 7 days apart), and monthly (observations 30 days apart).

Step 3 — Distribution analysis: The distribution of per-domain change rates is computed across the full corpus, identifying the proportions of fully stable, occasionally changing, and highly volatile domains.

Key metrics:

- Distribution of per-domain daily change rates
- Proportion of fully stable domains (0% change rate)
- Median change rate across the corpus
- Stability breakdown by temporal window (day/week/month)

3.5.3 Q3 — Assessment of RIPE Atlas Geographic Bias

Objective: Evaluate whether the geographic concentration of RIPE Atlas probes in Europe and North America significantly limits the ability to observe geographic DNS variation.

Method: The probe distribution actually achieved at deployment is compared to the target allocation from Section 3.3.1. For each continent, the coverage rate (proportion of domains observed by at least one probe) and the mean number of unique IP addresses observed per domain are computed.

A Mann-Whitney U test (non-parametric, for independent samples) compares the distribution of unique IPs observed per domain between Europe+North America probes and probes from all

other regions. A statistically significant difference ($p < 0.05$) indicates that the geographic stratification captures meaningfully different DNS responses from underrepresented regions.

Key metrics:

- Actual probe distribution by continent at deployment
- Per-continent domain coverage rate
- Mean unique IPs per domain per continent
- Mann-Whitney U statistic and p-value (EU+NA vs. other regions)

3.5.4 Q4 — Impact of Resolver Choice

Objective: Quantify the impact of resolver type on DNS responses, comparing local ISP resolvers with three major public DNS services.

This analysis uses the supplementary Q4 measurement campaign (Section 3.3.3), which provides four simultaneous observations for 100 domains from the same 50 probes.

Step 1 — Per-resolver statistics: For each resolver type (`isp_local`, `google`, `cloudflare`, `quad9`), the following metrics are computed: error rate (proportion of `RCODE != 0` responses), median RTT, and mean answer count.

Step 2 — Divergence analysis: For each domain and each probe, the `answer_ips` returned by the local ISP resolver is compared against each public DNS service. A divergence event is recorded when the IP sets differ. The per-comparison divergence rate is the proportion of (domain, probe) pairs showing divergence.

Step 3 — Statistical validation: For each pairwise comparison (ISP local vs. each public resolver), the Mann-Whitney U test is applied to the distribution of RTT values across all probes.

Key metrics:

- Error rate and median RTT per resolver type
- Pairwise IP divergence rate: ISP local vs. Google, Cloudflare, Quad9
- Geographic disaggregation of divergence rates by continent

3.6 Ethical Considerations and Reproducibility

3.6.1 Ethical Design

The measurement design follows the ethical framework of Kisteleki et al. [Kisteleki, 2016] and the principles of the Menlo Report (2012):

Probe host protection: All query targets are publicly accessible domains from the Tranco Top 10K corpus. Direct authoritative queries for A records are indistinguishable from normal client DNS traffic.

Infrastructure impact: The measurement frequency (once per domain per day, 50 probes, 500 domains) generates approximately 25,000 queries per day — negligible compared to OpenINTEL’s production campaign of several hundred million queries per day [van Rijswijk-Deij et al, 2016].

Transparency: All measurements are tagged with a project description in RIPE Atlas, allowing authoritative server operators to identify the measurement source. The measurement plan (`measurements.json`) records all active measurement IDs.

Data minimisation: Raw results are retained locally for 7 days only, then deleted after cloud synchronisation. The published dataset uses aggregated statistics.

3.6.2 Scientific Reproducibility

Fixed domain corpus: The Tranco list identifier recorded in `tranco_meta.json` for each measurement week enables exact reproduction of the domain corpus.

Measurement traceability: All RIPE Atlas measurement IDs are recorded in `measurements.json`. Any researcher with a RIPE Atlas account can retrieve the original raw results using these IDs.

Probe set reproducibility: The selected probe IDs and their metadata (country, ASN, hardware version) are saved in `selected_probes_<date>.json`.

Software versioning: The analysis pipeline is published at <https://github.com/ogautier-unam/dns-pipeline>. The `requirements.txt` file specifies exact package versions (Python 3.11, dnspython, pandas, pyarrow, scipy, matplotlib) to eliminate software version ambiguity.

Infrastructure reproducibility: The Docker container image built from the published `Dockerfile` provides a fully reproducible execution environment across x86_64 and ARM64 architectures, ensuring that the pipeline can be redeployed from source on any compatible host.

Chapter 4

Results

Note. This chapter presents the empirical results of the measurement campaign described in Chapter 3. All quantitative values appear as [VALUE] placeholders to be filled once data collection and analysis are complete. Figure references indicate planned visualisations. The interpretive framing of each section anticipates the expected results based on the literature reviewed in Chapter 2; where actual results diverge from expectations, the placeholders are accompanied by a [NOTE: unexpected -- discuss in Ch. 5] flag.

4.1 Dataset Overview

4.1.1 Collection Statistics

The measurement campaign ran from [DATE_START] to [DATE_END], covering [VALUE] calendar days of active data collection. Table 4.1 summarises the overall collection statistics.

Metric	Value
Measurement period	[DATE_START] – [DATE_END] ([VALUE] days)
Tranco list version	[TRANCO_ID] (retrieved [DATE])
Domains in Tranco Top 10K	10,000
Domains after pre-filtering	[VALUE] (~[VALUE]% of 10K)
Domains with $\geq 85\%$ valid results	[VALUE]
Total measurement instances scheduled	[VALUE]
Total results retrieved from RIPE Atlas API	[VALUE]
Results passing all validation filters	[VALUE] ([VALUE]%)
Days with complete collection	[VALUE] / [VALUE]
Days with partial collection	[VALUE]

Table 4.1: Campaign summary statistics

The [VALUE] domains excluded at the pre-filtering stage (Section 3.2.3) consisted primarily of domains returning NXDOMAIN at the reference resolver ([VALUE] domains, [VALUE]%) and

domains flagged by the Google Safe Browsing API ([VALUE] domains, [VALUE]%). The remaining [VALUE] domains formed the active measurement corpus for the duration of the campaign.

4.1.2 Probe Distribution

The measurement campaign used [VALUE] unique RIPE Atlas probes distributed across [VALUE] countries and [VALUE] Autonomous Systems. Table 4.2 summarises the geographic distribution of the probe set and compares it to the target allocation defined in Section 3.3.1.

Region	Target	Actual	Countries	ASes
Europe	30	[VALUE]	[VALUE]	[VALUE]
North America	25	[VALUE]	[VALUE]	[VALUE]
Asia-Pacific	20	[VALUE]	[VALUE]	[VALUE]
South America	10	[VALUE]	[VALUE]	[VALUE]
Africa	10	[VALUE]	[VALUE]	[VALUE]
Oceania	5	[VALUE]	[VALUE]	[VALUE]
Total	100	[VALUE]	[VALUE]	[VALUE]

Table 4.2: *Geographic probe distribution: target vs. actual*

Deviations from the target allocation reflect the availability of probes satisfying the system-tag requirements (Section 3.3.1) in specific regions. The largest deviation is observed in [REGION], where only [VALUE] probes satisfying all selection criteria were available, compared to the target of [VALUE].

4.1.3 Data Quality

Table 4.3 summarises the application of the four validation filters described in Section 3.4.3. The overall valid result rate of [VALUE]% is [above / below / consistent with] the 85–90% expected rate estimated in Chapter 3.

Filter	Excluded	%	Primary causes
Timeout (no abuf)	[VALUE]	[VALUE]%	Probe offline, network timeout
Scheduling deviation >4h	[VALUE]	[VALUE]%	Platform concurrent load [Holterbach et al, 2015]
RCODE $\neq$ NOERROR	[VALUE]	[VALUE]%	NXDOMAIN / SERVFAIL / REFUSED
Anomalous response flag	[VALUE]	[VALUE]%	Suspected in-path proxy or local cache
Total excluded	[VALUE]	[VALUE]%	
Valid results	[VALUE]	[VALUE]%	Used in all subsequent analyses

Table 4.3: *Validation filter outcomes on raw measurement results*

The [VALUE]% of results returning RCODE SERVFAIL were separately examined for DNSSEC correlation. [VALUE]% of SERVFAIL results were from domains with incomplete DNSSEC chains (detectable via the absence of a valid RRSIG record in the authority section), consistent with

the systemic DNSSEC key management failures documented by van Rijswijk-Deij et al. [van Rijswijk-Deij et al, 2016]. The remaining [VALUE]% of SERVFAIL responses may reflect transient authoritative server issues.

The [VALUE] results flagged as anomalous correspond to [VALUE] distinct (probe, domain) pairs. Of these, [VALUE] pairs show a persistent pattern across multiple measurement days — consistent with systematic in-path proxy behaviour rather than transient network anomalies. These pairs are excluded from all subsequent analyses.

4.1.4 Storage

Table 4.4 reports the storage volumes achieved by each format.

Format	Size	Compression ratio vs. raw JSON
Raw JSON (RIPE Atlas API output)	[VALUE] GB	1:1 (baseline)
Apache Avro (archival tier)	[VALUE] GB	1:[VALUE]
Apache Parquet (analytics tier)	[VALUE] GB	1:[VALUE]
Total	[VALUE] GB	

Table 4.4: *Storage volumes by format*

The Avro compression ratio of 1:[VALUE] is [consistent with / lower than / higher than] the 1:7.4 ratio reported by van Rijswijk-Deij et al. [van Rijswijk-Deij et al, 2016] for OpenINTEL, [likely due to the smaller average record size / the higher string cardinality in the probe metadata fields].

4.2 Results for Q1: Geographic Diversity of DNS Responses

4.2.1 Overall Distribution of Geographic Diversity

Research question: What proportion of Tranco Top 10K domains return geographically differentiated DNS responses, and what infrastructure mechanisms explain the differentiation?

For each domain in the active corpus, the geographic diversity metric — the median pairwise Jaccard similarity across continental probe pairs (Section 3.5.1) — was computed over the full measurement period. Table 4.5 reports the distribution of domains across the four diversity categories.

Category	Criterion	Domains	%
UNIFORM	Single IP across all probes	[VALUE]	[VALUE]%
LOW_DIVERSITY	Median pairwise Jaccard > 0.8	[VALUE]	[VALUE]%
MEDIUM_DIVERSITY	Median pairwise Jaccard 0.4–0.8	[VALUE]	[VALUE]%
HIGH_DIVERSITY	Median pairwise Jaccard < 0.4	[VALUE]	[VALUE]%
Total		[VALUE]	100%

Table 4.5: *Distribution of domains by geographic diversity category*

Figure 4.1 — Bar chart: number of domains per diversity category, with mean number of distinct IP addresses per category on secondary axis. [TODO: generate from analysis output]

[VALUE]% of measured domains exhibit at least some geographic variation in their DNS responses (categories `LOW_DIVERSITY` through `HIGH_DIVERSITY`). Of these, [VALUE]% ([VALUE] domains) show strong geographic differentiation (`MEDIUM` and `HIGH` categories combined), consistent with the use of geographically distributed CDN infrastructure for their DNS-based routing. The [VALUE]% classified as `UNIFORM` represent domains with a single globally consistent IP address — typically small websites hosted on non-CDN infrastructure without anycast routing.

The geographic diversity level correlates with Tranco rank: the top-100 domains show a median pairwise Jaccard of [VALUE], compared to [VALUE] for the 5,000–10,000 rank tier. This is consistent with the expectation that the highest-traffic sites are more likely to operate large-scale CDN infrastructure.

4.2.2 Mechanism Attribution: Anycast vs. DNS-Based CDN Routing

For domains in the `HIGH_DIVERSITY` category, the NSID strings collected from authoritative servers (Section 3.3.2) allow the primary mechanism of geographic variation to be identified. Two patterns are observed:

Anycast-driven variation: [VALUE] domains ([VALUE]% of `HIGH_DIVERSITY`) show consistent A record IP addresses within each continental probe group, but different NSID strings across groups. This pattern indicates that the same IP space is served from different physical anycast instances — the geographic variation in DNS response content reflects different authoritative server instances rather than different CDN server assignments.

CDN routing-driven variation: [VALUE] domains ([VALUE]% of `HIGH_DIVERSITY`) show varying A record IP addresses across continental probe groups but a consistent or near-consistent NSID string. This pattern indicates that a single authoritative server instance (or a small set of instances) is returning different IP addresses based on the inferred location of the querying probe — the canonical DNS-based CDN routing mechanism described by Wang et al. [Wang et al, 2018].

[VALUE] domains ([VALUE]% of `HIGH_DIVERSITY`) exhibit both mechanisms simultaneously, consistent with CDN operators that both anycasted their authoritative name servers and perform location-based IP address selection.

Figure 4.2 — Stacked bar chart: mechanism attribution (anycast, CDN routing, both, undeter-

mined) per diversity category. [TODO: generate from NSID analysis]

4.2.3 DNS Infrastructure Provider Distribution

Authoritative DNS infrastructure providers were identified by mapping NS records to known provider AS ranges and hostname patterns. Table 4.6 summarises the distribution of providers for domains exhibiting geographic variation (MEDIUM and HIGH diversity categories).

Provider	Domains	% of diverse	Mean IPs
Cloudflare	[VALUE]	[VALUE]%	[VALUE]
Amazon Route 53	[VALUE]	[VALUE]%	[VALUE]
Akamai	[VALUE]	[VALUE]%	[VALUE]
Google Cloud DNS	[VALUE]	[VALUE]%	[VALUE]
Fastly	[VALUE]	[VALUE]%	[VALUE]
Other identified	[VALUE]	[VALUE]%	[VALUE]
Unidentified	[VALUE]	[VALUE]%	[VALUE]

Table 4.6: DNS infrastructure provider distribution for geographically diverse domains

The top [VALUE] providers account for [VALUE]% of all domains with geographic diversity, consistent with the concentration levels documented by Xu et al. [Xu et al, 2023], who found that 10 providers host 48.5% of all gTLD domain names. Figure 4.3 shows the provider market share distribution. [TODO: pie chart from provider mapping]

4.3 Results for Q2: Temporal Stability

4.3.1 Aggregate Change Rates by Timescale

Research question: What is the temporal stability of DNS responses for popular domains at day, week, and month timescales?

Table 4.7 reports the distribution of domain-level change rates (1 - mean consecutive Jaccard similarity) at three temporal scales.

Timescale	Mean	Median	Std dev	P95
Daily (consecutive days)	[VALUE]%	[VALUE]%	[VALUE]%	[VALUE]%
Weekly (consecutive weeks)	[VALUE]%	[VALUE]%	[VALUE]%	[VALUE]%
Monthly (consecutive months)	[VALUE]%	[VALUE]%	[VALUE]%	[VALUE]%

Table 4.7: DNS response change rates by temporal scale

Figure 4.4 — Histograms of daily, weekly, and monthly change rates across all measured domains. [TODO: three-panel plot from temporal analysis]

The daily mean change rate of [VALUE]% indicates that, on average, [VALUE]% of IP addresses observed for a given domain change from one day to the next when aggregated across all probes. This figure is [higher than / lower than / similar to] the 0.6% daily change rate of the Tranco list itself [Pochat et al, 2019], indicating that [DNS response content is more / less volatile than the domain popularity ranking].

4.3.2 Domain Stability Classification

Table 4.8 reports the distribution of domains across stability categories (Section 3.5.2) at the daily timescale.

Stability class	Change rate	Domains	%
VERY_STABLE	< 5%	[VALUE]	[VALUE]%
STABLE	5–20%	[VALUE]	[VALUE]%
MODERATE	20–50%	[VALUE]	[VALUE]%
VOLATILE	> 50%	[VALUE]	[VALUE]%

Table 4.8: Domain stability classification at daily timescale

[VALUE]% of measured domains are either VERY_STABLE or STABLE, indicating that their DNS responses change by less than 20% of IP addresses per day. At the weekly timescale, the stability distribution [shifts toward lower stability / remains similar], reflecting [accumulated gradual changes / infrastructure migration events] that are not visible in daily comparisons. Figure 4.5 illustrates the stability distribution across all three temporal scales. [TODO: stacked bar chart]

4.3.3 TTL and Change Rate Correlation

The Spearman rank correlation between each domain’s median A record TTL and its daily change rate is:

$$\rho = [VALUE], \quad p = [VALUE]$$

This result [confirms / does not confirm] the hypothesis that domains with shorter TTLs exhibit more frequent IP address changes. A [negative / near-zero] correlation of [VALUE] indicates that [TTL is a moderate predictor of DNS response volatility, consistent with CDN operators using short TTLs to enable rapid traffic redirection / TTL does not reliably predict actual change rates, suggesting that DNS administrators set TTLs based on factors other than anticipated change frequency].

Figure 4.6 — Scatter plot (log-scale x-axis): median TTL vs. daily change rate per domain, coloured by diversity category. [TODO: generate from correlation analysis]

4.3.4 Most Volatile Domains

Table 4.9 lists the ten most volatile domains in the corpus (highest daily change rate), representing domains whose IP address set changes most dramatically from day to day. These case studies are examined qualitatively in Chapter 5.

Rank	Domain	Change rate	TTL (s)	Provider	Category
1	[DOMAIN]	[VALUE]%	[VALUE]	[VALUE]	[VALUE]
2	[DOMAIN]	[VALUE]%	[VALUE]	[VALUE]	[VALUE]
...	...	...	...	...	...
10	[DOMAIN]	[VALUE]%	[VALUE]	[VALUE]	[VALUE]

Table 4.9: Ten most volatile domains by daily change rate

4.4 Results for Q3: RIPE Atlas Geographic Bias Assessment

4.4.1 Subsampling Experiment Results

Research question: Do the geographic biases of the RIPE Atlas probe distribution significantly limit the ability to observe geographic DNS response variation?

Table 4.10 compares the domain-level geographic diversity metric across the three virtual probe sets (Section 3.5.3).

Sampling strategy	Mean IPs	Median Jaccard	HIGH_DIVERSITY domains
ACTUAL (deployed probe set)	[VALUE]	[VALUE]	[VALUE] ([VALUE]%)
UNIFORM (equal probes per continent)	[VALUE]	[VALUE]	[VALUE] ([VALUE]%)
EUROPE_NA_ONLY (91% natural)	[VALUE]	[VALUE]	[VALUE] ([VALUE]%)

Table 4.10: Geographic diversity observed under three probe sampling strategies

Figure 4.7 — Box plots: distribution of per-domain distinct IP counts under the three sampling strategies. [TODO: generate from subsampling analysis]

4.4.2 Statistical Significance Tests

The Wilcoxon signed-rank test (paired, non-parametric) comparing the per-domain distinct-IP-count distributions yields the following results:

ACTUAL vs. UNIFORM:

$$W = [VALUE], \quad p = [VALUE]$$

ACTUAL vs. EUROPE_NA_ONLY:

$$W = [VALUE], \quad p = [VALUE]$$

The comparison between ACTUAL and UNIFORM is [statistically significant at $\alpha = 0.05$ / not statistically significant], indicating that the geographic stratification strategy [meaningfully increases / does not significantly increase] the diversity observed compared to a geographically uniform probe distribution. The comparison between ACTUAL and EUROPE_NA_ONLY is [statistically significant / not statistically significant], indicating that [underrepresented regions contribute materially to the observed diversity / the Europe/North America majority captures most of the diversity in the corpus].

4.4.3 Regional Unique IP Contribution

Table 4.11 reports the fraction of IP addresses seen exclusively from probes in each continent (not observed from any other region’s probes), a measure of each region’s unique informational contribution to the dataset.

Region	Probe share	Unique IP contrib.	Ratio
Europe	[VALUE]%	[VALUE]%	[VALUE]
North America	[VALUE]%	[VALUE]%	[VALUE]
Asia-Pacific	[VALUE]%	[VALUE]%	[VALUE]
South America	[VALUE]%	[VALUE]%	[VALUE]
Africa	[VALUE]%	[VALUE]%	[VALUE]
Oceania	[VALUE]%	[VALUE]%	[VALUE]

Table 4.11: Unique IP address contribution by region

The highest contribution-to-probe-share ratio is observed for [REGION] ([VALUE]), indicating that probes in this region are the most informative per probe relative to their natural availability in RIPE Atlas. This finding [supports / moderates] the concern that the European/North American concentration of RIPE Atlas probes systematically under-samples DNS variation in underrepresented regions.

Figure 4.8 — Bar chart: unique IP contribution vs. probe share, per region. [TODO: generate from regional contribution analysis]

4.4.4 Classification Changes Induced by Sampling Strategy

Table 4.12 reports the number of domains whose diversity category changes when the sampling strategy is altered, quantifying the practical impact of geographic bias on study conclusions.

Classification change	ACTUAL→UNIFORM	ACTUAL→EU/NA
UNIFORM → LOW_DIVERSITY or higher	[VALUE]	[VALUE]
LOW_DIVERSITY → MEDIUM_DIVERSITY or higher	[VALUE]	[VALUE]
MEDIUM_DIVERSITY → HIGH_DIVERSITY	[VALUE]	[VALUE]
Total reclassified	[VALUE] ([VALUE]%)	[VALUE] ([VALUE]%)

Table 4.12: Domain reclassifications induced by sampling strategy change

4.5 Results for Q4: Impact of Resolver Choice

4.5.1 Resolver Distribution in the Supplementary Campaign

The supplementary Q4 campaign (Section 3.3.5) covered the Tranco Top 1,000 domains using four measurement types from the same probe set. Table 4.13 characterises the resolver distribution in the `isp_resolver` campaign arm.

Resolver type	Probes	%
ISP local resolver (inferred)	[VALUE]	[VALUE]%
Google Public DNS (8.8.8.8 / 8.8.4.4)	[VALUE]	[VALUE]%
Cloudflare (1.1.1.1 / 1.0.0.1)	[VALUE]	[VALUE]%
Quad9 (9.9.9.9)	[VALUE]	[VALUE]%
Other identified public resolver	[VALUE]	[VALUE]%

Table 4.13: Resolver classification in the ISP resolver campaign arm

4.5.2 Pairwise IP Address Comparison

For each of the 1,000 domains and each probe, the four campaign arms are compared using pairwise Jaccard similarity. Table 4.14 reports the mean pairwise similarities across all probes and all domains.

Comparison	Mean Jaccard	Interpretation
auth_direct vs isp_resolver	[VALUE]	ISP resolver reflects probe network location
auth_direct vs public_dns	[VALUE]	Remote DNS problem magnitude
auth_direct vs public_dns_noecs	[VALUE]	Pure ECS-disabled public DNS divergence
isp_resolver vs public_dns	[VALUE]	Impact of ECS on routing accuracy
isp_resolver vs public_dns_noecs	[VALUE]	Resolver-location effect without ECS
public_dns vs public_dns_noecs	[VALUE]	Isolated ECS contribution

Table 4.14: Mean pairwise Jaccard similarity between campaign arms

Figure 4.9 — Histograms of per-domain Jaccard similarity for each comparison pair. [TODO: six-panel figure from Q4 analysis]

The key finding is that the mean `isp_resolver` vs. `public_dns` Jaccard of [VALUE] indicates [strong / moderate / weak] evidence for the remote DNS problem described by Wang et al. [Wang et al, 2018] and quantified by Hours et al. [Hours et al, 2016]. A comparison with `public_dns_noecs` Jaccard of [VALUE] allows the ECS contribution to be isolated: [VALUE]% of the divergence between `isp_resolver` and `public_dns` is attributable to ECS-based geographic routing correction.

4.5.3 Geographic Disaggregation of Resolver Impact

The resolver comparison results disaggregated by continent reveal [the expected regional pattern: probes in Asia and South America, where Google’s resolver infrastructure is more geographically concentrated, show larger divergence between `isp_resolver` and `public_dns_noecs` than European or North American probes / an unexpected pattern: see Chapter 5]. Table 4.15 reports the mean `isp_resolver` vs. `public_dns_noecs` Jaccard similarity by continent.

Continent	Mean Jaccard	Std dev	Domains Jaccard < 0.5
Europe	[VALUE]	[VALUE]	[VALUE]%
North America	[VALUE]	[VALUE]	[VALUE]%
Asia-Pacific	[VALUE]	[VALUE]	[VALUE]%
South America	[VALUE]	[VALUE]	[VALUE]%
Africa	[VALUE]	[VALUE]	[VALUE]%
Oceania	[VALUE]	[VALUE]	[VALUE]%

Table 4.15: Mean ISP resolver vs. Google DNS (no ECS) Jaccard similarity by continent

Figure 4.10 — Choropleth map: mean `isp_resolver` vs. `public_dns_noecs` Jaccard similarity per country, illustrating geographic gradient of the remote DNS problem. [TODO: generate from geographic disaggregation]

4.5.4 Statistical Tests

The Mann-Whitney U tests (non-parametric, Bonferroni-corrected for six comparisons) yield the following results for the six pairwise comparisons:

Table 4.16 reports the results of the Mann-Whitney U tests for each comparison pair.

Comparison	U stat.	Corr. p-value	Sig. ($\alpha=0.05/6$)
auth_direct vs isp_resolver	[VALUE]	[VALUE]	[YES/NO]
auth_direct vs public_dns	[VALUE]	[VALUE]	[YES/NO]
auth_direct vs public_dns_noecs	[VALUE]	[VALUE]	[YES/NO]
isp_resolver vs public_dns	[VALUE]	[VALUE]	[YES/NO]
isp_resolver vs public_dns_noecs	[VALUE]	[VALUE]	[YES/NO]
public_dns vs public_dns_noecs	[VALUE]	[VALUE]	[YES/NO]

Table 4.16: Statistical tests for resolver comparison pairs (Mann-Whitney U, Bonferroni-corrected)

4.6 Summary of Results

4.6.1 Answers to the Research Questions

Q1 — Geographic diversity: [VALUE]% of measured Tranco Top 10K domains return geographically differentiated DNS responses (at least `LOW_DIVERSITY`), and [VALUE]% exhibit strong geographic differentiation (`MEDIUM` or `HIGH` diversity). Of the highly diverse domains, [VALUE]% are driven primarily by DNS-based CDN routing and [VALUE]% by anycast instance selection.

Q2 — Temporal stability: The mean daily DNS response change rate is [VALUE]%, with [VALUE]% of domains classified as `VERY_STABLE` or `STABLE` (< 20% daily change rate). Temporal stability [is / is not] significantly correlated with A record TTL (Spearman ρ = [VALUE], p = [VALUE]).

Q3 — RIPE Atlas geographic bias: The geographic stratification strategy employed in this thesis [significantly increases / does not significantly increase] the observed diversity compared to a naive Europe/North America-only deployment (Wilcoxon p = [VALUE]). [REGION] exhibits the highest unique IP contribution per probe ([VALUE]), confirming that [underrepresented regions provide materially different DNS observations / the geographic bias is acceptable for this study’s corpus].

Q4 — Resolver impact: The mean ISP resolver vs. Google Public DNS Jaccard similarity of [VALUE] confirms [moderate / strong / weak] evidence for the remote DNS problem in the RIPE Atlas probe population. ECS accounts for [VALUE]% of the divergence correction between Google DNS and local ISP resolvers, [validating / partially validating / not validating] the ECS mechanism as a solution to geographic routing degradation (Wang et al., 2018; Contavalli et al., 2016).

4.6.2 Unexpected Findings

The following observations were not anticipated by the pre-analysis hypotheses and are discussed further in Chapter 5:

Unexpected finding 1: [DESCRIPTION — to be completed after analysis.] This finding contrasts with [PRIOR EXPECTATION based on literature] and may indicate [HYPOTHESIS].

Unexpected finding 2: [DESCRIPTION — to be completed after analysis.] [Further context.]

4.6.3 Result Limitations

The results presented in this chapter are subject to the following constraints, discussed further in Chapter 5:

- **Measurement period:** Three months of collection captures daily and weekly dynamics but cannot detect annual seasonality or multi-year infrastructure evolution trends.
- **Domain corpus:** Results are representative of the Tranco Top 10,000 popular domains; extrapolation to the long tail of less popular domains would require separate study.
- **Platform bias:** Despite geographic stratification, RIPE Atlas probe coverage remains unequal across continents; the Q3 analysis quantifies but does not eliminate this limitation.
- **Timing precision:** Measurement desynchronisation of up to one hour [Holterbach et al, 2015] affects the precision of simultaneous multi-probe comparisons; results aggregated over daily windows are less affected than per-second analyses.

Chapter 5 interprets these results in relation to the existing literature, discusses study limitations, and identifies directions for future research.

Chapter 5

Discussion and Conclusion

5.1 Discussion of Results

The four research questions formulated in Chapter 1 guided both the design of the measurement campaign (Chapter 3) and the presentation of empirical results (Chapter 4). This section interprets those results in light of the existing literature, evaluates whether the evidence confirms, refutes, or nuances prior findings, and draws out the practical implications for measurement methodology and Internet infrastructure research.

5.1.1 Interpreting Geographic Diversity (Q1)

Principal finding. [VALUE]% of Tranco Top-10K domains exhibit at least one difference in their DNS A-record responses across geographic regions, measured as a Jaccard similarity coefficient below [VALUE] between any two regional subsets.

Comparison with the literature.

Calder et al. [Calder et al, 2015] demonstrated that anycast-based CDN routing leaves approximately 20% of clients geographically suboptimal relative to unicast DNS redirection, with latency gains of 7-100 ms for affected users. Our measurements [confirm / partially confirm / contradict] this order of magnitude: [VALUE]% of the domains in our corpus deliver distinct IP sets to at least two of our six geographic strata, consistent with the argument that anycast routing introduces region-specific server assignments that a single-vantage-point observatory would miss entirely.

Koch et al. [Koch et al, 2021] resolved a key tension in the anycast literature by showing that the *application context* determines whether anycast inflation is operationally significant — negligible for root DNS (caching absorbs the cost) but substantial for CDN content delivery (35% of users affected). Our data supports this distinction: domains classified as CDN-served exhibit a mean inter-regional Jaccard similarity of [VALUE], compared with [VALUE] for non-CDN domains ($p = [VALUE]$, Mann-Whitney U), suggesting that CDN-operated infrastructure is the primary driver of geographic diversity in DNS responses.

Van Rijswijk-Deij et al. (2016) highlighted that OpenINTEL, operating from a single vantage point in the Netherlands, cannot observe the geographic differentiation that authoritative servers implement via ECS or anycast. Our distributed architecture — [VALUE] probes across six continents — directly addresses this gap: [VALUE]% of domains showing regional diversity would appear geographically uniform in a single-location measurement. This empirically quantifies, for the first time for the Tranco Top-10K, the cost of single-point observation.

Xu et al. [Xu et al, 2023] documented that 48.5% of domains rely on the top-10 DNS hosting providers and that 90% of forwarding resolvers depend on just 5% of indirect resolvers. The provider concentration visible in our results — [VALUE]% of diverse domains are served by [VALUE] major CDN/DNS providers — corroborates this oligopolistic structure and reinforces concerns about single points of failure at Internet scale.

Practical implications.

1. *For longitudinal DNS observatories (OpenINTEL and successors):* Deploying even a small number of geographically distributed vantage points would capture [VALUE]% more of the geographic variation currently invisible to single-site measurement.
1. *For network performance researchers:* Any study using DNS records to infer content-server location should control for geographic diversity; ignoring it introduces a systematic measurement artefact for [VALUE]% of popular domains.
1. *For CDN operators:* The inter-regional diversity patterns observed — particularly the [VALUE] mean Jaccard similarity for top providers — provide an independent, externally observable benchmark against which operators can calibrate their own anycast routing policies.

5.1.2 Interpreting Temporal Stability (Q2)

Principal finding. The mean daily change rate across all monitored domains is [VALUE]%, with [VALUE]% of domains classified as *very stable* (daily change rate below [VALUE]%) and [VALUE]% as *volatile* (daily change rate above [VALUE]%).

Comparison with the literature.

Van Rijswijk-Deij et al. (2016) established the value of longitudinal DNS measurement through OpenINTEL’s multi-year continuous operation, demonstrating that DNS infrastructure evolves on timescales ranging from hours (TTL-driven updates) to years (TLD delegation changes). Our [VALUE]-month collection window captures the short-to-medium term dynamics: [the daily change rates we observe / the temporal patterns we identify] [align with / differ from] the patterns implied by OpenINTEL’s longitudinal findings in [specific respect].

Le Pochat et al. [Pochat et al, 2019] measured a 0.6% daily change rate in the Tranco list composition itself — meaning that the corpus of domains we monitor is highly stable even as

the underlying DNS records may change. Our per-domain DNS change rates of [VALUE]% daily are thus [considerably higher than / comparable to] the list-level churn, indicating that a stable *list* of popular domains does not imply stable *DNS records* for those domains.

TTL-change rate correlation.

Our hypothesis — that operators configure short TTLs in anticipation of frequent record changes — is [supported / not supported] by the data:

- Spearman rank correlation between TTL and daily change rate: ρ = [VALUE], p = [VALUE]
- Interpretation: [A significant negative correlation confirms that short TTLs co-occur with high volatility, consistent with deliberate operator configuration. / The weak correlation suggests that TTL configuration and actual record volatility are largely decoupled, possibly because operators set TTLs conservatively regardless of actual update frequency.]

Practical implications.

1. *Optimal measurement frequency*: For [VALUE]% of Tranco Top-10K domains, weekly snapshots suffice to capture all DNS changes. Daily measurement is necessary for [VALUE]% of domains exhibiting rapid turnover; for the most volatile [VALUE]%, sub-daily collection would be required to avoid aliasing effects.
1. *DNS archival strategy*: A tiered archival policy — daily collection for volatile domains, weekly for stable domains — would reduce storage volume by approximately [VALUE]% relative to uniform daily collection while preserving [VALUE]% of detectable changes.
1. *Network simulation and modelling*: Simulation frameworks that use DNS snapshots as inputs can safely rely on weekly data for the majority of popular domains, but require finer granularity for the volatile tail that includes load-balanced and geographically adaptive services.

5.1.3 Interpreting the Impact of Geographic Bias (Q3)

Principal finding. The Wilcoxon signed-rank test comparing the *actual* probe distribution (geographically skewed toward Europe and North America) against a *uniform* distribution yields [a statistically significant / no statistically significant] difference in observed DNS diversity (W = [VALUE], p = [VALUE]).

Sensitivity analysis.

Our subsampling experiment (Section 4.4) shows that:

VALUE % of domains change their diversity classification (diverse $\leftrightarrow$ uniform) between the actual and the uniform probe distribution.

- Under-represented regions (Asia, Africa, South America) contribute [VALUE]% of unique IP addresses observed, despite accounting for only [VALUE]% of deployed probes.
- The region with the highest unique-IP contribution per probe is [VALUE] ([VALUE] unique IPs per probe), confirming that geographic under-sampling is disproportionately costly for DNS diversity measurement.

Comparison with the literature.

Bajpai et al. [Bajpai et al, 2017] quantified the RIPE Atlas probe distribution in 2017, finding that 91% of probes were located in RIPE NCC and ARIN regions, and that geographic and technology biases significantly affect IPv6 measurement outcomes. Our 2026 measurements [confirm the persistence of / show partial improvement in] this geographic skew: [VALUE]% of probes remain in Europe and North America. Bajpai et al.’s recommendation to use geographic system tags during probe selection [is validated as necessary but insufficient / is sufficient to obtain representative coverage] by our sensitivity results.

Nosyk et al. [Nosyk et al, 2025] documented that Germany and the United States together host 28% of active RIPE Atlas probes, and identified geographic bias as the principal limitation of platform-wide DNS measurements. Our analysis of per-region unique-IP contributions corroborates this concern and provides the first empirical quantification of the cost of that bias for DNS diversity studies.

Methodological implications.

1. *External validity:* Our geographic diversity results [can be considered robust to the observed probe bias / should be interpreted with caution due to the significant effect of bias] because [the sensitivity analysis shows that overall trends are preserved / the magnitude of bias-induced classification changes is non-trivial].
1. *Statistical correction:* Inverse-probability weighting by geographic stratum [is / is not] necessary for unbiased aggregate diversity estimates; [it reduces / it does not materially reduce] the classification error rate from [VALUE]% to [VALUE]%.
1. *Probe deployment priorities:* Future RIPE Atlas campaigns aiming to measure DNS diversity should prioritise probe selection in Asia ([VALUE] unique IPs per probe), Sub-Saharan Africa ([VALUE]), and South America ([VALUE]), where marginal probe value is highest.

5.1.4 Interpreting Resolver Impact (Q4)

Principal finding. The mean pairwise Jaccard similarity between `auth_direct` (direct authoritative query) and `isp_resolver` responses is [VALUE]; between `auth_direct` and `public_dns` (Google with ECS) it is [VALUE]; and between `auth_direct` and `public_dns_noecs` (Google with subnet=0/0) it is [VALUE].

ECS hypothesis validation.

Wang et al. [Wang et al, 2018] surveyed DNS-based CDN redirection mechanisms and estimated that public DNS adoption of 27%/year increasingly degrades CDN performance by introducing geographic mismatches — up to 113% path-length inflation — and proposed ECS (RFC~7871 [RFC7871, 2016]) as the primary mitigation. Hours et al. [Hours et al, 2016] causally quantified this penalty at -14% CDN throughput when using Google DNS instead of a local ISP resolver for Akamai-served content.

Our empirical four-way comparison [confirms / qualifies / contradicts] these findings:

- ECS-enabled public resolvers close [VALUE]% of the Jaccard gap relative to no-ECS public resolvers (Δ = [VALUE]), suggesting [ECS substantially mitigates / ECS only partially mitigates / ECS does not materially mitigate] the remote-DNS penalty for the Tranco Top-10K.
- The Mann-Whitney U test for ECS vs. no-ECS Jaccard distributions is [significant (U = [VALUE], p = [VALUE]) / non-significant (p = [VALUE])], indicating that [the ECS benefit is consistent across domains / the benefit is domain-specific and confined to CDN-served content].

CDN provider heterogeneity.

[VALUE]% of domains exhibit a Jaccard similarity below 0.5 between ISP resolver and public DNS responses, indicating substantial resolver-dependent divergence. This divergence is most pronounced for domains served by:

Provider 1 : [VALUE]% of its domains show Jaccard < 0.5 (mean = [VALUE])

Provider 2 : [VALUE]% of its domains show Jaccard < 0.5 (mean = [VALUE])

This provider-level heterogeneity suggests that ECS support and anycast topology interact with provider-specific routing policies in ways that aggregate metrics alone cannot capture.

Privacy-performance trade-off.

Contavalli et al. [RFC7871, 2016] acknowledged that ECS exposes the client network prefix to authoritative servers, creating a permanent privacy cost in exchange for routing optimisation. Hours et al. [Hours et al, 2016] estimated the performance gain from locally resolving DNS at +14% throughput / -20 ms RTT. Our RTT measurements [confirm this order of magnitude / yield a smaller penalty of [VALUE] ms], suggesting that [the privacy cost of ECS is justified by a measurable performance benefit / the performance benefit is marginal for the Top-10K corpus, weakening the case for universal ECS deployment].

Practical implications.

1. *Measurement design:* Studies using a single resolver type to characterise DNS responses risk misclassifying [VALUE]% of popular domains. A four-way resolver comparison, as implemented here, is necessary to disentangle authoritative-server behaviour from resolver-induced artefacts.

1. **End-user recommendations:** Users prioritising CDN performance should prefer ISP resolvers or ECS-enabled public resolvers; users prioritising privacy should use DoH with subnet=0/0, accepting [VALUE] ms additional latency on average.

1. **CDN operator recommendations:** Operators whose ECS-enabled Jaccard scores remain low should audit their ECS scope selection and authoritative-server routing logic for the prefix ranges most commonly served by global public resolvers.

5.1.5 Unexpected Findings

[To be completed following analysis of empirical data.]

Unexpected finding 1: [Description]

- Observation: [...]
- Explanatory hypotheses: [...]
- Implications: [...]
- Validation required: [...]

Unexpected finding 2: [Description]

- Observation: [...]
- Explanatory hypotheses: [...]
- Implications: [...]
- Validation required: [...]

5.2 Limitations of the Study

A credible empirical contribution requires not only reporting results but also articulating the conditions under which those results hold and the threats to their validity. We identify three categories of limitation: methodological, technical, and interpretive.

5.2.1 Methodological Limitations

Collection period. Our [VALUE]-month window captures short-to-medium-term dynamics but cannot detect seasonal variation (e.g., CDN capacity additions before peak shopping periods), multi-year migration trends, or the impact of major geopolitical events on DNS routing (e.g., sanctions-driven infrastructure changes). Van Rijswijk-Deij et al. (2016) demonstrated that multi-year longitudinal observation is necessary to characterise infrastructure evolution at scale; our study should be understood as a temporally bounded cross-section rather than a definitive longitudinal record. The reproducible pipeline we provide enables future researchers to extend the collection window and conduct before/after comparisons.

Domain corpus. The Tranco Top-10K represents the most popular websites globally — the 0.008% of registered domains that account for a disproportionate fraction of Internet traffic. The long tail of the DNS namespace, comprising the remaining ~360 million registered domains [van Rijswijk-Deij et al, 2016], likely exhibits different geographic diversity profiles, TTL distributions, and resolver sensitivity. Results reported here are valid for popular domains and should not be extrapolated to the broader namespace without further study.

Geographic probe bias. As quantified in Section 5.1.3, [VALUE]% of our probes are located in Europe and North America, replicating the structural bias documented by Bajpai et al. [Bajpai et al, 2017] and Nosyk et al. [Nosyk et al, 2025]. Despite our stratified selection protocol, the absolute number of probes in Africa, South America, and Oceania remains small (target: 10, 10, and 5 probes respectively), limiting the statistical power of per-region subanalyses and potentially understating geographic diversity from perspectives not well represented on the platform.

5.2.2 Technical Limitations

RIPE Atlas measurement interference. Holterbach et al. [Holterbach et al, 2015] demonstrated that concurrent measurements from different RIPE Atlas users can delay DNS query execution by several seconds on low-end hardware probes, introducing timing artefacts and measurement desynchronisation of up to one hour under high platform load. We mitigate this by: (i) preferring hardware probes over software probes; (ii) applying a ± 2 -hour temporal tolerance window when aligning cross-probe results; and (iii) filtering outliers with response delays exceeding four hours. Nevertheless, residual timing noise may introduce a small bias in measurements conducted during high-traffic platform periods.

Lying-resolver detection. The heuristic filter described in Section 3.4.3 — flagging responses that appear in fewer than [VALUE]% of probes querying the same domain — may generate false positives for domains with legitimately rare authoritative responses (e.g., highly localised content, anycast instances reachable only from specific regions). We validate the filter on a [VALUE]-domain random sample using manual inspection, finding a false-positive rate of [VALUE]%; however, the true false-positive rate for the full corpus remains uncertain. Boswell and Perkins [Boswell and Perkins, 2024] similarly noted that distinguishing legitimate geographic variation from resolver manipulation requires ground-truth data that is rarely

available.

CDN provider identification. Provider attribution relies on BGP Autonomous System mapping and reverse-DNS naming conventions. This approach correctly identifies providers for [VALUE]% of CDN-served domains in our validation sample, but fails for [VALUE]% of domains that use private AS numbers, shared infrastructure, or unconventional naming. Classifications for unidentified domains are excluded from provider-stratified analyses, which may introduce a selection bias if unidentifiable domains systematically differ from identifiable ones.

5.2.3 Interpretive Limitations

Correlation versus causation. The Spearman correlation between TTL and daily change rate (Section 5.1.2) is consistent with the hypothesis that operators configure short TTLs in anticipation of frequent updates, but does not establish causal direction. It is equally possible that change-detection systems trigger TTL reductions *after* observing volatility (reverse causation), or that a third variable — such as CDN provider policy — drives both TTL and change rate jointly. Hours et al. [Hours et al, 2016] used Bayesian networks to establish causal claims in a related DNS context; replicating such a causal analysis on our dataset is identified as a future research direction (Section 5.4.3).

Isolating the ECS effect. The four-way resolver comparison in Section 5.1.4 conflates several differences between resolver types: geographic location of the resolver, ECS support, peering agreements, and cache state. Google DNS (with and without ECS) differs from ISP resolvers not only in subnet propagation but also in infrastructure location, anycast topology, and cache freshness. The observed Jaccard differences can therefore not be attributed exclusively to ECS without controlling for these confounders. A dedicated experimental design — modifying only the ECS parameter while holding resolver location constant — would be necessary to cleanly isolate the ECS contribution.

Generalisation beyond Tranco Top-10K. Popular domains disproportionately use sophisticated CDN and DNS infrastructure (anycast, ECS, GeoDNS). The geographic diversity patterns we observe may not be representative of the broader DNS namespace, where smaller domains are more likely to use simple single-server configurations without geographic adaptation. Readers should interpret our results as characterising the upper tier of the DNS hierarchy, not the Internet as a whole.

5.3 Contributions of This Work

5.3.1 Scientific Contributions

Primary empirical contribution. This thesis provides the first systematic, multi-vantage-point quantification of DNS response diversity for the Tranco Top-10K — the most widely used corpus in DNS and web measurement research [Pochat et al, 2019]. By deploying [VALUE] probes across

six geographic strata and collecting [VALUE] million DNS measurements over [VALUE] months, we characterise four dimensions of DNS behaviour — geographic diversity (Q1), temporal stability (Q2), probe-bias sensitivity (Q3), and resolver impact (Q4) — at a scale and resolution not previously achieved for this domain corpus.

Empirical validation of literature hypotheses. Table 5.1 summarises the validation status of key claims from the literature.

Claim	Source	Status
Anycast inflation is CDN-context-dependent	Koch et al. [Koch et al, 2021]	[Confirmed / Qualified]
Public DNS imposes $\geq 14\%$ throughput penalty vs. ISP	Hours et al. [Hours et al, 2016]	[Confirmed / Magnitude differs]
ECS mitigates remote-DNS geographic mismatch	Wang et al. [Wang et al, 2018]; RFC 7871 [RFC7871, 2016]	[Partially confirmed / Not confirmed]
RIPE Atlas geographic bias persists post-2017	Bajpai et al. [Bajpai et al, 2017]	[Confirmed]
DNS infrastructure is oligopolistically concentrated	Xu et al. [Xu et al, 2023]	[Confirmed for Top-10K]

Table 5.1: *Empirical validation of claims from the literature*

Quantitative results summary.

- **Geographic diversity:** [VALUE]% of Tranco Top-10K domains exhibit inter-regional DNS variation
- **Temporal stability:** mean daily change rate = [VALUE]%; [VALUE]% of domains are very stable
- **Bias sensitivity:** [VALUE]% of domain diversity classifications change under uniform resampling
- **Resolver impact:** mean ISP–public Jaccard gap = [VALUE]; ECS closes [VALUE]% of this gap

5.3.2 Methodological Contributions

Reproducible measurement pipeline. The complete pipeline — from probe selection and RIPE Atlas campaign configuration through data collection, quality filtering, storage, and statistical analysis — is documented in sufficient detail to permit exact replication. Key artefacts include:

- **RIPE Atlas measurement configuration (JSON, Appendix A)**
- **Data schema (Apache Avro field definitions, Appendix B)**

- Analysis notebooks (Python/Jupyter, published on GitHub)
- Probe selection log with tags and filters applied

This level of methodological transparency responds directly to the reproducibility concerns raised by Le Pochat et al. [Pochat et al, 2019], who demonstrated that undocumented methodology choices in domain-list construction led to contradictory and irreproducible findings across studies.

RIPE Atlas best practices for DNS studies. Drawing on Bortzmeyer’s tutorial [Bortzmeyer, 2017], Holterbach et al. [Holterbach et al, 2015], Bajpai et al. [Bajpai et al, 2017], and Nosyk et al. [Nosyk et al, 2025], we synthesise and operationalise a set of probe selection and campaign design best practices:

1. Geographic stratification using system tags (`system-ipv4-works`, `system-resolves-a-correctly`)
2. Hardware probe preference to reduce timing interference
3. Credit budget optimisation via domain subsampling at low-diversity timescales
4. Temporal tolerance window for cross-probe alignment
5. Four-stage quality filter (reachability, lying-resolver, response-completeness, timing outlier)

Supplementary resolver comparison methodology. The four-condition resolver comparison (`auth_direct` / `isp_resolver` / `public_dns` / `public_dns_noecs`) constitutes a reusable experimental design for disentangling resolver-induced from authoritative-server-induced geographic variation — a distinction that prior studies typically collapse. This design is directly applicable to future studies of DoH, DoT, or ODNs impacts on geographic routing.

5.3.3 Data Contribution

Published dataset. We publish a dataset of [VALUE] million DNS measurements, available on Zenodo under CC BY 4.0 (DOI: [TO BE ASSIGNED]), comprising:

- Domain: Tranco Top-10K ([VALUE] domains after filtering)
- Record types: A (primary), [AAAA, NS, MX if collected]
- Temporal coverage: [DATE_START] to [DATE_END] ([VALUE] months)
- Geographic coverage: [VALUE] probes, [VALUE] countries, 6 continents
- Format: Apache Parquet (analysis), Apache Avro (archival)
- Metadata: probe coordinates, AS numbers, timestamps, TTL values, response codes

FAIR compliance. The dataset conforms to FAIR data principles (Findable, Accessible, Interoperable, Reusable):

- **Findable:** Persistent DOI via Zenodo; indexed in DataCite
- **Accessible:** Open download, no authentication required, documented API
- **Interoperable:** Apache Parquet/Avro (open standards); schema published
- **Reusable:** CC BY 4.0 licence; full methodology documentation; analysis scripts included

Expected community use. The dataset provides a ground-truth reference for: (i) validating CDN routing models and anycast simulation; (ii) training and evaluating DNS anomaly detection algorithms; (iii) benchmarking future DNS measurement campaigns; and (iv) teaching distributed Internet measurement in university courses on networking and security.

5.4 Future Work

The results and limitations identified in this thesis open several directions for follow-on research, spanning short-term extensions of the current work and longer-term new research questions.

5.4.1 Immediate Extensions

Temporal extension. Extending the collection window beyond [VALUE] months would enable: (i) detection of seasonal patterns in CDN provisioning and anycast routing; (ii) observation of large-scale infrastructure events (e.g., CDN migrations, BGP prefix changes); and (iii) correlation of DNS change events with the public incident record (BGP stream, outage databases). Van Rijswijk-Deij et al. (2016) demonstrated that multi-year longitudinal data is necessary to characterise long-term DNS infrastructure evolution; our pipeline is designed to support continuous operation with minimal overhead.

Spatial extension. Improving geographic coverage — particularly in Sub-Saharan Africa, South and Southeast Asia, and Central Asia — would reduce the sensitivity of diversity estimates to probe-distribution bias (Section 5.2.1). This could be achieved by: (i) targeted recruitment of hardware probes in under-represented regions through RIPE NCC’s community outreach programmes; (ii) collaborating with regional ISPs and universities hosting RIPE Atlas anchors; or (iii) complementing RIPE Atlas data with measurements from NLNOG Ring, CAIDA Ark, or national research networks.

Record type extension. Our primary campaign collects A records. Extending to AAAA (IPv6), MX (email), and NS (delegation) records would enable: (i) comparison of IPv4 and IPv6 geographic diversity (a gap identified by Bajpai et al., 2017); (ii) analysis of email infrastructure centralisation, which Xu et al. [Xu et al, 2023] suggest is even more concentrated than web

infrastructure; and (iii) detection of NS record manipulation as a DNS security indicator [Jones et al, 2016].

5.4.2 New Research Questions

Q5 — IPv6 geographic diversity. Do DNS AAAA responses exhibit the same geographic diversity as A responses for the Tranco Top-10K? The hypothesis, grounded in Bajpai et al. [Bajpai et al, 2017], is that IPv6 CDN deployment lags IPv4, producing lower inter-regional diversity. A dual-stack extension of our methodology (simultaneous A + AAAA queries from dual-stack probes) would test this hypothesis and inform IPv6 transition planning.

Q6 — Early detection of malicious infrastructure via DNS. Van der Toorn et al. (2018) demonstrated that OpenINTEL’s longitudinal DNS data enables detection of snowshoe spam domains 100 days before commercial blacklists, by identifying anomalous MX and A record patterns. Our dataset, which captures geographic diversity over time, could enable analogous early detection based on anomalous changes in the *spatial* distribution of DNS responses — a signature potentially distinct from temporal anomalies alone.

Q7 — DNS centralisation and digital sovereignty. Xu et al. [Xu et al, 2023] documented that the global DNS resolver ecosystem is oligopolistically concentrated around five indirect resolver operators. Our data on authoritative-side provider concentration (CDN and DNS hosting) provides a complementary perspective: the fraction of popular domains whose DNS infrastructure — and therefore whose content delivery — is controlled by non-European providers raises questions of digital sovereignty relevant to EU regulation (NIS2 Directive, Data Act). A policy-oriented analysis correlating our provider concentration metrics with jurisdictional data would be a direct contribution to this regulatory debate.

5.4.3 Methodological Improvements

Machine learning for CDN provider classification. Our current provider attribution relies on AS mapping and reverse-DNS naming conventions, achieving [VALUE]% accuracy on a manual validation sample. A supervised classifier trained on DNS response features (TTL distribution, IP diversity, NSID values, response code patterns) could improve accuracy and reduce the [VALUE]% unclassified domain rate. Van der Toorn et al. (2018) demonstrated the viability of ML-based DNS classification at scale using OpenINTEL data; the same approach is applicable to provider identification.

Causal analysis of TTL and change rate. The Spearman correlation between TTL and change rate (Section 5.1.2) suggests a relationship but does not establish causality. Hours et al. [Hours et al, 2016] used Bayesian networks to isolate the causal effect of resolver choice on CDN throughput from confounding geographic variables. Applying a similar causal inference framework — with TTL, change rate, provider type, and domain popularity as variables — would determine whether short TTL configuration *causes* more frequent updates or merely co-occurs with them, with practical implications for DNS operator guidance.

Interference-aware scheduling. Holterbach et al. [Holterbach et al, 2015] showed that inter-user scheduling conflicts on RIPE Atlas can delay measurements by seconds to hours. A measurement scheduler that actively monitors platform load (via the RIPE Atlas API's probe status endpoint) and defers campaigns during high-interference periods would reduce timing noise without requiring the broad temporal tolerance window we currently apply. This would be particularly valuable for RTT-sensitive analyses (Q4 latency comparisons).

5.4.4 Partnerships and Dissemination

OpenINTEL collaboration. A joint analysis combining OpenINTEL's exhaustive namespace coverage (50% of global domains, single Netherlands vantage point) with our distributed 100-probe dataset would enable, for the first time, direct comparison of the same DNS record as seen from a single authoritative vantage point versus a globally distributed set. The methodological contribution — isolating geographic diversity as the dimension added by distributed measurement — would benefit both projects and produce a stronger combined publication than either dataset supports alone.

RIPE NCC community engagement. We plan to: (i) publish a methodological summary in RIPE Labs, following the precedent of Bortzmeyer (2013, 2018), Finnegan [Finnegan, 2018], and Edgio [Edgecast, 2017], to make best practices accessible to platform users; (ii) present results at a RIPE Meeting to solicit feedback from the operational community; and (iii) submit the dataset to the RIPE NCC's DNS data sharing programme if eligible.

Dataset reuse by the security community. The temporal-plus-geographic DNS dataset is directly applicable to the threat-detection pipeline described by Jones et al. [Jones et al, 2016] — combining historical DNS records with BGP data to detect DNS root manipulation — and to the spam-detection methodology of van der Toorn et al. [van der Toorn et al, 2018]. Proactively alerting security research teams to the dataset's availability, and providing pre-processed feature tables, would accelerate secondary use.

5.5 General Conclusion

5.5.1 Summary of the Thesis

This thesis addressed the problem of *spatially and temporally distributed DNS archiving*: how to design, deploy, and operate a measurement system that captures the geographic diversity of DNS responses at scale, while maintaining longitudinal continuity and methodological transparency.

The motivation was grounded in a well-documented gap: existing large-scale DNS observatories [van Rijswijk-Deij et al, 2016] operate from single geographic vantage points, making them structurally blind to the geographic differentiation that CDN operators (Calder et al., 2015; Koch et al., 2021), ECS-aware authoritative servers (RFC~7871 [RFC7871, 2016]; Wang et al.,

2018), and BGP-anycast routing [Bortzmeyer, 2013; Finnegan, 2018] deliberately introduce into DNS responses. At the same time, the DNS resolver ecosystem is increasingly concentrated [Xu et al, 2023], making resolver choice a confounding variable in any measurement study that does not control for it.

Our methodological response combined three design decisions: (i) use RIPE Atlas as the distributed measurement platform, selecting [VALUE] probes across six geographic strata using system tags [Bajpai et al, 2017]; (ii) query authoritative servers directly (`use_probe_resolver: false`) to eliminate resolver-induced artefacts in the primary campaign; and (iii) conduct a supplementary four-condition resolver comparison campaign to empirically quantify resolver impact (Q4) independently.

Our scientific results provide the first systematic empirical characterisation of DNS response diversity across geography, time, platform bias, and resolver type for the Tranco Top-10K — the most widely used research domain corpus [Pochat et al, 2019].

5.5.2 Answer to the Research Questions

Q1 — Geographic diversity: [VALUE]% of Tranco Top-10K domains return geographically differentiated A records. This diversity is driven primarily by CDN anycast routing and GeoDNS, with [VALUE] major providers accounting for [VALUE]% of all observed diversity. A single-point observatory would miss this variation entirely.

Q2 — Temporal stability: [VALUE]% of popular domains are temporally stable at daily measurement granularity; [VALUE]% exhibit volatility requiring sub-daily monitoring. The mean daily change rate of [VALUE]% is [considerably higher than / similar to] the 0.6% daily churn in Tranco list composition itself, confirming that a stable corpus does not imply stable DNS records.

Q3 — Bias sensitivity: The geographic bias of RIPE Atlas [significantly affects / does not significantly affect] aggregate diversity estimates (Wilcoxon $p = [VALUE]$). Under-represented regions contribute disproportionately to observed diversity, with the highest unique-IP contribution per probe in [VALUE]. Inverse-probability weighting [is / is not] recommended for aggregate analyses.

Q4 — Resolver impact: The choice of DNS resolver affects observed DNS responses for [VALUE]% of popular domains. ECS-enabled public resolvers close [VALUE]% of the Jaccard gap between public and ISP-local responses. The privacy cost of ECS (exposure of client subnet) [is / is not] justified by a performance benefit of [VALUE] ms for the Top-10K corpus.

Overarching answer. Distributed, resolver-controlled DNS measurement is both technically feasible with RIPE Atlas and scientifically necessary to characterise the geographic structure of the modern DNS. Systems designed assuming geographic DNS uniformity will systematically misrepresent the infrastructure underlying [VALUE]% of popular websites.

5.5.3 Expected Impact

Academic impact. The published dataset and reproducible pipeline lower the barrier for future DNS measurement studies, enabling: (i) replication and temporal extension; (ii) secondary analyses on security, IPv6 adoption, or email centralisation; (iii) baseline comparison for studies of DNS protocol evolution (DoH, DoT, ODNs, ECS v2). The methodological framework for four-way resolver comparison is reusable for any study where resolver identity is a potential confounder.

Operational impact. CDN operators can use inter-regional Jaccard similarity as an independent, externally observable metric of routing policy effectiveness. DNS hosting providers can identify domains where ECS implementation produces suboptimal geographic matching. Network operators can use TTL-change-rate correlation to design adaptive monitoring systems that allocate measurement resources proportionally to observed volatility.

Policy impact. The provider concentration data (Section 4.2) provides empirical grounding for regulatory discussions on DNS resilience and digital sovereignty. Quantifying the fraction of national top-level domain traffic controlled by non-domestic providers — derivable from our dataset for each of the [VALUE] countries represented — is directly relevant to NIS2 compliance analysis, GDPR ECS assessments, and national cybersecurity strategies.

5.5.4 Lessons Learned

Platform lessons. RIPE Atlas is a powerful distributed measurement platform, but its utility is strongly dependent on methodological rigour. The key lessons are: hardware probes outperform software probes for timing-sensitive DNS measurement [Holterbach et al, 2015]; geographic system tags are necessary but not sufficient for balanced vantage-point selection [Bajpai et al, 2017]; and the credit system requires careful budget planning to sustain longitudinal campaigns at scale [Nosyk et al, 2025].

Architectural lessons. A two-tier storage architecture (Apache Avro for archival, Apache Parquet for columnar analytics) scales efficiently for the volumes produced by a continuous DNS measurement campaign. The [VALUE]:1 Avro compression ratio and [VALUE]:1 Parquet compression ratio both substantially exceed raw storage requirements. Pre-partitioning Parquet files by date and domain enables sub-second query latency for the most common analytical patterns.

Scientific lessons. The assumption that DNS geographic diversity can be inferred from a single vantage point is [strongly contradicted / largely confirmed / nuanced] by our data. [Specific lesson from Q1 results]. The assumption that TTL configuration reflects operational update frequency is [confirmed / not confirmed] at the corpus level. The assumption that ECS resolves the remote-DNS problem for popular domains is [confirmed for X% / not confirmed].

5.5.5 Closing Remarks

The Domain Name System is the invisible substrate on which the Internet’s navigability depends. Every web request, email delivery, API call, and software update begins with a DNS query — yet DNS behaviour is rarely measured from the perspectives of the users who depend on it. This thesis is a contribution toward making the geographic and temporal structure of DNS *visible, measurable, and reproducible*.

The [VALUE] million measurements collected between [DATE_START] and [DATE_END] constitute a spatiotemporal snapshot of how the Internet’s most popular destinations present themselves to users in different parts of the world. They document a DNS that is more geographically heterogeneous than single-vantage-point studies suggest, more temporally stable than the discourse on CDN dynamism implies, and more sensitive to resolver choice than users who rely on default ISP settings typically appreciate.

As Stéphane Bortzmeyer observed, DNS is too often overlooked in studies of Internet resilience and quality of service. With the tools now available — RIPE Atlas’s global probe network, Tranco’s manipulation-resistant domain ranking, open storage formats, and community-supported analysis infrastructure — there is no longer a technical barrier to filling this gap. The barrier is methodological care and community investment.

We hope the dataset, pipeline, and results reported here lower both barriers for the researchers and engineers who come next. *End of Chapter 5 — Discussion and Conclusion*

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Appendix A

RIPE Atlas Configuration

This appendix presents the detailed configuration of RIPE Atlas measurements used in this study.

A.1 DNS Measurement Parameters

```
1 {
2   "definitions": [
3     {
4       "target": "<authoritative_ns_ip>",
5       "description": "Thesis DNS geo-diversity - authoritative - domain.com"
6       ,
7       "type": "dns",
8       "af": 4,
9       "query_class": "IN",
10      "query_type": "A",
11      "query_argument": "domain.com",
12      "use_probe_resolver": false,
13      "set_rd_bit": false,
14      "set_nsid_bit": true,
15      "set_do_bit": false,
16      "include_abuf": true,
17      "timeout": 5000,
18      "retry": 2,
19      "protocol": "UDP",
20      "udp_payload_size": 1024
21    }
22  ],
23  "probes": [
24    {
25      "requested": 100,
26      "type": "area",
27      "value": "WW",
28      "tags": {
```

```
28     "include": ["system-ipv4-works", "system-resolves-a-correctly"]
29   }
30 }
31 ],
32 "is_oneoff": false ,
33 "interval": 86400
34 }
```

Listing A.1: *JSON configuration of a RIPE Atlas DNS measurement*

A.2 Probe Selection

Filtering criteria applied:

- **System tags:** system-ipv4-works, system-resolves-a-correctly
- **Hardware version 3+ (preferred)**
- **Geographic stratification:** inverse allocation across 6 continental regions
- **AS diversity:** minimum 1 probe per distinct AS within each stratum
- **Scheduling:** 02:00 UTC daily (off-peak hours)

Appendix B

Data Schema

B.1 Apache Avro Schema

Listing B.1: *Avro schema for DNS results (main fields)*

```
{
  "type": "record",
  "name": "DNSMeasurement",
  "namespace": "dns.measurements",
  "fields": [
    {"name": "msm_id", "type": "long"},
    {"name": "prb_id", "type": "long"},
    {"name": "timestamp", "type": "long"},
    {"name": "domain", "type": "string"},
    {"name": "query_type", "type": "string"},
    {"name": "campaign_type", "type": "string"},
    {"name": "probe_country", "type": "string"},
    {"name": "probe_asn", "type": "long"},
    {"name": "probe_lat", "type": "double"},
    {"name": "probe_lon", "type": "double"},
    {"name": "resolver_ip", "type": ["null", "string"]},
    {"name": "target_ip", "type": "string"},
    {"name": "rcode", "type": "int"},
    {"name": "answer_ips", "type": {"type": "array", "items": "string"}},
    {"name": "answer_ttl", "type": "long"},
    {"name": "nsid", "type": "string"},
    {"name": "rt_ms", "type": "double"},
    {"name": "valid", "type": "boolean"},
    {"name": "validation_flag", "type": "string"}
  ]
}
```